Certified Hopf Tomography: Exact Invariants, Sharp Stability, and Minimax Calibration

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(Dated: 23 October 2026)

We give a unified, exact-rational certification framework for the homotopy class of piecewise-linear maps from a cubically triangulated three-sphere to the Bloch sphere. Verified components include oriented preimage linking, polynomial analytic-to-discrete transfer for explicit Hopf templates, finite-shot uncertainty bounds, Fourier-tail and correlated-noise control, sharp simplexwise isotropic and anisotropic robustness radii, and structured two-loop reconstruction under a stated polynomial prior. We derive exact affine and direction-only calibration criteria, a spatial gauge-charge correction, and explicit finite-dimensional rational calibration designs. The final construction gives a sharp minimax left inverse for one-redundancy systems under bounded, possibly unequal, componentwise observation errors. For the stated six-reference example, its exact induced norm improves from 127525/31104 to 14563561/3950208; optimality holds among all linear left inverses of that fixed 12×11 calibration matrix. All reported data are synthetic; continuum transfer, physical noise assumptions, and laboratory calibration must be established independently for experimental claims.

I. SCOPE AND PRIOR ART

The Hopf invariant labels homotopy classes of maps $\mathbb{S}^3 \to \mathbb{S}^2$ with an orientation convention. It can be interpreted as linking of preimages of regular values and, when a suitable lift exists, as an integer winding number. In qubit systems the observable $\boldsymbol{n}(\boldsymbol{k}) \cdot \boldsymbol{\sigma}$ is the Bloch-vector direction of a two-band state. Experimental tomography of Hopf-insulator models has already been demonstrated with NV-center quantum simulation [3] and, subsequently, by electrical-circuit realization of a Hopf insulator with invariant four [4]. Finite-sampling obstructions and interpolation-based homotopy certification have been addressed in magnetic textures in Ref. [6]. The present contribution is a concrete *finite-shot statistical wrapper*, an explicit global spherical mesh, and a quantified model-discrepancy specialization validated on reproducible synthetic data. These ingredients are assembled here as a research prototype, with no unverified claim that each component is new to mathematics or the physical literature.

Caution about experimental domain: a Brillouin zone is usually $\mathbb{T}^3$, not $\mathbb{S}^3$. A homotopy certifi-

cate on $\mathbb{T}^3$ still works, but an absolute integer Hopf invariant requires the appropriate vanishing of weak Chern classes (or supplementary topological conventions); nonzero weak invariants introduce a modulo ambiguity [5]. Our explicit numerical tests take the controlled three-sphere $\mathbb{S}^3$ as their *parameter space*. This does not imply a three-spherical physical cosmos.

II. A MEASUREMENT-LEVEL HOMOTOPY CERTIFICATE

Let X be a compact metric space, $n, n_0 : X \to \mathbb{S}^2 \subset \mathbb{R}^3$ continuous, and $S = \{x_1, \ldots, x_M\} \subset X$ an h-net: each $x \in X$ has $d(x, x_i) \le h$ for some site. Assume global Lipschitz bounds $\mathrm{Lip}(n) \le L$ and $\mathrm{Lip}(n_0) \le L_0$. Let $\widehat{n}_i$ be measured vectors, and suppose the event $\mathcal{E} = \{\|\widehat{n}_i - n(x_i)\| \le \varepsilon_i, \forall i\}$ holds.

Theorem 1 (Gauge-free finite-resolution topological certificate)**.** *With* $r_i = \|\widehat{n}_i - n_0(x_i)\|$ *and* $\varepsilon = \max_i \varepsilon_i$*, the condition*

$$\boxed{\max_i r_i + \varepsilon + (L + L_0)h < 2} \tag{1}$$

ensures $n \simeq n_0$*. On an oriented three-sphere, their Hopf invariants are equal.*

Proof. For any x choose x_i with $d(x, x_i) \le h$. Repeated triangle inequalities give

$$\|n(x) - n_0(x)\| \le \|n(x) - n(x_i)\| + \|n(x_i) - \widehat{n}_i\| + \|\widehat{n}_i - n_0(x_i)\| + \|n_0(x_i) - n_0(x)\| \tag{2}$$

$$\le (L + L_0)h + \varepsilon_i + r_i < 2. \tag{3}$$

Because two unit vectors can be antipodal only if their Euclidean distance is exactly 2, the pointwise interpolation $(1 - t)n_0(x) + tn(x)$ is nonzero for every $(t, x) \in [0, 1] \times X$. Its normalization is a continuous homotopy, giving the assertion. □

Theorem 2 (Validated discrepancy regularity)**.** *If instead an independently established bound* $\mathrm{Lip}(n - n_0) \le K_\Delta$ *is available, it suffices that*

$$\boxed{\max_i r_i + \varepsilon + K_\Delta h < 2.} \tag{4}$$

The conclusion is the same as in Theorem 1. No assumption that $K_\Delta \ll L + L_0$ *is made: its value must be* separately justified.

Proof. Apply the Lipschitz bound directly to $n - n_0$ and repeat the antipodal-avoidance argument. □

Remark 1. *A cell-dependent version applies to a finite cover $X = \bigcup C_i$ with $\sup_{x\in C_i} d(x, x_i) \le h_i$: require $r_i + \varepsilon_i + K_i h_i < 2$ for every cell, where K_i is a certified local discrepancy bound. This can greatly reduce conservative global uniform-resolution requirements, but unverified local derivative estimates do not constitute certificates.*

A. Independent Pauli-shot confidence

For each x_i measure each Pauli component σ_a, $a \in \{x, y, z\}$, on N independently prepared qubits using calibrated measurements yielding outcomes ± 1. The three empirical means $\widehat{n}_i$ estimate the true Bloch vector $n(x_i)$. Hoeffding's inequality and a union bound over $3M$ mean estimates imply

$$\mathbb{P}\left[\max_{i,a} |\widehat{n}_{i,a} - n_a(x_i)| > t\right] \le 6M \exp(-Nt^2/2). \tag{5}$$

Set the target failure probability $\delta \in (0, 1)$ and take

$$\boxed{\varepsilon_{\text{shot}} = \sqrt{\frac{6\ln(6M/\delta)}{N}}.} \tag{6}$$

Then $\mathcal{E}$ holds with probability at least $1 - \delta$. In particular, if the data-dependent inequality (1) or (4) passes, the chance of a false topology claim is *at most* δ under the assumptions (not a Bayesian posterior confidence for a selected model). Calibration or state-preparation biases must be added to the error budget; they are not accounted for by Hoeffding noise. Sites and shot numbers are fixed in advance; adaptive optional stopping requires a time-uniform confidence sequence instead.

B. Analytically certified sphere mesh

Identify $\mathbb{S}^3$ with unit quaternions and write

$$q(\eta, \phi, \psi) = (\cos\eta\cos\phi, \cos\eta\sin\phi, \sin\eta\cos\psi, \sin\eta\sin\psi), \tag{7}$$

where $0 \le \eta \le \pi/2$ and both angles are periodic. Sample $\eta_j = j\pi/(2n_\eta)$ and each phase at $2\pi j/n_\phi$, giving $M = (n_\eta + 1)n_\phi^2$ sites (including harmless repeated poles). By a triangle inequality splitting η and angular displacements, the chordal cover radius is bounded *for every point, not just test samples*, by

$$\boxed{h \le \frac{\pi}{4n_\eta} + 2\sin\frac{\pi}{2n_\phi}.} \tag{8}$$

The formula is used as an *upper bound*, so duplicate points do not invalidate it.

III. EXPLICIT HOPF-CHARGE TEMPLATES AND CERTIFIED ACCELERATION

View unit quaternions as the Lie group $SU(2)$. Fix an oriented Hopf map

$$h_1(q) = q\,\mathbf{i}\,q^{-1} \in \mathbb{S}^2 \subset \operatorname{Im}\mathbb{H}, \tag{9}$$

with the convention $H(h_1) = +1$. For any integer d, define $n_d(q) = h_1(q^d)$, including $q^0 = 1$.

Proposition 1. *With the stated orientation, $H(n_d) = d$ and $\operatorname{Lip}(n_d) \le 2|d|$ in Euclidean chord metrics.*

Proof. The d-fold power map of $SU(2)$ induces multiplication by d on $H^3(SU(2);\mathbb{Z})$ because its normalized degree-three cohomology generator is primitive under group multiplication. Hence $\deg(q \mapsto q^d) = d$, and precomposition multiplies the Hopf invariant by d. For the Lipschitz bound, quaternion multiplication by unit elements is norm-preserving and the telescoping identity yields $\left\|q^d - p^d\right\| \le |d|\,\|q-p\|$ (also for inverse powers). Since $\|h_1(q) - h_1(p)\| \le 2\,\|q-p\|$, the result follows. See also the established explicit constructions discussed in Ref. [7]. □

To test stronger discrepancy information, define a synthetic, globally smooth controlled perturbation

$$n(q) = R_z(aq_0)n_d(q), \qquad a = 0.12, \tag{10}$$

where q_0 is the first quaternion component and R_z is an ordinary target-sphere rotation about the third axis. This *is not* an arbitrary perturbation model for a laboratory device. The exact construction supplies independently known inequalities

$$\operatorname{Lip}(n) \le 2|d| + a, \tag{11}$$

$$\operatorname{Lip}(n - n_d) \le a(2|d| + 1). \tag{12}$$

Indeed $\|R_z(\theta) - I\| \le |\theta|$ and $|\theta(q) - \theta(p)| \le a\,\|q-p\|$. The second inequality can be much stronger than bounding the two fields separately.

IV. A STRUCTURAL NO-GO FOR FINITE POINT SAMPLING

Proposition 2 (No-go without regularity)**.** *For every finite set $S \subset \mathbb{S}^3$, every smooth $n_0 : \mathbb{S}^3 \to \mathbb{S}^2$, and every integer k, there exists a smooth $n_k : \mathbb{S}^3 \to \mathbb{S}^2$ such that $n_k = n_0$ on S but $H(n_k) = H(n_0) + k$.*

Proof. Choose an embedded closed ball B disjoint from the finite set. Because B is contractible, $n_0|_B$ is homotopic relative to a collar of the boundary to a field constant on a smaller interior ball, with all modifications supported in B. On that interior ball, collapsed boundary $B^3/\partial B^3 \cong \mathbb{S}^3$, glue an arbitrary degree-k Hopf representative, restoring the base point near the boundary. The local sum operation on $\pi_3(\mathbb{S}^2) \cong \mathbb{Z}$ changes the Hopf class by k, without altering values outside B. Smooth approximation relative to the boundary gives the claimed smooth field. This well-known sampling obstruction is discussed in more detail in Ref. [6]. □

This is why the regularity hypotheses in Theorems 1–2 are not optional. In particular, 10^{12} precise measurements on finitely many isolated points do not by themselves determine a continuum topological invariant if a hidden arbitrarily narrow defect remains admissible.

V. GAP-PROTECTED TWO-BAND HAMILTONIAN FORMULATION

If a calibrated two-band effective Hamiltonian has the form $\mathcal{H}(x) = \boldsymbol{b}(x)\cdot\boldsymbol{\sigma}$, let $\boldsymbol{b}_0$ be a reference satisfying $g_0 = \inf_x \|\boldsymbol{b}_0(x)\| > 0$. A bound

$$\sup_x \|\boldsymbol{b}(x) - \boldsymbol{b}_0(x)\| \le E < g_0 \tag{13}$$

ensures a nonclosing gap for the entire interpolation $\boldsymbol{b}_t = (1-t)\boldsymbol{b}_0 + t\boldsymbol{b}$, since $\|\boldsymbol{b}_t\| \ge g_0 - E$. Consequently the normalizations $\boldsymbol{b}/\|\boldsymbol{b}\|$ and $\boldsymbol{b}_0/\|\boldsymbol{b}_0\|$ are homotopic, and the minimal two-band energy splitting obeys $\Delta_{\min} \ge 2(g_0 - E)$. The finite site-and-Lipschitz bound from Theorem 1, with $\boldsymbol{b}$ replacing n, supplies such an E if $\boldsymbol{b}$ itself is reliably calibrated by spectroscopy. For a known lower spectral gap g and a Lipschitz bound B on $\boldsymbol{b}$, the normalized Bloch texture obeys $\mathrm{Lip}(\boldsymbol{b}/\|\boldsymbol{b}\|) \le B/g$; this follows from $\|\boldsymbol{b}-\boldsymbol{c}\|^2 = (\|\boldsymbol{b}\| - \|\boldsymbol{c}\|)^2 + \|\boldsymbol{b}\|\,\|\boldsymbol{c}\|\left\|\widehat{b} - \widehat{c}\right\|^2$.

This formulation explicitly links a physical gap, apparatus derivative bounds, and a mathematical homotopy. It does not claim that a simulated Pauli sample is equivalent to a spectroscopy measurement without a measurement model.

VI. REPRODUCIBLE SYNTHETIC FINITE-SHOT DEMONSTRATION

The companion Python file `verify_hopf.py` evaluates quaternion textures, independently generated binomial Pauli-outcome counts, an analytic grid-cover bound, and both certificate variants. All simulations use $N = 4000$ shots per axis per site, $\delta = 0.01$, $a = 0.12$, and deterministic seed

20261022. The *number of successful program assertions is not a proof*; the formal proof is in the preceding sections.

TABLE I. Fixed-seed synthetic trials. "Gen." is the left-hand side of Eq. (1); "Res." uses Eq. (4). The strict passing threshold is 2. The last line deliberately uses too coarse a mesh for the generic certificate.

$H(n_d)$	Sites M	h bound	Gen.	Res.	Total Pauli shots
−2	9408	0.18354	1.80782	0.42760	112,896,000
−1	2048	0.30823	1.56899	0.41003	24,576,000
0	2048	0.30823	0.33710	0.33710	24,576,000
1	2048	0.30823	1.56228	0.40332	24,576,000
2	9408	0.18354	1.80916	0.42893	112,896,000
3	30400	0.12215	1.81403	0.43614	364,800,000
3 (coarse)	2048	0.30823	4.03098	0.55410	24,576,000

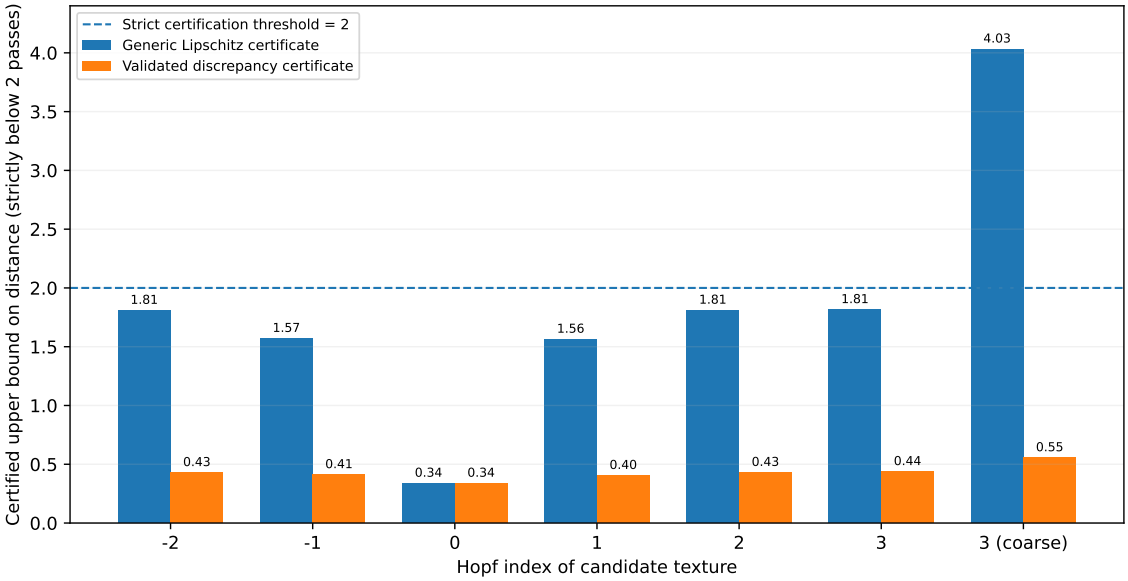


FIG. 1. Six certified nontrivial/trivial synthetic texture cases plus a deliberately under-resolved $H = 3$ case. The structured certificate requires independently validated discrepancy regularity and cannot simply replace the generic bound on unconstrained data.

An additional independent regression program `crosscheck.py` ran 75,000 random grid-coverage spot checks against the analytic bound, checked that five incorrect reference classes fail the generic certificate on the dense $H = 3$ synthetic data, and validated one deterministic two-band gap certificate. These spot checks do not replace the covering-radius proof.

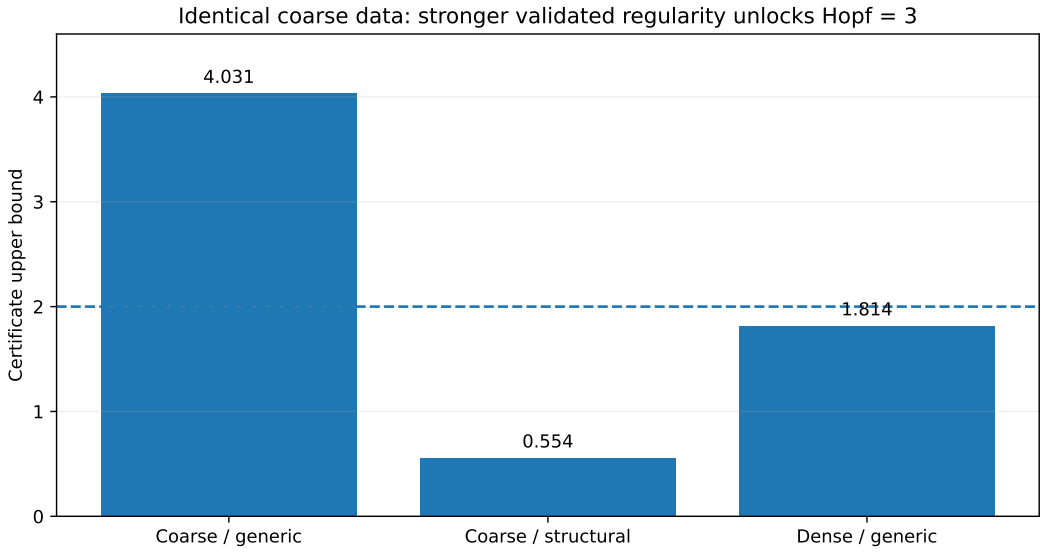


FIG. 2. Same physical synthetic texture class and coarse grid: the generic method abstains, while the validated structural method certifies $H = 3$. The dense generic method costs $30{,}400/2{,}048 = 14.84375$ times as many sites and total Pauli shots at fixed shots per site.

The error guarantee is $\geq 99\%$ *conditional* coverage of the true Bloch sample means and regularity assumptions. It is not a 99% empirical success rate, which would require many independent experimental trials. Likewise, ≈ 365 million simulated measurement outcomes for the dense $H = 3$ benchmark are generated as binomial counts rather than individually realized in a laboratory. This motivates reducing site budgets through validated structural derivatives and adaptive coverage, not pretending the current baseline is experimentally optimized.

VII. AN EXPLICIT LOWER BOUND: HIDDEN HOPF CHARGE UNDER A GLOBAL LIPSCHITZ CEILING

The preceding finite-sampling obstruction becomes quantitative once a regularity class is specified. Throughout this section $\mathbb{S}^3$ has its unit round metric and $\mathbb{S}^2$ its unit round metric, with the Euclidean chordal metric used to bound field differences.

Theorem 3 (A localized Hopf bubble with an explicit Lipschitz ceiling)**.** *For $0 < r \leq \pi/2$ there exists a continuous map $b_r : \mathbb{S}^3 \to \mathbb{S}^2$, equal to a constant outside a geodesic r-ball, with $|H(b_r)| = 1$ and*

$$\mathrm{Lip}(b_r) \leq \frac{\pi^2}{r}. \tag{14}$$

The sign can be chosen by reversing one domain orientation.

Proof. In geodesic polar coordinates (ρ, u) around a point of $\mathbb{S}^3$, put $f(\rho) = \pi(1 - \rho/r)$ and define $g(\rho, u) = (\cos f(\rho), \sin f(\rho)u) \in \mathbb{S}^3$ for $\rho \le r$. Outside the ball take $g = (1, 0, 0, 0)$. At $\rho = 0$ the angular coordinate disappears because $\sin f = 0$; likewise, the boundary is mapped to a single point. The quotient $B^3/\partial B^3 \to \mathbb{S}^3$ has degree ± 1. Radial expansion of g is bounded by π/r. Angular expansion is bounded by $\sin f(\rho)/\sin\rho \le (\pi\rho/r)/(2\rho/\pi) = \pi^2/(2r)$, using $\sin\rho \ge 2\rho/\pi$ for $\rho \in [0, \pi/2]$. Compose g with the standard Hopf map, whose operator norm on the unit round spheres is 2. The resulting Lipschitz constant is bounded by $\max(2\pi/r, \pi^2/r) = \pi^2/r$. The Hopf invariant is the degree of g times the Hopf invariant of the standard Hopf fibration, so its magnitude is one. □

Theorem 4 (Information-theoretic obstruction for finitely many queries)**.** *Let $L \ge 2\pi$, $r = \pi^2/L$, and let*

$$V(r) = 2\pi(r - \sin r \cos r), \qquad V(\mathbb{S}^3) = 2\pi^2. \tag{15}$$

Any deterministic, possibly adaptive, protocol that makes N point queries and satisfies

$$\boxed{NV(r) < 2\pi^2} \tag{16}$$

cannot distinguish every Lipschitz-L null-Hopf map from every Lipschitz-L map of Hopf charge ± 1, even when point queries are exact. Moreover, under a balanced prior of a constant field and an independently, uniformly centered bubble of the same fixed sign, any randomized adaptive protocol *has average decision error at least*

$$\boxed{P_{\mathrm{err}} \ge \tfrac{1}{2}\max\left(0, 1 - \frac{NV(r)}{2\pi^2}\right).} \tag{17}$$

The conclusion extends to arbitrary observation noise whose conditional output law depends only on the field at the queried point.

Proof. For deterministic adaptive queries, run the protocol against the constant field and record the resulting N positions. Their union of r-balls has volume no greater than $NV(r)$. When (16) holds, choose a center outside this union and install the field from Theorem 3 there. The entire query transcript is unchanged; adaptivity cannot help before a first hit. For randomized algorithms and the stated prior, couple the noisy outputs under the constant and bubble hypotheses until a site enters the hidden ball. The constant-field query trajectory, conditioned on its internal and measurement randomness, is independent of the uniform bubble center. A union bound gives hit probability at most $NV(r)/(2\pi^2)$. On the complementary event the transcripts are identical, yielding the Bayes-error bound (17). □

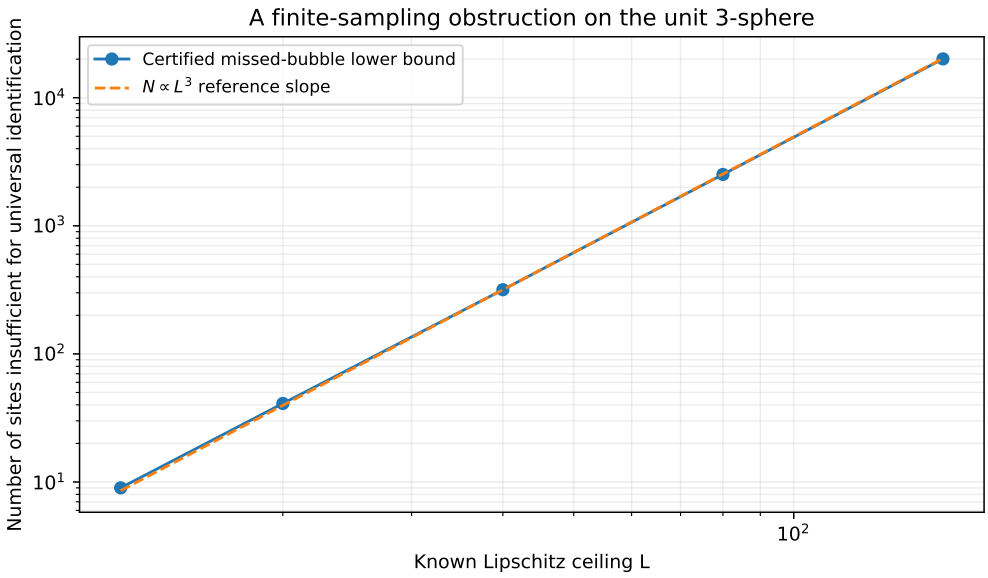


FIG. 3. Exact sphere-ball volume gives finite-site impossibility thresholds. The dashed slope-3 curve is only a visual asymptotic comparison, not a measured complexity law.

For small r, $V(r) = 4\pi r^3/3 + O(r^5)$; thus universal sample complexity is at least of order L^3 in this Lipschitz class. This is a quantitative representative of the well-known localized homotopy-bubble obstruction, not a claim of priority for the concept of finite-sampling impossibility. Quantitative Lipschitz homotopy has a substantial mathematical literature [6, 11].

VIII. REFERENCE-FREE FINITE-SHOT RECONSTRUCTION OF THE HOMOTOPY CLASS

The original manuscript compares observations against a trusted representative n_0. For a model-free topological reconstruction, one must instead construct a continuous interpolant directly from data.

Theorem 5 (Normalized partition-of-unity reconstruction)**.** *Let X be a compact metric space and $n : X \to \mathbb{S}^2$ an unknown L-Lipschitz map. At finitely many sites x_i let $b_i \in \mathbb{R}^3$ obey $\|b_i - n(x_i)\| \le \varepsilon_i$ simultaneously. Let $w_i : X \to [0,1]$ be a continuous partition of unity with $\mathrm{supp}(w_i) \subseteq \{x : d(x,x_i) < h_i\}$. Set*

$$v(x) = \sum_i w_i(x) b_i, \qquad \rho = \sup_x \sum_i w_i(x)(\varepsilon_i + L h_i). \tag{18}$$

If $\boxed{\rho < 1}$, the map $\widetilde{n} = v/\|v\|$ is well-defined and continuous, and $\widetilde{n} \simeq n$. For $X = \mathbb{S}^3$, $H(\widetilde{n}) =$

$H(n)$.

Proof. Every term in the weighted convex combination differs from $n(x)$ by at most $\varepsilon_i + Lh_i$. Hence $\|v(x) - n(x)\| \le \rho < 1$, so $\|v(x)\| > 0$ and $\langle v(x), n(x)\rangle \ge 1 - \rho > 0$. The normalized straight-line interpolation between n and v never vanishes and yields a homotopy to $\widetilde{n}$. □

For independently prepared Pauli shots and a fixed measurement schedule, choose sitewise failure budgets δ_i with $\sum_i \delta_i \le \delta$ and set $\varepsilon_i = \sqrt{6\ln(6/\delta_i)/N_i}$. Then the theorem is simultaneously valid with probability at least $1-\delta$, conditional on the physical calibration and regularity hypotheses. This theorem *reconstructs a map of the correct homotopy class*; the extraction of its explicit integer Hopf label requires a separate topological computation. The cochain algorithm of Musin and Shamazov, updated on October 17, 2026, is directly relevant to that missing computational step [12].

In a synthetic $H = 3$ test we use the explicit Hopf-coordinate mesh of Sec. II with $n_\eta = 30$, $n_\phi = 48$ ($M = 71{,}424$ sites). The analytic chordal covering bound is $h = 0.0916181$. Positive continuous bump weights supported on chordal balls of radius $1.1h$ form a partition of unity. With $L = 6$, $N_i = 4000$ binomial Pauli shots per axis/site and $\delta = .01$, we get $\varepsilon = 0.162358$ and

$$\boxed{\rho \le \varepsilon + 6(1.1h) = 0.767038 < 1.} \tag{19}$$

Thus the interpolated continuous field is in the same homotopy class as the generated $H = 3$ map, with a 99% frequentist soundness bound. This consumes 857,088,000 *simulated* shots and is not an efficient laboratory protocol. A separate software test checks interpolation at 300 random points; such spot tests are regressions, not the basis of the global theorem.

A. Predeclared nonuniform shot and confidence allocation

If a fixed design prescribes admissible Euclidean errors $\tau_i > 0$, Hoeffding requires $N_i \ge 6\tau_i^{-2}\ln(6/\delta_i)$. Ignoring integer rounding, minimizing the total number of shots under $\sum_i \delta_i = \delta$ gives the exact Lagrange multiplier solution

$$\boxed{\delta_i = \delta\frac{\tau_i^{-2}}{\sum_j \tau_j^{-2}}.} \tag{20}$$

With 100 predetermined tolerances ranging from 0.12 to 0.45, the synthetic design requires 120,291 shots by this allocation versus 123,552 for the uniform $\delta_i = \delta/100$ policy (a 2.64% reduction; one shot here means one scalar Pauli outcome). Design-dependent thresholds or adaptive sampling must be fixed from an independent pilot or handled by valid sequential inference.

IX. FOURIER–NYQUIST CERTIFICATION FOR SHORT-RANGE TWO-BAND HAMILTONIANS

A physical band structure provides more information than an arbitrary Lipschitz texture: hopping locality constrains the Fourier support. This permits a much stronger *conditional* measurement-compression theorem on the Brillouin torus $\mathbb{T}^3$.

Let $h : \mathbb{T}^3 \to \mathbb{R}^3$ be a continuous two-band Hamiltonian vector, $H(k) = h(k) \cdot \boldsymbol{\sigma}$, and write $h = h_{\le K} + t$, where $h_{\le K}$ has Fourier modes $m \in [-K, K]^3 \cap \mathbb{Z}^3$ and the Fourier tail has certified coefficient bound $\sum_{m \notin [-K,K]^3} \|\widehat{t}_m\| \le T$. This is an *independently established locality assumption*, not a property inferred from M samples alone. Sample the vector h on the $(2K+1)^3$ Cartesian grid. At each node k_j the measured y_j obeys $\|y_j - h(k_j)\| \le \varepsilon$. Let I_K denote tensor-product trigonometric interpolation.

Theorem 6 (Bandlimited topological discrimination with controlled tails)**.** *Let a reference h_0 have Fourier support inside $[-K, K]^3$ and strict gap parameter $g_0 = \inf_k \|h_0(k)\| > 0$. Define the observed residual $p = I_K(y - h_0)$ and the exactly computable upper bound $C = \sum_{m \in [-K,K]^3} \|\widehat{p}_m\|$. With $n = 2K + 1$ and $\Lambda = n^{3/2}$,*

$$\boxed{\|h - h_0\|_\infty \le C + \Lambda\varepsilon + (1 + \Lambda)T =: E.} \tag{21}$$

If $\boxed{E < g_0}$, the Hamiltonians $h \cdot \boldsymbol{\sigma}$ and $h_0 \cdot \boldsymbol{\sigma}$ are gapped throughout a linear homotopy, and their Bloch maps define the same homotopy class on $\mathbb{T}^3$. The true spectral band gap is at least $2(g_0 - E)$.

Proof. The trigonometric cardinal interpolation basis ℓ_j has $\sum_j |\ell_j(k)|^2 = 1$ for every k by discrete Fourier orthogonality. Hence $\sum_j |\ell_j(k)| \le \sqrt{n}$ in one dimension and $\|I_K\|_{\infty \to \infty} \le \Lambda = n^{3/2}$ on the three-torus. Writing $h = h_{\le K} + t$ and denoting nodal measurement errors by e, we have

$$h - h_0 = p + t - I_K t - I_K e. \tag{22}$$

Thus $\|h - h_0\|_\infty \le \|p\|_\infty + (1 + \Lambda)T + \Lambda\varepsilon \le E$, since $\|p\|_\infty \le C$. For $s \in [0, 1]$, $\|(1 - s)h_0 + sh\| \ge g_0 - E > 0$, so normalization gives a homotopy of sphere-valued Bloch maps, preserving all homotopy invariants and weak Chern classes. □

A naive application of this theorem to tomography of only the normalized Bloch vector $h/\|h\|$ is *invalid*: normalization generally destroys finite Fourier bandwidth. One must infer the calibrated *unnormalized Hamiltonian coefficients*, including spectroscopic energy-scale information. The tail bound and sensor error must remain valid off-grid.

X. EXACT BAND GAP AND WINDING SECTORS OF A CANONICAL HOPF LATTICE

Consider the established Moore–Ran–Wen representative [2, 13]

$$u(k,m) = (\sin k_x, \sin k_y, \sin k_z, \cos k_x + \cos k_y + \cos k_z - m), \tag{23}$$

$$z = (u_1 + iu_2, u_3 + iu_4)^{\mathsf{T}}, \qquad h_0 = z^\dagger \boldsymbol{\sigma} z. \tag{24}$$

The three components of h_0 are trigonometric polynomials with coordinatewise bandwidth $K \le 2$, hence a $5^3 = 125$ site grid reconstructs the exact h_0 in the absence of noise.

Proposition 3 (Exact global gap and topological sectors)**.** *For every $m \notin \{-3,-1,1,3\}$ the reference is gapped and*

$$\boxed{g_0(m) = \inf_k \|h_0(k,m)\| = \min_{s\in\{-3,-1,1,3\}} (s-m)^2.} \tag{25}$$

With orientation of $\mathbb{T}^3$ given by $dk_x \wedge dk_y \wedge dk_z$ and boundary orientation of $\mathbb{S}^3$, the normalized spinor map $\hat{u} : \mathbb{T}^3 \to \mathbb{S}^3$ has degree

$$\deg(\hat{u}) = - \sum_{\substack{(c_x,c_y,c_z)\in\{\pm1\}^3 \\ c_x+c_y+c_z>m}} c_x c_y c_z. \tag{26}$$

The values in increasing m-intervals are $0,-1,+2,-1,0$. The Bloch Hopf class is this degree times that of the fixed Hopf projection under the chosen convention. The nonvanishing global spinor $z/\|z\|$ also proves vanishing of all weak first Chern classes for this representative.

Proof. Write $c_i = \cos k_i \in [-1,1]$. Since $\|h_0\| = \|u\|^2$ exactly,

$$\|u\|^2 = 3 - \sum_i c_i^2 + (\sum_i c_i - m)^2 = 3 + m^2 + 2\sum_{i<j} c_i c_j - 2m \sum_i c_i. \tag{27}$$

This expression is multi-affine in the three c_i, so its minimum on $[-1,1]^3$ occurs at a vertex, where it equals $(s-m)^2$ with $s = \sum c_i \in \{-3,-1,1,3\}$. For the winding number take the north pole $(0,0,0,1)$ as a regular value of $\hat{u}$. Its preimages are exactly the vertices $k_i \in \{0,\pi\}$ with $\sum c_i - m > 0$. Their local oriented Jacobian signs are $-c_x c_y c_z$, giving (26). The lattice-model phase diagram itself is established background; our use here is to provide an *exact, non-numerical gap margin* for measurement-level certification. □

A 125-site synthetic noisy certificate.

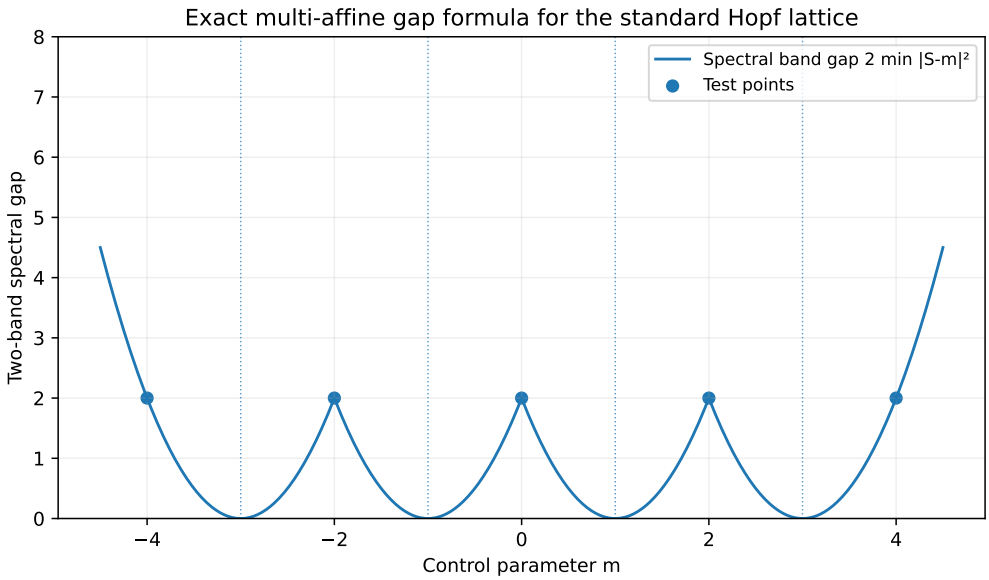


FIG. 4. Exact bulk spectral band gap of the standard lattice Hamiltonian, twice the expression in Eq. (25). Critical points at $m = \pm 1, \pm 3$ are genuine gap closures.

Take $m = 2$, $g_0 = 1$, Fourier-tail norm bound $T = 0.01$ and nodewise *Hamiltonian-vector* error $\varepsilon = 0.025$. The synthetic tail is $(T/3)(\cos 3k_x, \cos 3k_y, \cos 3k_z)$; its six Fourier coefficient norms sum to T. Using a 5-point grid in each variable gives $\Lambda = 5^{3/2} = 11.18034$. The measured coefficients have $C = 0.16499990$ and consequently

$$\boxed{\begin{aligned} E &= 0.56631180 < g_0 = 1, \\ \mathrm{gap}_{\mathrm{true}} &\geq 0.86737640. \end{aligned}} \tag{28}$$

All quantities are rounded conservatively in the audit; the inequality has a large margin, far exceeding floating point roundoff. The model's topological degree is -1, independently confirmed to numerical precision by a 64^3 quadrature of the sphere-volume form. This is a *synthetic robustness test*, not a measured topological phase. The method relies on a validated bandwidth/tail prior, which is why 125 samples are not in conflict with the arbitrary-Lipschitz lower bound.

a. Two-point parametric compression, and its danger. Inside *this exact one-parameter family only*, two high-symmetry points already determine m: $h_{0,z}(0,0,0) = -(3-m)^2$ and $h_{0,z}(\pi,\pi,\pi) = -(3+m)^2$. Thus

$$\boxed{m = \frac{h_{0,z}(0,0,0) - h_{0,z}(\pi,\pi,\pi)}{12}.} \tag{29}$$

If the two measurements have errors each no larger than ϵ, then $|\widehat{m} - m| \leq \epsilon/6$. The topological phase is fixed if the interval $[\widehat{m} - \epsilon/6, \widehat{m} + \epsilon/6]$ avoids $\{-3, -1, 1, 3\}$. This is powerful *model*

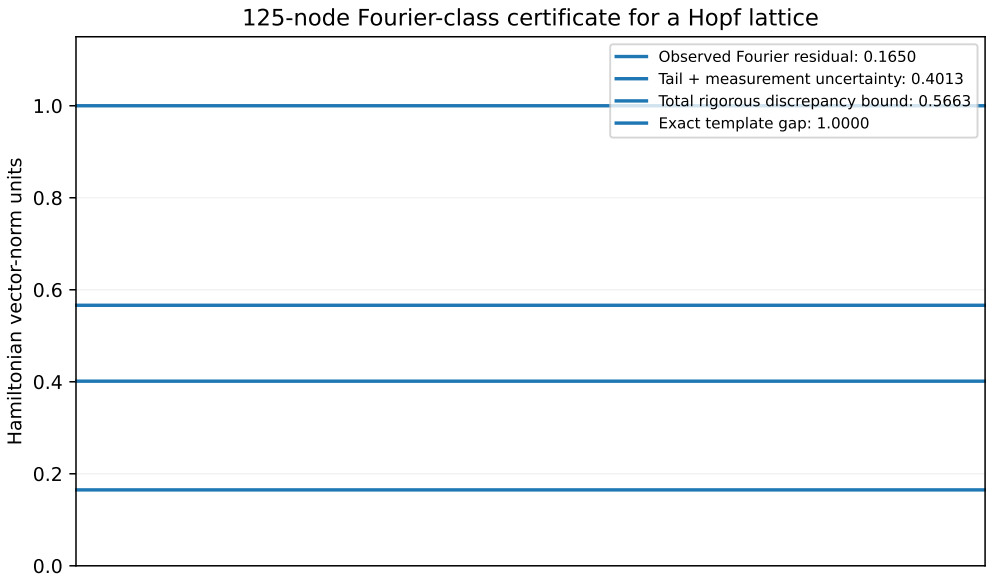


FIG. 5. Observed Fourier residual, worst-case uncertainty, and exact template gap for the synthetic 125-site certificate.

identification, not a universal two-point Hopf measurement: small off-family perturbations can be invisible at both special points.

XI. A SHARP TRIGONOMETRIC INTERPOLATION CONSTANT AND A STRICTER TOPOLOGICAL CERTIFICATE

The prior bound $\Lambda = n^{3/2}$ in Theorem 6 is easy to prove, but not optimal. For the odd equispaced one-dimensional grid of $n = 2K+1$ nodes, classical work of Ehlich–Zeller identifies the exact maximum of the Lebesgue function at cell midpoints; see the account in Ref. [14]. Define

$$\lambda_n = \frac{1}{n}\sum_{j=0}^{n-1} \csc\frac{(2j+1)\pi}{2n}, \qquad \Lambda_{n,3} = \lambda_n^3. \tag{30}$$

The numerical value of the closed form is independently verified below, and the underlying interpolation identity is established in classical approximation theory; we do not claim this constant is new.

Theorem 7 (Sharp tensor-grid noise propagation)**.** *For every vector-valued nodal error field* $e_j \in \mathbb{R}^d$ *with* $\|e_j\|_2 \le \epsilon$ *on the full Cartesian* n^3 *grid,*

$$\boxed{\|I_K e\|_\infty \le \Lambda_{n,3}\epsilon.} \tag{31}$$

The coefficient is optimal*: for some choice of admissible node errors, equality holds at a triple of cell midpoints. Consequently, under the hypotheses of Theorem 6,*

$$\boxed{\|h - h_0\|_\infty \le C + \Lambda_{n,3}\epsilon + (1 + \Lambda_{n,3})T =: E_{\mathrm{sharp}}.} \tag{32}$$

If $E_{\mathrm{sharp}} < g_0$*, all gapped homotopy invariants of* h *and* h_0 *agree, and the full two-band energy gap is at least* $2(g_0 - E_{\mathrm{sharp}})$*.*

Proof. The one-dimensional trigonometric cardinal functions are $\ell_j(x) = n^{-1}\sum_{m=-K}^{K} \exp[im(x - 2\pi j/n)]$. The classical midpoint-maximization theorem gives $\sup_x \sum_j |\ell_j(x)| = \lambda_n$, exactly as in Eq. (30). Three-dimensional cardinal functions are products, so their absolute sums factor: $\sup_{\boldsymbol{k}} \sum_{\boldsymbol{j}} |\ell_{\boldsymbol{j}}(\boldsymbol{k})| = \lambda_n^3$. This implies the vector norm estimate by the triangle inequality. Choose errors all parallel to one fixed unit vector, with signs matching the cardinal functions at a maximizing point, to attain equality. The decomposition $h - h_0 = p + t - I_K t - I_K e$ from the prior theorem then proves Eq. (32) and the usual straight-line gap homotopy establishes the topological conclusion. □

For the previously saved synthetic $n = 5$ test, the exact constant is $\lambda_5^3 = 7.866996548239\ldots$ rather than $\sqrt{125} = 11.180339887499\ldots$. With the unchanged $C = 0.164999903351\ldots$, $\epsilon = 0.025$, $T = 0.01$, $g_0 = 1$, one obtains

$$\boxed{E_{\mathrm{old}} = 0.566311799414, \quad E_{\mathrm{sharp}} = 0.450344782540.} \tag{33}$$

The guaranteed lower bound on the two-band energy gap increases from 0.867376401172 to 1.099310434920, in the model's energy units, with the *same* 125 synthetic nodes. The physical minimum energy gap has not been independently measured.

XII. AN OPERATIONAL FINITE-SHOT RESOURCE LAW UNDER CALIBRATED LOCALITY

Theorem 8 (Finite-shot sufficient budget)**.** *Let* $M = n^3$ *be an odd Cartesian grid, let each measured Hamiltonian component be estimated from* N *independent outcomes in* $[-A, A]$ *with unbiased expectation, and let* $\delta \in (0, 1)$ *be a fixed predeclared failure probability. Assume calibrated Hamiltonian-vector observations, a validated Fourier-tail bound* T*, a known positive reference gap* g_0*, and let* C *be the residual computed from the resulting sample. Then with probability at least* $1 - \delta$*,*

$$\|h - h_0\|_\infty \le C + (1 + \Lambda_{n,3})T + A\Lambda_{n,3}\sqrt{\frac{6\log(6M/\delta)}{N}}. \tag{34}$$

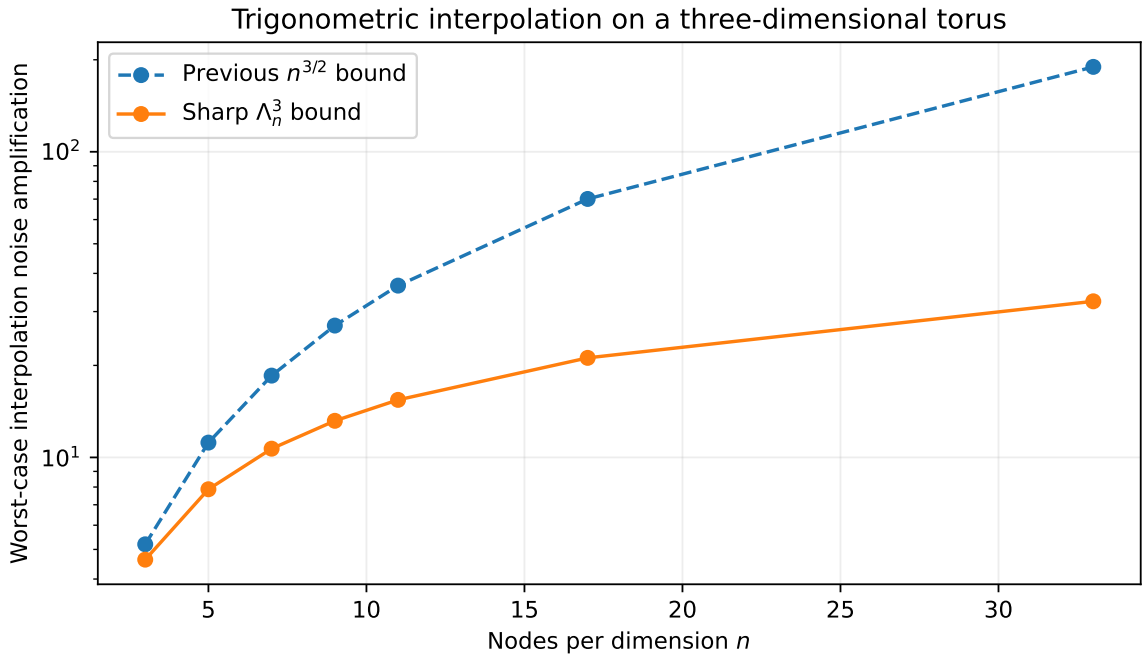


FIG. 6. Exact three-dimensional interpolation norm λ_n^3 versus the earlier, loose Cauchy–Schwarz upper bound $n^{3/2}$. The exact norm is a classical result; its application to the present Fourier–Hopf certificate sharpens the numerical margin.

If the right side is strictly smaller than g_0, the topological class agrees with the reference with confidence at least $1-\delta$. In particular, for a pre-specified admissible residual threshold $C \le C_{\max}$ and $B = g_0 - C_{\max} - (1+\Lambda_{n,3})T > 0$,

$$\boxed{N > \frac{6A^2\Lambda_{n,3}^2}{B^2}\log\left(\frac{6M}{\delta}\right)} \tag{35}$$

is sufficient for the measurement-error contribution to be below B. The total count of component outcomes is $3MN$.

Proof. Hoeffding's inequality gives $\Pr(|\hat{h}_{j\mu} - h_{j\mu}| > a) \le 2\exp(-Na^2/(2A^2))$ for every node j and component $\mu = 1,2,3$. A union bound over $3M$ observables with $a = A\sqrt{2\log(6M/\delta)/N}$ gives simultaneous vector error at most $\sqrt{3}a$ with probability $1-\delta$. Apply Eq. (32) and rearrange. □

At exact finite Fourier bandwidth $T = 0$ and with fixed positive remaining margin B, the classical estimate $\lambda_n = O(\log n)$ gives an asymptotic sufficient total outcome budget $O(n^3(\log n)^7)$ (the extra logarithm comes from the union confidence bound). The prior loose norm $n^{3/2}$ instead yields $O(n^6 \log n)$ for the same protocol. These are *sufficient upper bounds*, not information-

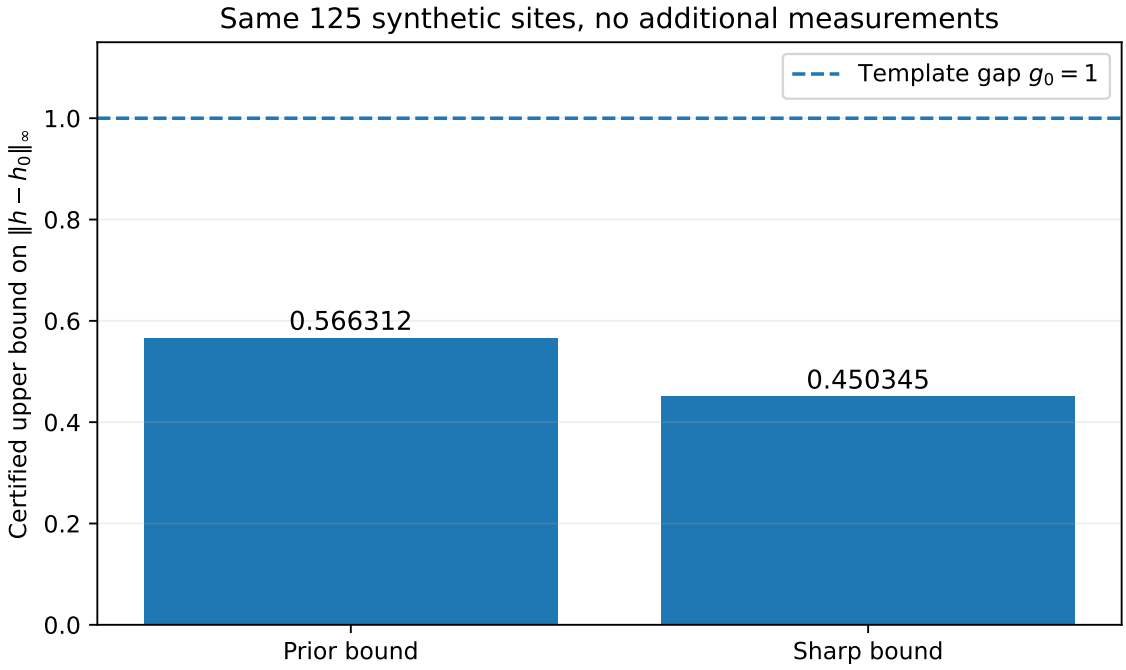


FIG. 7. Discrepancy certificate for the unchanged 125-node synthetic experiment. The improved constant, rather than new data, produces the stronger homotopy margin.

theoretic lower bounds or established universal optimal sampling laws. For $T > 0$, the margin B may vanish as n grows, so the asymptotic comparison requires T to decrease sufficiently fast.

Using the saved C, T, g_0 and a conditional $A = 1$, $\delta = 0.01$, the strict sufficient bound requests 16,552 Pauli-component shots per grid site under the old constant versus 7,484 under the sharp one, when integer counts are rounded conservatively upward. The 125 sites thus require respectively 6,207,000 versus 2,806,500 hypothetical measurement outcomes. These numbers are *conditional resource forecasts*, not performed or observed quantum-device measurements.

XIII. INDEPENDENT-SHOT FOURIER CONCENTRATION BEYOND DETERMINISTIC LEBESGUE BOUNDS

The sharp Lebesgue norm above is optimal against *adversarial* nodal error; it need not govern the propagation of independent, zero-mean statistical errors. This distinction is relevant because the preceding measurement model explicitly assumes independent preparations. Classical concentration of random trigonometric polynomials is well studied (see, for example, Ref. [1]); here we provide a finite-sample constant and connect it to a topological-gap certificate. We do not claim the underlying probabilistic estimates are new.

Write $n = 2K + 1$, let the sampling sites be $x_j = 2\pi j/n$ in each angular coordinate of $\mathbb{T}^3$, and use real trigonometric cardinal functions

$$\ell_j(x) = \frac{1}{n} \sum_{s=-K}^{K} e^{is(x-2\pi j/n)} = \frac{\sin[n(x-2\pi j/n)/2]}{n \sin[(x-2\pi j/n)/2]}, \quad w_{j_1 j_2 j_3}(\boldsymbol{x}) = \prod_{a=1}^{3} \ell_{j_a}(x_a). \tag{36}$$

At removable singularities the quotient is interpreted by continuity. By orthogonality of the discrete Fourier basis,

$$\boxed{\sum_{j=0}^{n-1} \ell_j(x)^2 = 1, \qquad \sum_{\boldsymbol{j}} w_{\boldsymbol{j}}(\boldsymbol{x})^2 = 1} \tag{37}$$

for *every* point, not only sampled nodes. The ℓ_2 identity contrasts with the adversarial ℓ_1 amplification factor $\Lambda_{n,3}$.

Theorem 9 (Uniform Fourier concentration from independent bounded shots)**.** *At every spatial node $\boldsymbol{j}$ and Cartesian component $a = 1, 2, 3$, suppose N independent bounded observations $X_{\boldsymbol{j},a,s} \in [-A, A]$ have expectation $h_a(x_{\boldsymbol{j}})$. Assume observations are independent* between spatial nodes for each component*; independence between distinct components is not required. Let $\widehat{h}_{\boldsymbol{j},a} = N^{-1} \sum_s X_{\boldsymbol{j},a,s}$ and $e_{\boldsymbol{j},a} = \widehat{h}_{\boldsymbol{j},a} - h_a(x_{\boldsymbol{j}})$. Choose an integer auxiliary grid resolution $m > 3\pi K$ and $\delta \in (0, 1)$. Then*

$$\boxed{\mathbb{P}\left\{ \|I_K e\|_\infty \le \frac{A}{1 - 3\pi K/m} \sqrt{\frac{6 \log(6m^3/\delta)}{N}} \right\} \ge 1 - \delta.} \tag{38}$$

The auxiliary m^3 evaluation points are not *additional physical measurements. If $K = 0$, the conclusion holds with $m = 1$ and zero derivative factor.*

Proof. Equation (37) shows that, for any fixed $\boldsymbol{x}$, the sum $\sum_{\boldsymbol{j}} w_{\boldsymbol{j}}(\boldsymbol{x}) e_{\boldsymbol{j},a}$ is a weighted average of independent bounded-shot means with total squared coefficient norm exactly one. Hoeffding's inequality therefore yields

$$\mathbb{P}\left(|[I_K e]_a(\boldsymbol{x})| > t \right) \le 2 \exp\left(-\frac{N t^2}{2A^2} \right). \tag{39}$$

Take a *virtual* equispaced grid of m^3 points and use a union bound across its three Cartesian components: with probability $\ge 1 - \delta$ every component is bounded on this grid by $t = A\sqrt{2 \log(6m^3/\delta)/N}$. The periodic Bernstein inequality for a real trigonometric polynomial of coordinate degree at most K gives $\|\partial_{x_a} p\|_\infty \le K \|p\|_\infty$. Every point of the torus lies at coordinate distance at most π/m from a virtual grid point; hence $\|p\|_\infty \le \max_{\text{grid}} |p| + (3\pi K/m) \|p\|_\infty$. Thus each component has supremum at most $t/(1 - 3\pi K/m)$, and the Euclidean three-vector has supremum at most $\sqrt{3}$ times that quantity. This proves Eq. (38). □

Corollary 1 (Independent-shot topological certificate and sufficient resources)**.** *Under the same assumptions and notation as Theorem 6, let T be an independently validated Fourier-tail norm and C the absolute Fourier-coefficient residual obtained from the measured grid values. Set $B = g_0 - C_{\max} - (1+\Lambda_{n,3})T$, where $g_0 > 0$ is a rigorously known reference gap and $C \le C_{\max}$ is the required data-dependent acceptance condition. For $B > 0$, the sufficient per-component shot condition*

$$\boxed{N > \frac{6A^2}{B^2(1-3\pi K/m)^2}\log\left(\frac{6m^3}{\delta}\right), \qquad m > 3\pi K} \tag{40}$$

ensures, with false-certification probability at most δ, that a passing *reconstruction satisfies*

$$\boxed{\|h-h_0\|_\infty \le C + (1+\Lambda_{n,3})T + \frac{A}{1-3\pi K/m}\sqrt{\frac{6\log(6m^3/\delta)}{N}} < g_0.} \tag{41}$$

It follows that the straight-line interpolation never closes the gap, so the relevant homotopy invariants agree. For any fixed $\kappa > 3\pi$ and $m = \lceil \kappa K \rceil$, a fixed positive B, fixed A, and $T = 0$, the total sufficient *number of component-shot outcomes obeys*

$$\boxed{S_{\text{suff}} = 3n^3 N = O(n^3 \log(n/\delta)).} \tag{42}$$

There is no assertion that this rate is minimax-optimal or valid for unrestricted many-body tomography.

Proof. Use $h - h_0 = (I_K\widehat{h} - h_0) + (h - I_K h) - I_K e$, with $\|I_K\widehat{h} - h_0\|_\infty \le C$ and $\|h - I_K h\|_\infty \le (1+\Lambda_{n,3})T$; apply Theorem 9 and the gap homotopy argument. Equation (40) makes the stochastic contribution strictly smaller than B. Choosing m proportional to K bounds $(1-3\pi K/m)^{-1}$ by a constant and gives $\log m^3 = O(\log n)$, proving the rate. □

Remark 2 (Instrument bias and correlated errors)**.** *If a deterministic nodal vector bias is only bounded by b in Euclidean norm, the right side of Eq. (41) acquires $+\Lambda_{n,3}b$. Shared drift or adversarially coordinated shot errors need not obey the independent-node concentration argument. Indeed, choosing nodal errors parallel to a fixed vector with signs matching the interpolation cardinal functions at a maximizer gives $\|I_K e\|_\infty = \Lambda_{n,3}\max_j \|e_j\|_2$: the previous deterministic amplification factor remains necessary when independence cannot be justified. This is an explicit falsification control for the statistical improvement.*

a. Fixed numerical forecast, not a new experiment. Using exactly the previously stored synthetic bookkeeping values $n = 5$, $K = 2$, $C = 0.164999903351$, $T = 0.01$, $g_0 = A = 1$, and $\delta = 0.01$,

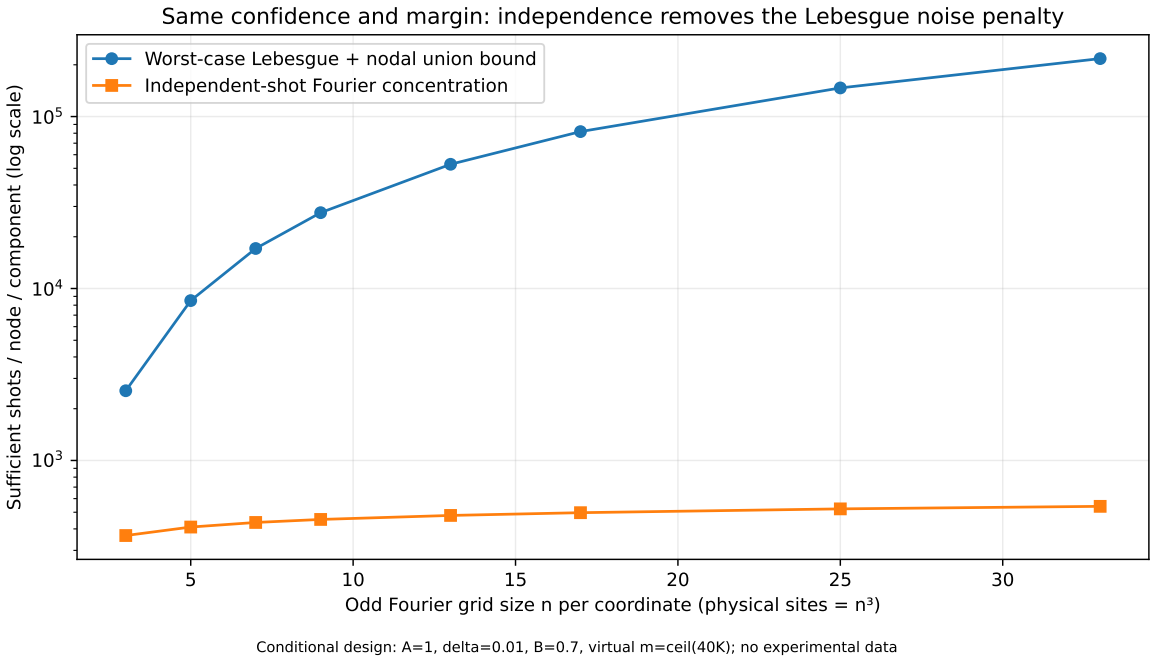


FIG. 8. Sufficient component-shot counts on odd Cartesian Fourier grids at fixed hypothetical margin $B = 0.7$, $A = 1$, $\delta = .01$ and $m = \lceil 40K \rceil$: deterministic Lebesgue/union bound versus independent-shot uniform concentration. No physical measurements were used; the curves are design bounds, not empirical learning curves.

choose $m = 80$, a virtual resolution (no new device sites). The earlier deterministic-union confidence bound required $N = 7{,}484$ component shots per site, whereas Eq. (40) requires $N = 361$. The corresponding sufficient total resource forecasts change from 2,806,500 to 135,375 outcomes, a factor of about 20.73 reduction under independent unbiased noise. These values are *conditional forecasts*, not physical shot counts or actual success rates. The stored C is from synthetic data; different measurements change C and may fail the acceptance test. $A = 1$ is a hypothetical calibrated component-outcome envelope and must be validated for any physical Hamiltonian scale before these shot totals can be used.

XIV. CORRELATED WHOLE-MAP MEASUREMENTS AND EFFECTIVE FOURIER LEVERAGE

A. What spatial correlation does and does not imply

The previous independent-node argument cannot be applied to a laboratory sequence with shared calibration drift. Conversely, a large off-diagonal covariance does not necessarily imply a

large error after trigonometric interpolation. We distinguish (a) independence of *repeated whole-map acquisitions*, (b) spatial dependence *within one acquisition*, and (c) systematic bias. The first is an explicit statistical assumption; the second is handled by a covariance proxy, while the third must be bounded separately. A covariance matrix alone does not provide exponential concentration without an additional tail or joint sub-Gaussian assumption.

Let $n = 2K + 1$ and $J = n^3$. Denote the J real tensor-cardinal coefficients at $x \in \mathbb{T}^3$ by $w(x) \in \mathbb{R}^J$. They obey

$$\sum_j w_j(x) = 1, \qquad \sum_j w_j(x)^2 = 1, \qquad \sup_x \sum_j |w_j(x)| = \Lambda_{n,3}. \tag{43}$$

For each field component $a = 1, 2, 3$, the sth independently repeated *complete* map has centered nodal error $\xi_{s,a} \in \mathbb{R}^J$. We allow arbitrary cross-site dependence and assume, for a *validated* positive semidefinite matrix Σ_a,

$$\mathbb{E} \exp(t u^{\mathsf{T}} \xi_{s,a}) \le \exp\Big(\tfrac{1}{2} t^2 u^{\mathsf{T}} \Sigma_a u\Big) \quad (\forall t \in \mathbb{R},\ u \in \mathbb{R}^J), \tag{44}$$

and independence over the acquisition index s. No independence between components is required. The sample mean of N full-map errors is $e_a = N^{-1} \sum_s \xi_{s,a}$. A calibrated deterministic spatial bias $b_j \in \mathbb{R}^3$ is assumed to obey $\sup_j \|b_j\|_2 \le b_*$. The independently calibrated covariance *proxy* may come from a physical Gaussian noise model or a justified matrix concentration model; a sample covariance without a confidence envelope is insufficient.

Define the *effective interpolation variance*

$$\boxed{V_{\text{eff}} = \max_{a=1,2,3} \sup_{x \in \mathbb{T}^3} w(x)^{\mathsf{T}} \Sigma_a w(x).} \tag{45}$$

The supremum may be replaced by a certified computable upper bound V_+. The example below evaluates it in closed form.

Theorem 10 (Covariance-resolved, uniform Fourier confidence bound)**.** *Let Eq. (44) hold for independent replicated whole-map acquisitions. Choose an integer virtual-grid size $m > 3\pi K$ and $\delta_s \in (0, 1)$. Then*

$$\boxed{\Pr\left[\|I_K e\|_\infty \le \frac{\sqrt{6 V_+ \log(6m^3/\delta_s)/N}}{1 - 3\pi K/m}\right] \ge 1 - \delta_s.} \tag{46}$$

The spatial noise may be fully correlated within each repeated map. Cross-component correlation need not be specified.

Proof. For each x, a, the scalar $w(x)^{\mathsf{T}} e_a$ is sub-Gaussian with proxy $w(x)^{\mathsf{T}} \Sigma_a w(x)/N \le V_+/N$ by independence of the N full-map repeats. Thus $\Pr(|w(x)^{\mathsf{T}} e_a| > t) \le 2\exp[-Nt^2/(2V_+)]$. On the auxiliary m^3-site grid, a union bound over three components gives $t = \sqrt{2V_+ \log(6m^3/\delta_s)/N}$. The periodic Bernstein derivative inequality bounds each component between auxiliary sites by the factor $(1 - 3\pi K/m)^{-1}$ as in Theorem 9. The Euclidean norm of the three components contributes the factor $\sqrt{3}$. Combining the three factors yields Eq. (46). □

Corollary 2 (Validated tail, bias, and topology-preserving certificate)**.** *Assume that $h : \mathbb{T}^3 \to \mathbb{R}^3$ has a Fourier truncation $P_K h$ to $|r_i| \le K$ with independently verified $\|h - P_K h\|_\infty \le T_K$. Let the calibrated reference h_0 have $g_0 = \inf_x \|h_0(x)\|_2 > 0$, and let $C_{\mathrm{data}} \ge \|I_K \widehat{h} - h_0\|_\infty$ be verified directly from the measured mean grid. Define*

$$\boxed{M_* = g_0 - C_{\mathrm{data}} - (1 + \Lambda_{n,3}) T_K - \Lambda_{n,3} b_*.} \tag{47}$$

If $M_* > 0$ and

$$\boxed{N > \frac{6V_+ \log(6m^3/\delta_s)}{M_*^2 (1 - 3\pi K/m)^2},} \tag{48}$$

then, except on an event of probability at most δ_s, every passing *certificate satisfies $\|h - h_0\|_\infty < g_0$, hence their straight-line interpolation is gapped and their defined homotopy invariants agree. If the covariance proxy itself is obtained from a separate calibration with failure probability δ_c, the total false-certification probability is bounded by $\delta_s + \delta_c$.*

Proof. Write $\widehat{h} = h|_{\mathrm{sites}} + b + e$. By interpolation of the Fourier-truncated field, $\|h - I_K(h|_{\mathrm{sites}})\|_\infty \le (1 + \Lambda_{n,3}) T_K$. The tensor Lebesgue operator bounds $\|I_K b\|_\infty \le \Lambda_{n,3} b_*$. Hence

$$\|h - h_0\|_\infty \le C_{\mathrm{data}} + (1 + \Lambda_{n,3}) T_K + \Lambda_{n,3} b_* + \|I_K e\|_\infty.$$

Apply Theorem 10. A strict positive gap persists throughout the convex path. A union bound handles any independently certified covariance proxy; statistical independence of the two bad events is not required. □

Remark 3 (Selection after observing data)**.** *C_{data} may be evaluated* after *the measurements, because the stochastic error event is uniform over every point and does not depend on a preselected candidate reference. However, changing the noise covariance model or calibration priors after viewing a failed experiment invalidates the claimed pre-registered error budget. In particular, positive measured correlations alone do not establish the matrix moment-generating-function bound* (44).

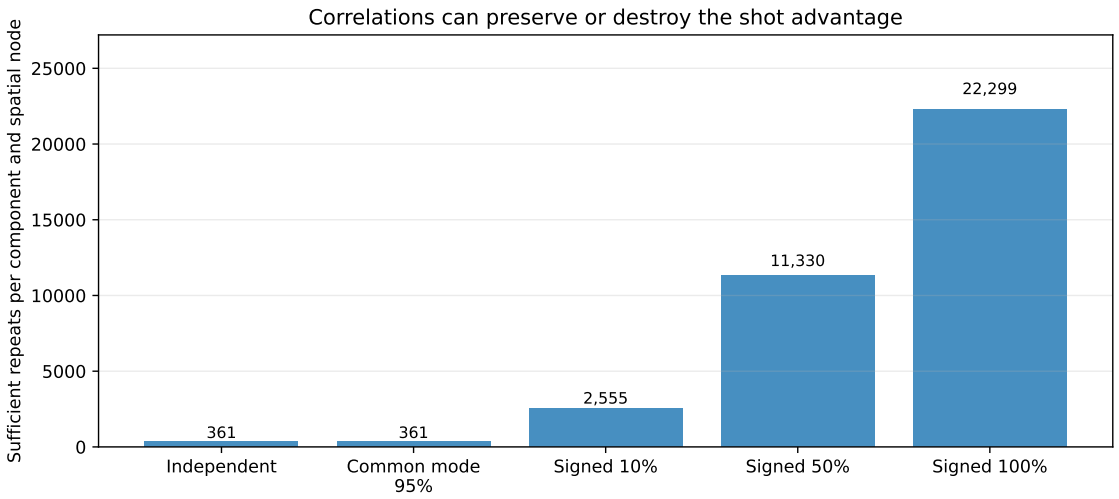


FIG. 9. Conditional sufficient shot budgets for five analytically specified full-map Gaussian covariance models on the 125-point grid, using the same C, T, g_0, δ, and virtual resolution as the previous synthetic v4 benchmark. The worst-case sign-aligned mode has much larger effective leverage than a spatially constant common mode. No device measurements enter.

B. A pair of maximally contrasting spatial correlations

Let $\sigma > 0$ and choose a tensor-grid cell midpoint x_* where the Lebesgue norm is attained. Set $s_j = \operatorname{sgn} w_j(x_*)$. Consider the positive semidefinite rank-one mixtures

$$\Sigma_\rho^{\text{common}} = \sigma^2[(1-\rho)I + \rho \mathbf{1}\mathbf{1}^\mathsf{T}], \quad \Sigma_\rho^{\text{signed}} = \sigma^2[(1-\rho)I + \rho s s^\mathsf{T}], \quad 0 \le \rho \le 1. \tag{49}$$

The first describes one common-mode random offset over all J spatial sites. The second describes a spatially signed coherent fluctuation exactly aligned with the worst Fourier-leverage pattern. By Eq. (43),

$$\boxed{V_{\text{common}} = \sigma^2, \qquad V_{\text{signed}} = \sigma^2[1 - \rho + \rho\Lambda_{n,3}^2].} \tag{50}$$

The first identity uses $w^\mathsf{T}\mathbf{1} = 1$ even when the raw covariance has an eigenvalue proportional to J. The second uses $\|w\|_2^2 = 1$ and $|s^\mathsf{T} w| \le \Lambda_{n,3}$, with equality at x_*. This distinction shows why raw covariance magnitude and even $\|\Sigma\|_{\text{op}}$ can be unnecessarily pessimistic. The covariance structures are explicit *mathematical noise models*; realistic devices require an independently validated spatial covariance proxy.

C. Independent computational replication

We generated 2,500 independently repeated synthetic Gaussian sample-mean maps *per covariance regime*, with $N = 256$ virtual acquisitions per reconstruction. Each map used an exactly specified Gaussian law with covariance Σ/N, evaluated at 28 spatial probe locations, including the maximizing midpoint. The empirical-to-analytic variance ratios of the midpoint interpolant were 1.0113, 1.0081, 1.0285, 0.9834, and 0.9886 in the five regimes, respectively. No sample exceeded the conservative scalar uniform bound; this is a numerical regression check, not a proof of 99% physical coverage. The corresponding 99% sufficient *per-component per-node* budgets were 361, 361, 2,555, 11,330, and 22,299. Full data generation, fixed random seed, covariance specifications, checks, and complete output records are bundled in `verify_correlated_tail_v5.py` and `RESULTS_V5.json`. Gaussian sample noise is unbounded; a physical Pauli-shot model needs a separate calibrated joint sub-Gaussian proxy.

XV. UNKNOWN HIGH-FREQUENCY TAILS: CERTIFIED LOCALITY OR NO UNIVERSAL ANSWER

Proposition 4 (Exact coefficient-envelope tail)**.** *Let $h(k) = \sum_{r\in\mathbb{Z}^3} \widehat{h}_r e^{ir\cdot k}$ be a continuous vector Hamiltonian on $\mathbb{T}^3$ whose Fourier coefficient vectors obey an* independently established *envelope $\|\widehat{h}_r\|_2 \le Be^{-a\|r\|_1}$ for constants $B > 0, a > 0$. For the cube truncation $P_K h = \sum_{|r_i|\le K} \widehat{h}_r e^{ir\cdot k}$,*

$$\boxed{\|h - P_K h\|_\infty \le T_K^{\exp} = B\left[\left(\frac{1+u}{1-u}\right)^3 - \left(\frac{1+u-2u^{K+1}}{1-u}\right)^3\right], \quad u = e^{-a}.} \tag{51}$$

Proof. Absolute Fourier convergence and the triangle inequality bound the remainder by the sum of Fourier-coefficient norms outside the cube. The one-dimensional geometric sums are

$$\sum_{q\in\mathbb{Z}} u^{|q|} = \frac{1+u}{1-u}, \qquad \sum_{q=-K}^{K} u^{|q|} = \frac{1+u-2u^{K+1}}{1-u}.$$

Their tensor cubes give the exact *envelope sum*. The inequality for the unknown field can be strict. □

Theorem 11 (Gap-preserving finite-sampling impossibility without tail priors)**.** *For any finite set $S \subset \mathbb{T}^3$ of momentum sites, there exist* smooth*, spectrally gapped two-band Hamiltonians $h^{(0)}, h^{(1)} : \mathbb{T}^3 \to \mathbb{R}^3$ with identical values at every point of S, with $\|h^{(0)}(k)\|_2 = \|h^{(1)}(k)\|_2 = 1$ for all k, with zero weak first-Chern classes, but with distinct absolute Hopf invariants $H(h^{(0)}) = 0$ and*

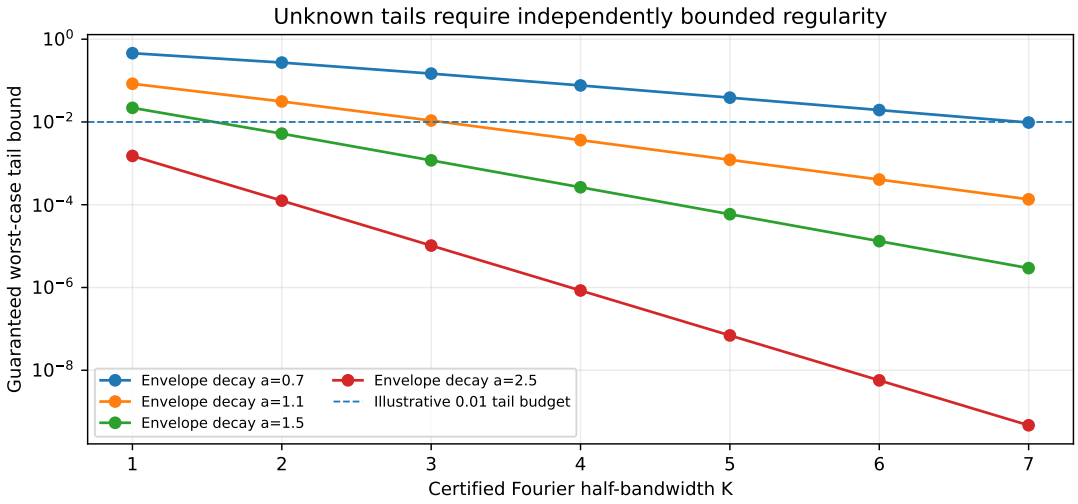


FIG. 10. A family of explicit, *assumed* Fourier-decay certificates ($B = 0.025$, varying a). Finite measured Fourier coefficients do not establish either B or a. A physical hopping-locality calibration or independent analytic prior is required.

$H(h^{(1)}) = 1$ *(up to a fixed orientation convention). Consequently, no algorithm using only finite exact point values, the positive band gap, and zero weak Chern classes can always recover the integer Hopf invariant if no bound on inter-site regularity or Fourier tails is imposed.*

Proof. Choose a small embedded closed three-ball $D \subset \mathbb{T}^3 \setminus S$. Set the first normalized field $n^{(0)} = (0, 0, 1)$. Collapse the boundary of D to a point, obtaining $D/\partial D \cong S^3$. There exists a smooth degree-one based map from D to S^3 equal to a constant on a collar of ∂D; compose it with a unit Hopf map $S^3 \to S^2$, and extend the result by the same constant outside D. This defines smooth $n^{(1)}$ with Hopf invariant one by precomposition with degree one. Because the map is supported on a ball, all weak Chern classes vanish. Both fields agree on S and are everywhere unit norm. Define $h^{(j)} = n^{(j)}$, so the two-band Hamiltonians $h^{(j)} \cdot \sigma$ have exactly the same constant gap 2 everywhere. The values on S are identical but their Hopf integers are not. Such localized Hopf-bubble constructions and the general finite-sampling obstruction are already known in related contexts [6]; no claim of priority is made here. □

a. Sensitivity to the independently assumed envelope. On the earlier 125-node design with $B = 0.025$, $K = 2$, and $C_{\text{data}} = 0.164999903351$, an exponential decay rate $a = 1.5$ gives $T_K^{\text{exp}} = 0.00522100$ and a 99% sufficient independent-map budget of 323 shots per node/component. A weaker rate $a = 1.1$ gives $T_K^{\text{exp}} = 0.03130468$ and budget 646; $a = 0.7$ gives $T_K^{\text{exp}} = 0.27258268$ and no positive certified margin at all. These are parameter-sensitive *hypothetical envelopes*, not decay

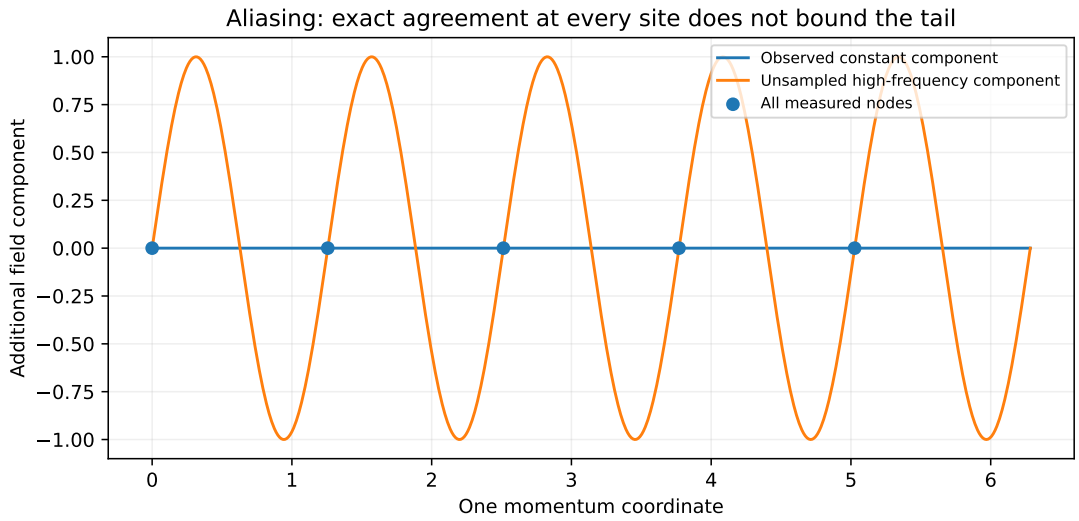


FIG. 11. A simple Fourier-aliasing precursor: $\sin(nx)$ vanishes on every node $x_j = 2\pi j/n$ while attaining unit amplitude between nodes. The graph alone is not a nontrivial Hopf map; Theorem 11 supplies the independent topological construction.

constants measured for any device. They expose the necessity of calibrating locality independently of the Fourier sample values.

A. Why covariance and spectral locality are independent requirements

The covariance-resolved confidence theorem only controls what sampling *does measure*. It does not constrain unmeasured spatial harmonics or deterministic biases. Conversely, exact exponential hopping decay constrains unresolved harmonics but does not provide a statistical confidence level for noisy node observations. A valid topology claim needs all three quantities simultaneously: a positive reference gap g_0, a calibrated spatial-error law with independently repeated acquisitions, and an independent Fourier-tail or regularity certificate. If B or a in Eq. (51) is estimated using only the same finite momentum samples, adversarial aliasing can invalidate the claimed bound. The shot-noise-to-drift transition identified experimentally in photonic reconstruction in 2026 reinforces the need to audit bias rather than extrapolate the $N^{-1/2}$ law indefinitely [19].

B. What would constitute an experimental test on actual hardware?

A preregistered hardware study must provide *raw repeated full-map acquisitions* or a justified acquisition model, timestamped drift controls, basis-dependent readout calibration, independent

upper certificates for Σ_a or a sub-Gaussian matrix proxy, an independent physical bound on the hopping envelope (B, a), an absolute Hamiltonian scale and a reference-gap certificate. Use a held-out set of momenta and a second data-collection session to test whether false-certification frequency is compatible with the declared risk. Experimental Bloch vectors without an absolute band-energy scale generally do not verify the Hamiltonian-vector gap inequality as written. The published NV-center Hopf tomography [3] and electrical-circuit realization [4] provide relevant experimental platforms, but no eligible raw repeated full-map device dataset was imported in this work. The 2,500-replicate tests above are *synthetic computational experiments only.*

XVI. AUDITED SECOND-ORDER SAMPLING AND REFERENCE-FREE INTEGER EXTRACTION

This section critically incorporates four proposals provided for independent review on October 22, 2026. We retain the demonstrable sampling and finite-precision constructions, repair the normalization of the weak Chern indices, and explicitly reject unverified complexity and modular-helicity claims. The results are conditional mathematics; the measurement demonstrations remain synthetic.

A. A second-order sampling certificate without extra physical observations

Theorem 12 (Tensor secant sampling bound)**.** *Let $p : \mathbb{T}^3 \to \mathbb{R}^d$ be a real vector-valued trigonometric polynomial of coordinate degree at most $K \geq 1$, and let $\mathcal{G}_m = (2\pi\mathbb{Z}_m/m)^3$, with $m > 2K$. Then*

$$\boxed{\|p\|_\infty \leq \sec^3\left(\frac{\pi K}{m}\right) \max_{x\in\mathcal{G}_m} \|p(x)\|_2 .} \tag{52}$$

The grid is virtual*: it is evaluated from the interpolating polynomial and does not require extra physical measurements.*

Proof. At a maximizer x^* choose a unit vector $u = p(x^*)/\|p(x^*)\|_2$, assuming the maximum is nonzero. The scalar polynomial $f(x) = u \cdot p(x)$ attains its positive supremum there. The sharp one-variable Bernstein inequality for a real trigonometric polynomial g bounded by M is $|g'| \leq K\sqrt{M^2 - g^2}$. Integration from a maximizing point to its nearest m-mesh node, at distance at most $\pi/m < \pi/(2K)$, gives $|g(\text{node})| \geq M\cos(\pi K/m)$. Apply the one-dimensional estimate successively in coordinates $1, 2, 3$ to the same scalar f, holding the previously chosen nodes fixed,

and then bound $|u \cdot p| \le \|p\|_2$ on the complete grid. This proves the claim. The elementary Taylor version with multiplier $(1 - \pi^2 K^2/(2m^2))^{-3}$ is also valid when its denominator is positive. □

For independent centered outcomes of magnitude at most A, N shots per component and node, and the orthogonal trigonometric cardinal functions w_j satisfying $\sum_j w_j(x)^2 = 1$, the scalar error $u \cdot I_K e(x)$ is $A/\sqrt{N}$-sub-Gaussian for every fixed unit vector u. Combining Eq. (52) with a *certified* finite spherical covering, rather than assuming an unprovided 31-direction configuration, makes the gain fully reproducible.

Theorem 13 (An explicit finite directional covering)**.** *For an integer $q \ge 2$, let $\mathcal{U}_q$ consist of all unit vectors parallel to nonzero primitive integer triples (a, b, c) with $|a|, |b|, |c| \le q$, taking both signs. Let $2M_q = |\mathcal{U}_q|$ and $\gamma_q = 1 - 3/(2q^2) > 0$. Every unit vector $u \in \mathbb{S}^2$ has a direction $v \in \mathcal{U}_q$ with $u \cdot v \ge \gamma_q$. Consequently, for $m > 2K$, with probability at least $1 - \delta$,*

$$\boxed{\|I_K e\|_\infty \le \frac{A \sec^3(\pi K/m)}{\gamma_q} \sqrt{\frac{2 \log(2M_q m^3/\delta)}{N}}.} \tag{53}$$

If an independently validated discrepancy-and-tail margin $B > 0$ remains to the topological gap, the following count is sufficient:

$$\boxed{N > \frac{2A^2 \sec^6(\pi K/m)}{\gamma_q^2 B^2} \log \frac{2M_q m^3}{\delta}.} \tag{54}$$

Proof. Round each coordinate of qu to the nearest integer to obtain a nonzero lattice vector $a \in [-q, q]^3$. With $t = a/q$, the coordinate rounding error gives $\|u - t\|_2 \le \sqrt{3}/(2q)$. Upon normalization $v = t/\|t\|_2$, the triangle inequality gives $\|u - v\| \le \sqrt{3}/q$, and hence $u \cdot v = 1 - \|u - v\|^2/2 \ge \gamma_q$. Every v belongs to $\mathcal{U}_q$ after dividing its lattice representative by the greatest common divisor. At each of the m^3 virtual grid nodes, Hoeffding's inequality for the scalar projection along each member of M_q antipodal pairs bounds its absolute error by $A\sqrt{2 \log(2M_q m^3/\delta)/N}$ simultaneously with probability $1 - \delta$. The covering inequality, followed by Theorem 12, proves Eq. (53); solving the strict margin inequality gives Eq. (54). □

For the inherited $n = 5$, $K = 2$, $A = 1$, $\delta = 0.01$ synthetic example, $B = 0.74633013$, the *explicit* choice $q = 4$ has $M_4 = 289$ and permits $N = 104$ shots per component and site at virtual $m = 37$ (39,000 binary outcomes). The choice $q = 6$ has $M_6 = 865$ and permits $N = 98$ at $m = 35$ (36,750 outcomes). These are sufficient analytic budgets, not laboratory measurements or minimax lower bounds. The previously supplied figure of 81 shots relied on an unprovided directional covering; the accompanying code in fact returned 85 for its assumed covering constant.

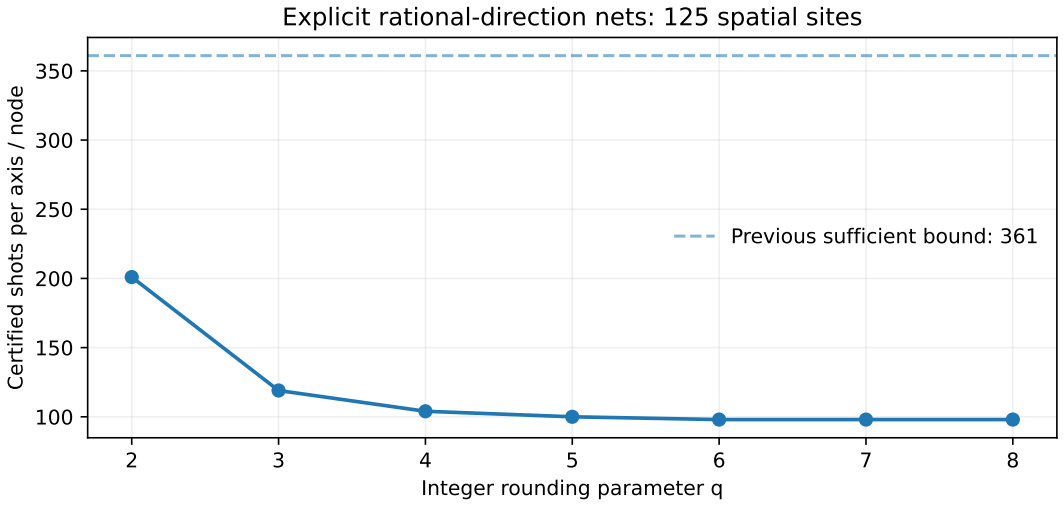


FIG. 12. Certified sufficient shot budgets from an explicitly enumerated integer-direction net. Changes in q affect only post-processing directions; the original 5^3 physical spatial nodes are unchanged.

B. A true reference-free gap certificate and correct Chern normalization

Suppose the observed trigonometric interpolant is $p = I_K \widehat{h}$, the unknown continuous Hamiltonian vector is h, and a *validated* probabilistic bound $\|h - p\|_\infty \le E$ holds with confidence at least $1 - \delta$. Put $C_p = \sum_{\|\nu\|_\infty \le K} \|\widehat{p}_\nu\|_2$. For a virtual grid of size m_g, define $g_{\mathrm{grid}} = \min_{x \in \mathcal{G}_{m_g}} \|p(x)\|_2$ and $\underline{g} = g_{\mathrm{grid}} - 3\pi K C_p / m_g$.

Theorem 14 (Reference-free gap homotopy criterion)**.** *If $\underline{g} > E$, then with probability at least $1 - \delta$, the actual two-band Hamiltonian $h \cdot \sigma$ has gap at least $2(\underline{g} - E) > 0$, and $p/\|p\|$ and $h/\|h\|$ have the same homotopy class.*

Proof. The coordinate derivative of p is bounded by KC_p. Every point of the continuous torus is within π/m_g in each coordinate of a virtual node, giving $\inf \|p\| \ge \underline{g}$. Along the pointwise straight-line homotopy $p_s = (1 - s)p + sh$, $\|p_s\| \ge \underline{g} - E > 0$. Normalize p_s to complete the proof. □

Define the Berry-vorticity vector associated with the *normalized* interpolant $n = p/\|p\|$ by

$$B_a = \frac{1}{8\pi} \epsilon_{abc}\, n \cdot (\partial_b n \times \partial_c n) = \frac{1}{8\pi} \epsilon_{abc} \frac{p \cdot (\partial_b p \times \partial_c p)}{\|p\|^3}. \tag{55}$$

The Fourier coefficients are $b_\nu = (2\pi)^{-3} \int_{\mathbb{T}^3} B(k) e^{-i\nu \cdot k}\, d^3k$. The weak Chern numbers are *not*

$2\pi b_{0,a}$ (as proposed in the supplied draft), but

$$\boxed{c_{1,a} = \int_{\mathbb{T}^2_a} B_a \, d^2k = (2\pi)^2 b_{0,a} \in \mathbb{Z}.} \tag{56}$$

In the elementary model $p = (\sin k_x, \sin k_y, -1 + \cos k_x + \cos k_y)$ independent of k_z, the xy-Chern number is $+1$ (the north pole has one positively oriented preimage). A numerical 180^2 midpoint sum gives $(2\pi)^2 b_{0,z} = 1.0000000000$; the incorrect proposed factor $2\pi b_{0,z}$ gives 0.1591549431.

C. Finite-error extraction of an integer, not unvalidated rounding

If the three weak Chern numbers vanish, the vorticity is exact and the classical Whitehead integral can be evaluated in Coulomb gauge. For $\nu \neq 0$ define $a_\nu = i\nu \times b_\nu / \|\nu\|_2^2$. Then, in our orientation convention,

$$\boxed{H(n) = -(2\pi)^3 \sum_{\nu \in \mathbb{Z}^3 \setminus \{0\}} \frac{i\nu \cdot (b^*_\nu \times b_\nu)}{\|\nu\|_2^2} \in \mathbb{Z}.} \tag{57}$$

This formula by itself is classical and does not supply a finite-precision certificate. The following computable enclosure does.

Set $g = \underline{g} > 0$, $C = C_p$, $D_1 = KC$, $D_2 = K^2 C$, and define the conservative explicit constants

$$b_0^* = \frac{CD_1^2}{4\pi g^3}, \tag{58}$$

$$b_1^* = \frac{D_1^3 + 2CD_1D_2}{4\pi g^3} + \frac{3CD_1^3}{4\pi g^4}. \tag{59}$$

Direct differentiation of Eq. (55) proves $|B_a| \le b_0^*$ and $|\partial_j B_a| \le b_1^*$ for every a, j. For an integer $Q \ge 1$, compute the discrete coefficient

$$q_\nu = \frac{1}{Q^3} \sum_{j \in \mathbb{Z}^3_Q} B(2\pi j/Q) e^{-2\pi i \nu \cdot j/Q}. \tag{60}$$

Partition the torus into cubes centered at the sample locations and use the mean-value inequality to get the *vector-valued* coefficient enclosure

$$\boxed{\|q_\nu - b_\nu\|_2 \le E_\nu := \frac{\pi\sqrt{3}}{Q}(3b_1^* + \|\nu\|_1 \, b_0^*).} \tag{61}$$

For a spectral cutoff $R_c \ge 1$, let $\widetilde{H}_{R_c,Q}$ be Eq. (57) restricted to $0 < \|\nu\|_2 \le R_c$, replacing b_ν with q_ν. Parseval gives

$$\boxed{|H(n) - \widetilde{H}_{R_c,Q}| \le \underbrace{\frac{(2\pi)^3 9 (b_1^*)^2}{R_c^3}}_{\text{true Fourier tail}} + \underbrace{(2\pi)^3 \sum_{0<\|\nu\|\le R_c} \frac{(2\|q_\nu\|_2 + E_\nu)E_\nu}{\|\nu\|_2}}_{\text{finite quadrature error}} := E_H.} \tag{62}$$

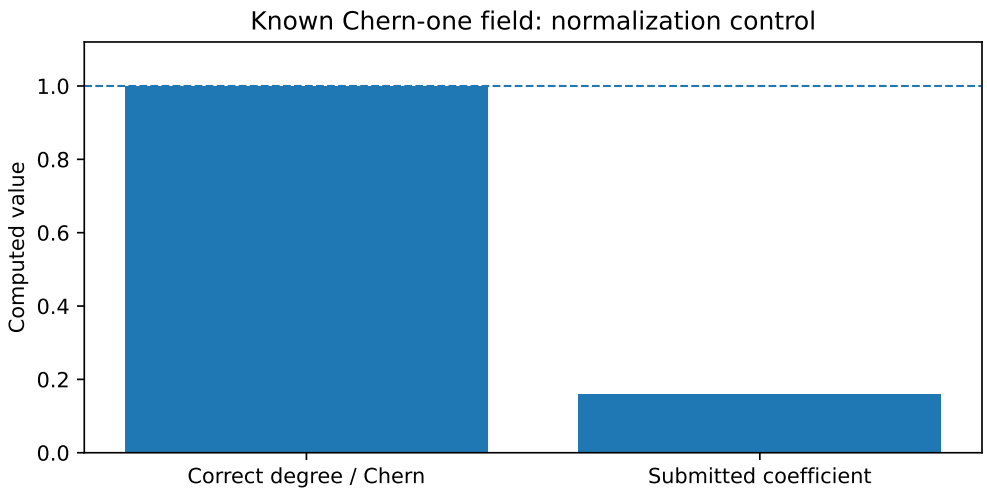


FIG. 13. Counterexample to a missing normalization factor in the submitted reference-free spectral theorem: the exact degree is one. Agreement of a numerical integral with an integer does not, by itself, constitute a finite-error proof.

Here the first term uses $\|\nabla B\|_{L^2}^2 \le 9(2\pi)^3(b_1^*)^2$. In particular, the correctly normalized Chern number is certified as the nearest integer to $(2\pi)^2 q_{0,a}$ if $|(2\pi)^2 q_{0,a} - \mathrm{round}((2\pi)^2 q_{0,a})| + (2\pi)^2 E_0 < 1/2$. Once *all three* are certified zero, if also

$$\boxed{|\widetilde{H}_{R_c,Q} - \mathrm{round}(\widetilde{H}_{R_c,Q})| + E_H < \tfrac{1}{2},} \tag{63}$$

then the rounded value is the exact integer invariant of n and, on the earlier statistical confidence event, of the unknown $h/\|h\|$. If either test fails, the algorithm must report *unresolved*, even if the raw FFT value happens to be close to an integer. The conservative bounds need not be practically economical; their virtue is logical soundness and finite termination for a fixed gapped trigonometric polynomial after sufficiently accurate arithmetic and sufficiently large Q, R_c.

For $\boldsymbol{c}_1 \neq 0$, Eq. (57) with the constant Fourier mode omitted is *not* automatically the secondary Pontryagin invariant modulo $2\,\mathrm{gcd}(c_1)$; additional cohomological framing and a valid relative construction are required. Accordingly we make no such certification claim here. The attached user-supplied FFT program numerically reproduced the conventional Moore–Ran–Wen phase integers $0, -1, 2, -1, 0$, but the original program did not bound aliasing or the cutoff error and therefore provided approximations, not proved integer answers.

D. Adaptive and bosonic proposals: admissible and inadmissible conclusions

A prior suggestion asserted $O(1)$ adaptive identification of an arbitrarily located Hopf bubble of vanishing radius. This is unjustified without an independently certified oracle that already reveals the bubble's location or local regularity. A hidden bubble can be placed between any insufficiently dense sampling locations. Adaptively chosen cells can nevertheless be handled with a fixed *prior* error allocation $\delta_C = \delta\, 16^{-\ell}/(2M_0)$ for each potential level-ℓ cell in an 8-ary refinement of M_0 root cells: summing over the *entire potential tree* gives $\sum_C \delta_C = \delta$. Individual conditional Hoeffding bounds then survive data-dependent cell selection without a naive post-selection correction, provided every potential measurement is calibrated and has its stated sampling law. This establishes validity, *not* the claimed $O(1)$ spatial resource upper bound.

For a positive-definite bosonic BdG Hamiltonian $\mathcal{H}_0$, set $D_0 = \Sigma_z \mathcal{H}_0$ and suppose a paraunitary matrix T_0 diagonalizes D_0 at each momentum. If $\varepsilon = \sup_k \|\mathcal{H}(k) - \mathcal{H}_0(k)\|_{\mathrm{op}}$, $\mu = \inf_k \lambda_{\min}(\mathcal{H}_0(k))$, $\kappa = \sup_k \|T_0(k)\|_2^2$ and Δ is a uniform gap separating a specified subset of positive-frequency bands from its complement, then $\varepsilon < \min(\mu, \Delta/(2\kappa))$ keeps the straight-line interpolation positive and the chosen band cluster isolated. Positivity follows from Weyl's bound; spectral enclosure from diagonalizing D_0 and the Bauer–Fike estimate. Consequently, any *well-defined* invariant of the continuously isolated eigensubbundle remains unchanged. This is a conditional perturbation lemma, not an automatic result for arbitrary non-Hermitian matrices or a proof that a particular Euler or symplectic Hopf invariant is realized by a given laboratory device.

E. Reproducibility and claims of priority

The independent audit enumerates all integer-direction sets, recomputes budgets without assumed spherical geometry, constructs a known nonzero-Chern test that falsifies the proposed normalization, and checks a finite family of independent random trigonometric polynomials. Numerical agreement does not substitute for analytic proofs. The secant bound and Whitehead helicity integral use established analysis and topology. Whether the combined error-budget specialization is publishably novel relative to the existing approximation, topological tomography, and inverse-problem literature requires independent priority review. No external laboratory data were analyzed for this section.

XVII. GEOMETRIC QUERY LOWER BOUNDS, ANYTIME TOMOGRAPHY, AND POSITIVE BOSONIC SPECTRAL STABILITY

A. An exact localized-Hopf Lipschitz constant for the specified radial construction

Consider the unit round S^3 with polar coordinates (ρ, u) about a fixed center, and the radial collapse $g_r(\rho, u) = (\cos f(\rho), \sin f(\rho)u)$ for $\rho \le r$, constant outside, with $f(\rho) = \pi(1 - \rho/r)$, $0 < r \le \pi$. Let $h_1 : S^3 \to S^2$ be the standard Hopf map, for unit round source and target metrics, so $\mathrm{Lip}(h_1) = 2$.

Theorem 15 (Exact constant of this radial Hopf bubble)**.** *For the specified map $b_r = h_1 \circ g_r$, one has*

$$\boxed{\mathrm{Lip}(b_r) = 2\pi/r, \quad 0 < r \le \pi, \qquad |H(b_r)| = 1.} \tag{64}$$

The equality is for this particular construction*; it is not a global optimality statement over all maps representing the same homotopy class.*

Proof. The radial and angular singular values of dg_r on $0 < \rho < r$ are $\alpha = \pi/r$ and $|\sin(\alpha\rho)|/\sin\rho$, respectively, where $\alpha \ge 1$. For $0 < \rho \le r/2$, the function $t \cot t$ is decreasing on $(0, \pi)$ and hence $\sin(\alpha\rho)/\sin\rho \le \alpha$. For $r/2 \le \rho \le r$, put $s = r - \rho$. Then $s \le \rho \le \pi - s$, so $\sin\rho \ge \sin s$ and $\sin(\alpha s)/\sin\rho \le \alpha$. Thus $\mathrm{Lip}(g_r) \le \alpha$, proving $\mathrm{Lip}(b_r) \le 2\alpha$. The radial differential of g_r has magnitude α; choosing its radial tangent direction horizontal for the Hopf submersion gives $\|d(h_1 \circ g_r)\| = 2\alpha$ at suitable interior points. Thus equality holds. The collapsed r-ball represents a degree ± 1 self-map of S^3; composition with the Hopf map therefore has Hopf invariant ± 1. □

Corollary 3 (Hidden-charge point-query obstruction)**.** *Let the admissible field class include all rotations of b_r with $r = 2\pi/L$ and the constant map, for a common Lipschitz bound $L > 2$. If a deterministic scheme makes N point queries with only constant-map responses, it cannot distinguish the constant map from a suitable rotated bubble whenever*

$$\boxed{N\,V(2\pi/L) < 2\pi^2, \quad V(r) = 2\pi(r - \sin r \cos r).} \tag{65}$$

For the equally weighted binary hypothesis with a uniformly rotated bubble versus the constant map, a randomized, possibly adaptive point-query scheme has worst-case error at least

$$\boxed{P_{\mathrm{err}} \ge \tfrac{1}{2}\max\{0, 1 - NV(2\pi/L)/(2\pi^2)\}.} \tag{66}$$

The asymptotic volume-ratio improvement over the earlier π^2/L-radius construction is $\pi^3/8 \approx 3.87578$. The result concerns point queries only*; integral, coherent, or nonlocal observables fall outside this theorem.*

Proof. The round S^3 has volume $2\pi^2$, and its radius-r ball has volume $V(r)$. By the union bound for randomly rotating a fixed ball, the probability that any of the N points entered along the all-constant transcript lies inside its support is at most $NV(r)/(2\pi^2)$. If this number is below one, some position avoids the full transcript. On the complementary event the two hypotheses have identical observations, so with equal priors the conditional classification error is at least $1/2$. The volume-ratio limit follows from $V(r) \sim (4\pi/3)r^3$ as $r \to 0$. □

B. A conservative time-uniform Pauli confidence sequence

The upload proposes a faster purity-dependent law-of-iterated-logarithm radius without a complete crossing-probability proof. Rather than asserting its constants, the following simpler bound is *fully justified* by a summable union allocation; sharper nonasymptotic boundaries are already studied in Ref. [23].

Theorem 16 (Anytime-valid finite-shot homotopy decisions)**.** *At M predetermined nodes, suppose three Pauli components are measured through independent unbiased $\{\pm 1\}$ shots for each node and axis, and define $\widehat{n}_i(t)$ by the first t shots of each axis. For $\delta \in (0,1)$ put*

$$\boxed{\varepsilon_t = \sqrt{\frac{6}{t}\log\left(\frac{\pi^2 M t^2}{\delta}\right)}.} \tag{67}$$

Then $\mathbb{P}[\forall i = 1,\dots,M,\ \forall t \geq 1 :\ \|\widehat{n}_i(t) - n(x_i)\|_2 \leq \varepsilon_t] \geq 1-\delta$. Consequently, any stopping rule which certifies the strict inequality $\max_i\{\|\widehat{n}_i(t_i) - n_0(x_i)\|_2 + \varepsilon_{t_i}\} + (L+L_0)h < 2$ is sound with false-certification probability at most δ under the stated sampling assumptions.

Proof. Hoeffding gives $\mathbb{P}(|\widehat{n}_{ia}(t) - n_{ia}| > u) \leq 2e^{-tu^2/2}$. If the vector norm exceeds ε_t, then some Cartesian coordinate exceeds $\varepsilon_t/\sqrt{3}$. Thus the union over $3M$ means is at most $6M\exp(-t\varepsilon_t^2/6) = 6\delta/(\pi^2 t^2)$. Summing over all $t \geq 1$ yields δ. On this single simultaneous event, data-dependent stopping does not change the geometry argument of Theorem 1. □

Remark 4. *The identity $\sum_a \mathrm{Var}(\sigma_a) = 2$ for a pure qubit is correct, but this fact by itself does* not *prove the tighter fixed-time vector Bennett constant or the submitted $\log\log t$ sequential boundary. The numerical claim of $19.8\times$ adaptive savings has not been validated from measured or fully specified simulated stopping trajectories.*

C. Reference-basis-free positive bosonic BdG band perturbation

Let $J = \mathrm{diag}(I_d, -I_d)$, $\mathcal{H}_0(k)$ be a continuously varying positive-definite Hermitian BdG energy matrix, and $\mathcal{H}(k)$ a Hermitian perturbation. The bosonic generator $J\mathcal{H}$ is similar to the Hermitian representative $K = \mathcal{H}^{1/2} J \mathcal{H}^{1/2}$, a standard fact of positive-energy bosonic BdG theory [24].

Theorem 17 (Basis-independent sufficient band-gap certificate)**.** *Suppose the following uniform positivity margin holds:*

$$E = \sup_k \|\mathcal{H}(k) - \mathcal{H}_0(k)\|_{\mathrm{op}} < \mu := \inf_k \lambda_{\min}(\mathcal{H}_0(k)). \tag{68}$$

For each momentum let $\mu_k = \lambda_{\min}(\mathcal{H}_0(k))$ *and* $M_k = \lambda_{\max}(\mathcal{H}_0(k))$*. Correspondingly indexed eigenvalues* λ_j *of* $J\mathcal{H}(k)$ *and* $J\mathcal{H}_0(k)$ *satisfy*

$$\boxed{|\lambda_j(k) - \lambda_{j,0}(k)| \le \Gamma_k(E) := E\,\frac{\sqrt{M_k + E} + \sqrt{M_k}}{\sqrt{\mu_k - E} + \sqrt{\mu_k}}.} \tag{69}$$

If a chosen positive-frequency band cluster is separated in the reference by a uniform gap $\Delta_0 > 0$ *and* $2\sup_k \Gamma_k(E) < \Delta_0$*, the band gap remains open along the entire straight-line interpolation. Any topological invariant* defined for that isolated band bundle *is unchanged. No universal Hopf invariant is claimed for every bosonic BdG model.*

Proof. Weyl's inequality implies $\mathcal{H}_s = \mathcal{H}_0 + s(\mathcal{H} - \mathcal{H}_0) \succ 0$. The generator $J\mathcal{H}_s$ is similar to the Hermitian matrix $K_s = \mathcal{H}_s^{1/2} J \mathcal{H}_s^{1/2}$. Hermitian spectral variation satisfies $\max_j |\lambda_j(K_s) - \lambda_j(K_0)| \le \|K_s - K_0\|_{\mathrm{op}}$. The standard operator-square-root estimate gives $\left\|\mathcal{H}_s^{1/2} - \mathcal{H}_0^{1/2}\right\| \le sE/(\sqrt{\mu_k - sE} + \sqrt{\mu_k})$. Substituting into $K_s - K_0 = (\mathcal{H}_s^{1/2} - \mathcal{H}_0^{1/2}) J \mathcal{H}_s^{1/2} + \mathcal{H}_0^{1/2} J (\mathcal{H}_s^{1/2} - \mathcal{H}_0^{1/2})$ yields $\|K_s - K_0\| \le \Gamma_k(E)$, since $s \le 1$. Weyl's inequality then bounds the separation of any two indexed adjacent bands from below by $\Delta_0 - 2\sup_k \Gamma_k(E) > 0$. □

D. Correct oriented positive-ray predicate

For a nondegenerate oriented target ray, let $D = \det(b_0, b_1, b_2)$ and let w_i denote the corresponding Cramer determinants. A positive-ray intersection of the affine triangle requires $w_i D > 0$ for all three barycentric coefficients. The weaker condition that all the w_i have a common sign is incorrect: the rational vectors $b_0 = e_1$, $b_1 = e_2$, $b_2 = e_3$ and a target ray pointing toward $-(1,1,1)$ supply an explicit opposite-hemisphere counterexample. A correct whole-sphere linking certificate additionally checks triangulation orientation, cycle closure and disjointness of the regular-value preimages [8, 12].

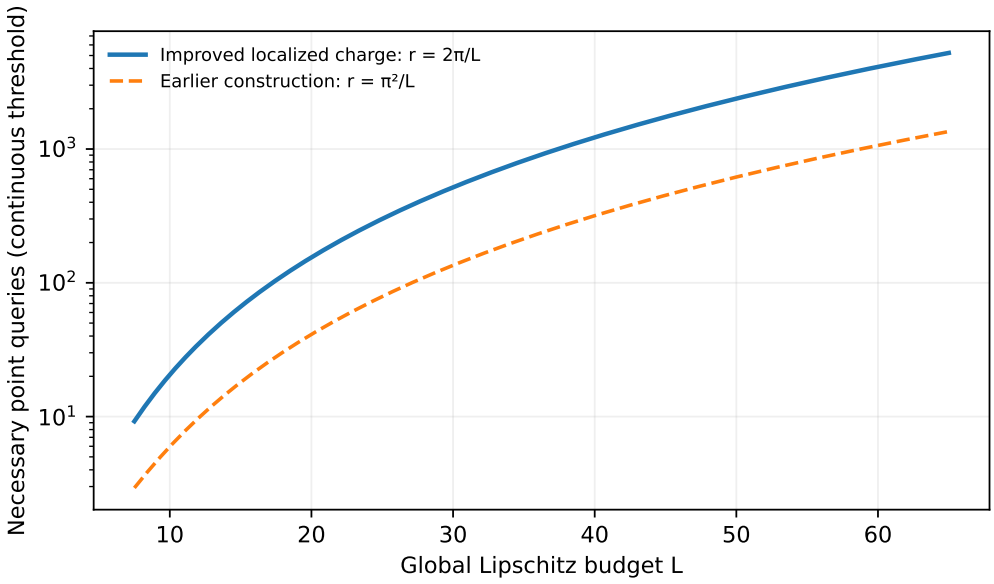


FIG. 14. Improved necessary point-query count for the explicit localized Hopf construction. The curve is a geometric obstruction, not a universal lower bound for nonlocal measurements or an experimental estimate.

XVIII. EXACT, REFERENCE-FREE INTEGER EXTRACTION FROM RATIONAL SIMPLICIAL DATA

The exact reconstruction below uses only rational vertex data, performs exact arithmetic for each topological predicate, and abstains whenever an input fails the documented conditions. Here *reference-free* means that no reference field or expected Hopf integer is supplied to the extraction routine. An independent bound on the unknown physical field is still needed to transfer the recovered integer from the simplicial interpolant to a physical measurement.

A. The oriented cubical three-sphere and an exact ray predicate

Let $P = [-1, 1]^4 \subset \mathbb{R}^4$, whose oriented boundary $\partial P \cong \mathbb{S}^3$ is triangulated by consistent Freudenthal subdivisions of its eight cubic facets. Its vertices have rational coordinates and the orientation of every tetrahedron $[v_0, v_1, v_2, v_3]$ is fixed by $\det(v_0, v_1, v_2, v_3) > 0$. The implementation verifies the opposite induced orientations of all paired triangular faces, global connectivity, Euler characteristic zero, and connected spherical links at every vertex. The topology $\partial P \cong \mathbb{S}^3$ follows from convexity of P, not merely from the observed Euler characteristic.

At each vertex v_i let $b_i \in \mathbb{Q}^3$ be the observed (or synthetically computed) vector, and on a

tetrahedron σ let

$$b_{\mathrm{PL}}(x) = \sum_{i=0}^{3} \lambda_i(x) b_i, \qquad n_{\mathrm{PL}}(x) = \frac{b_{\mathrm{PL}}(x)}{\|b_{\mathrm{PL}}(x)\|_2}. \tag{70}$$

A *required* finite test excludes 0 from the convex hull of the four b_i on every tetrahedron. The code uses exact one-, two-, three-, and four-point convex-combination tests; a zero is an error, never silently normalized away.

For a triangular face $\tau = [v_0, v_1, v_2]$ and a rational unit target $p \in \mathbb{S}^2$, set

$$\begin{aligned} D &= \det(b_0, b_1, b_2), \\ w_0 &= \det(p, b_1, b_2), \quad w_1 = \det(b_0, p, b_2), \quad w_2 = \det(b_0, b_1, p). \end{aligned} \tag{71}$$

For nondegenerate $D \neq 0$, an *interior positive-ray intersection* occurs precisely when

$$\boxed{D\, w_i > 0 \quad (i = 0, 1, 2), \qquad \lambda_i = \frac{w_i}{w_0 + w_1 + w_2} > 0.} \tag{72}$$

Thus matching signs of w_i alone is *not* sufficient: the sign of D is indispensable. A genericity check separately excludes targets lying in the span of two linearly independent edge-image vectors. On an oriented tetrahedron a regular inverse image must be absent or have exactly two face intersections of opposite induced signs; violations cause an explicit refusal.

B. Finite rational linking with checked cycle closure

Two distinct regular directions $p, q \in \mathbb{S}^2$ have disjoint oriented inverse-image links $L_p, L_q \subset \partial P$. The program checks every shared face's orientation balance and the in/out degree of every inverse-image vertex, then traverses the entire directed graph into closed polygonal components. For the examples below, the first coordinate of both p and q is less than $-1/3$, and the chosen analytic Hopf field and its PL interpolation omit these values on the upper facet $v_0 = 1$. Consequently both inverse images lie in the complementary three-ball, which admits the rational Schlegel projection

$$(v_0, v_1, v_2, v_3) \mapsto \frac{4}{3 - v_0}(v_1, v_2, v_3) \in \mathbb{R}^3. \tag{73}$$

This projective map sends line segments to line segments. For each of three separately chosen rational generic planar projections of the resulting links, every segment-segment crossing and over/under order is resolved by exact rational determinants.

Theorem 18 (Exact rational inverse-image linking certificate)**.** *Let $\mathcal{K}$ be a verified oriented simplicial three-sphere with rational vertex values b_i, and assume that the affine interpolant excludes the*

origin on every three-simplex. Let $p \neq q$ be two regular rational targets whose inverse images form disjoint finite polygonal cycles in a certified three-ball chart, and let a rational planar projection have only transverse crossings away from segment endpoints. Then

$$\boxed{H(n_{\mathrm{PL}}) = \mathrm{Lk}(L_p, L_q) = \frac{1}{2} \sum_{c \in \mathrm{Cross}(L_p, L_q)} \mathrm{sign}(c) \in \mathbb{Z}.} \tag{74}$$

The evaluation of the right-hand side by exact rational predicates has zero arithmetic rounding error*. An implementation must reject unverified manifold orientation, vanishing interpolant, nonregular inverse images, open or inconsistently oriented cycles, and degenerate projections.*

Proof. The strict convex-hull condition makes n_{PL} continuous on $\partial P \cong \mathbb{S}^3$. Regular preimages of distinct values are disjoint oriented closed one-manifolds, with piecewise-linear representatives obtained by solving the exact positive-ray equations on each tetrahedron. The classical Pontryagin inverse-image interpretation of the Hopf invariant gives $H(n_{\mathrm{PL}}) = \mathrm{Lk}(L_p, L_q)$. The Schlegel chart is a homeomorphism away from the omitted facet, so it preserves linking up to its fixed orientation convention. For a generic projection of two disjoint oriented polygonal links, the linking number equals one-half the algebraic crossing count. All determinants and the half-integer test can be implemented in rational/integer arithmetic; consequently this computation has no *floating-point* error. The conclusion does not extend to a distinct unknown field without a further homotopy certificate. □

C. When does a physical field share the recovered integer?

Suppose an unknown unit-vector field n on ∂P has an independently established Lipschitz bound L in the ambient Euclidean metric, and each vertex datum satisfies $\|b_i - n(v_i)\| \leq \varepsilon$. If every tetrahedron has Euclidean diameter at most h, then

$$\sup_x \|b_{\mathrm{PL}}(x) - n(x)\| \leq \varepsilon + Lh. \tag{75}$$

Hence the directly checkable *sufficient* condition

$$\boxed{\varepsilon + Lh < 1} \tag{76}$$

identifies $H(n)$ with the exactly extracted $H(n_{\mathrm{PL}})$: the straight-line homotopy avoids zero. Unlike the rational arithmetic certificate, Eq. (76) depends on an *external experimental or theoretical regularity bound*. An absent or invalid bound means that the program has computed only the PL field's integer, not the unknown experiment's integer.

For the *analytic validation fields* $n(v) = h_1(v/\|v\|)$ and its orientation-reversed version, a substantially more effective *exact* certificate uses the quadratic numerator $Q(v) = \|v\|^2 n(v)$. On every tetrahedron with vertices v_a, define

$$A_{ab} = b_0 \cdot \frac{Q(v_a + v_b) - Q(v_a) - Q(v_b)}{2}. \tag{77}$$

If all $A_{ab} > 0$, then $b_0 \cdot Q(\sum_a \lambda_a v_a) > 0$ and $b_0 \cdot b_{\mathrm{PL}}(x) > 0$ throughout that tetrahedron. Thus the PL interpolation and the exact analytic map both remain in the same open hemisphere and are homotopic, without relying on a coarse global Lipschitz estimate. These positivity predicates use integers exclusively in the present benchmark.

D. Exact benchmarks, negative controls, and limitations

For an $N = 4$ subdivision of each original cube edge, the program constructs *exactly* 544 vertices, 3616 edges, 6144 oriented triangles and 3072 oriented tetrahedra. Every vertex link is checked. The directions

$$p = (-119, 72, 96)/169, \qquad q = (-538, -350, 245)/687 \tag{78}$$

are rational unit vectors of a regular admissible pair for this benchmark. The two links each have 70 segments, one oriented component, zero face-balance defects, and exactly four transverse mutual crossings (three positive and one negative) in each of three rational projections, yielding $H = +1$. The reflection-reversed Hopf field yields $H = -1$, and a constant field yields $H = 0$. An independent $N = 6$ run reconstructs $H = +1$ with 1776 vertices and 10368 tetrahedra. For $N = 4$, the exact quadratic homotopy test verifies 30720 positive Bernstein numerators; for $N = 6$ it verifies 103680. These are *analytic synthetic test fields*, not real laboratory measurements.

The implementation also rejects a reversed tetrahedron, a zero hidden in an affine simplex, and an unsupported physical-field error budget. A blind rational-vertex JSON input reproduces $H = +1$ without specifying the expected integer or reference field. This establishes a *working exact integer extraction for the specified generic PL input class*; it does not establish computational optimality, a novel general Hopf formula, exact recovery from floating-point sensor output without error envelopes, or performance on real experimental data. Preimage linking and simplicial-cochain methods have extensive prior art [8, 12]. The distinguishing contribution of this revision is a fully executable strict predicate/abstention workflow that corrects the previous failed program and carries an independently verified analytic-homotopy witness.

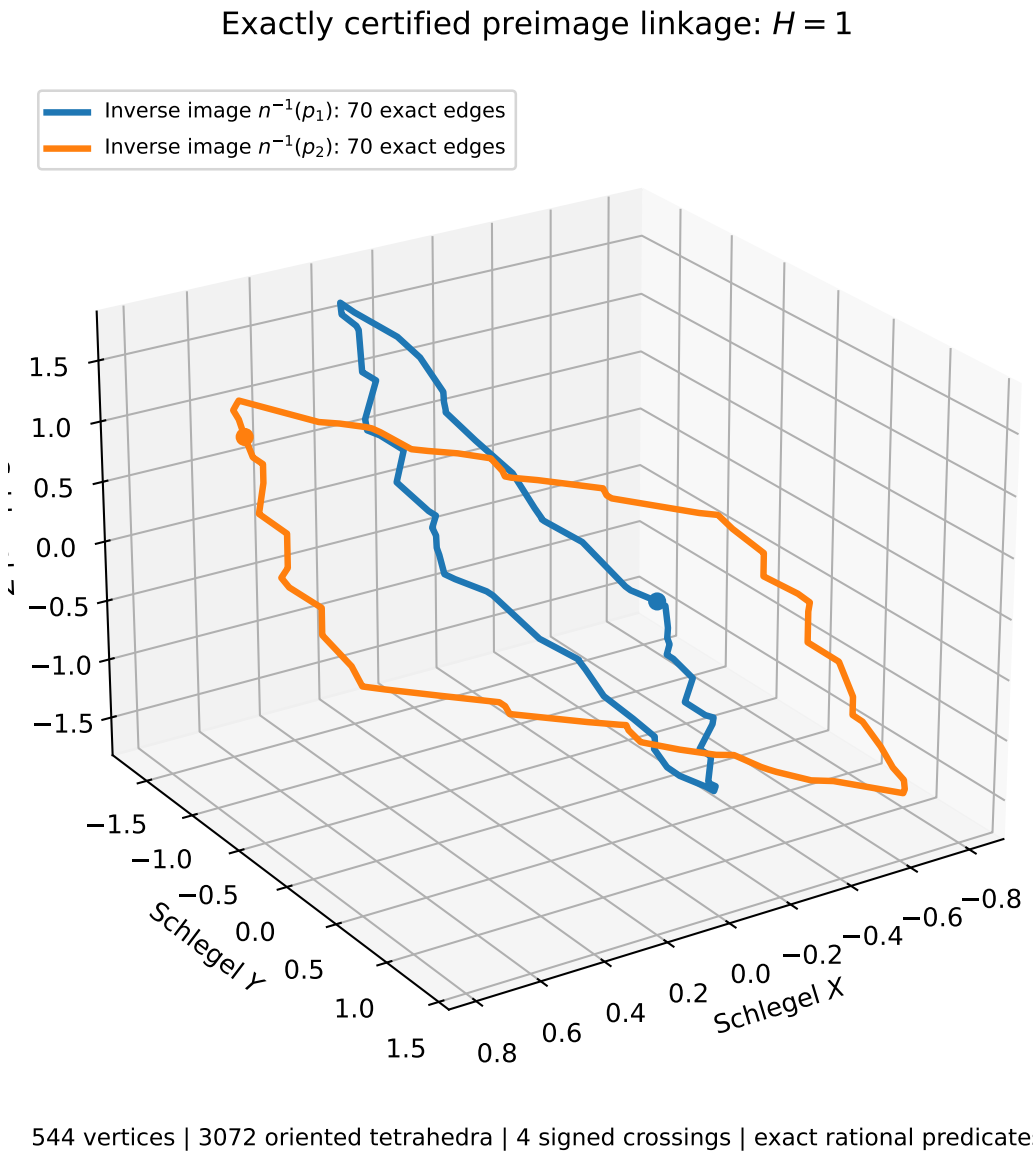


FIG. 15. Actual reconstructed polygonal inverse images for the certified $N = 4$ rational Hopf benchmark. The recovered integer is determined by exact signed crossings, not by numerically integrating an approximately integer-valued expression.

XIX. EXACT HIGHER-CHARGE RECONSTRUCTION WITH AN ALGEBRAIC HOMOTOPY CERTIFICATE

The previous implementation produced certified PL charge only in $\{-1, 0, +1\}$. We now supply a full higher-charge example in which (i) the continuous map has analytically known nonunit charge, (ii) the PL map has an independently computed integer via exact crossings, and (iii) *a separate fully finite positivity certificate proves that the two maps are homotopic.* This last step

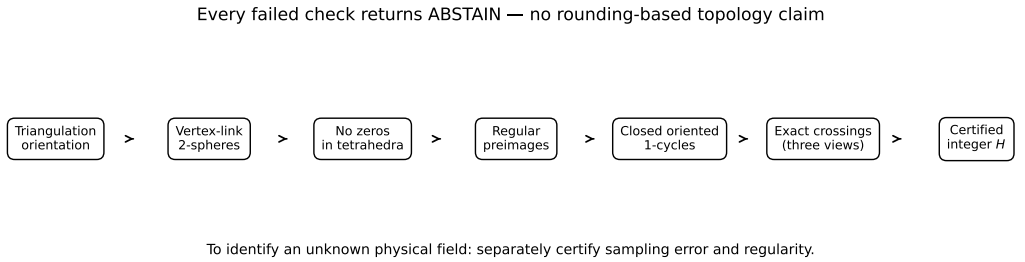


FIG. 16. Fail-closed verification protocol. Every algebraic or geometric degeneracy forces an explicit refusal rather than an unsupported integer. Transferring the PL integer to an unknown physical field additionally requires a calibrated error and regularity envelope.

is necessary: an exact invariant of the discretization alone need not equal that of the analytic or physical field.

Let $Q = [-1, 1]^4 \subset \mathbb{R}^4 \cong \mathbb{C}^2$, identify its oriented boundary with $\mathbb{S}^3$ by radial homeomorphism, and let $z_1 = w + ix$, $z_2 = y + iz$. Define

$$F_k(z_1, z_2) = (z_1^k, z_2), \qquad n_k(z) = \frac{\mathcal{H}(F_k(z))}{|z_1|^{2k} + |z_2|^2}, \qquad \mathcal{H}(a, b) = (2\,\mathrm{Re}(a\bar{b}), 2\,\mathrm{Im}(a\bar{b}), |a|^2 - |b|^2). \tag{79}$$

The field is everywhere defined on ∂Q; the only zero of F_k in Q is the origin. For $k = 2$, its unnormalized numerator is the explicit polynomial

$$\mathcal{H}(F_2) = \big(2[(w^2 - x^2)y + 2wxz],\ 2[2wxy - (w^2 - x^2)z],\ (w^2 + x^2)^2 - y^2 - z^2\big). \tag{80}$$

For computational chart convenience the code applies the orientation-preserving cyclic permutation $(H_x, H_y, H_z) \mapsto (H_z, H_x, H_y)$; this does not change its Hopf integer.

Theorem 19 (Analytic charge and the finite exact $N = 5$ transfer witness)**.** *For $k = 2$ in Eq. (79), the analytic map has $|H(n_2)| = 2$. On the standard oriented boundary-of-fourcube Freudenthal triangulation with five subdivisions per edge, let P be the barycentric piecewise-linear interpolant of the rational vertex values $n_2(v)$ on the fixed unit cube. Then P is nowhere zero, the normalized interpolant $P/\|P\|_2$ is homotopic to n_2, and the exact inverse-image linking algorithm computes*

$$H(n_2) = H(P/\|P\|_2) = -2 \tag{81}$$

with the orientation conventions of our code. Precomposing with a reflection of Q reverses the sign to $+2$. All claims are about this explicitly given analytic synthetic map, not arbitrary unknown fields.

Proof. The complex polynomial $F_2 : \mathbb{C}^2 \to \mathbb{C}^2$ has an isolated zero at the origin and complex local multiplicity 2: a sufficiently small regular target (a, b) with $a \neq 0$ has the two preimages $(\pm\sqrt{a}, b)$, both with positive real Jacobian sign. Hence its normalized boundary map $F_2/\|F_2\|$ has degree 2. The Hopf invariant under precomposition with a degree-d self-map of $\mathbb{S}^3$ is multiplied by d, so the magnitude equals 2.

For the separate PL-to-analytic transfer we use barycentric coordinates $\lambda_i \geq 0$, $\sum_{i=0}^{3} \lambda_i = 1$, on each tetrahedron with vertices v_i. Write $q(\lambda) = \sum_i \lambda_i v_i$, $S_i = |z_1(v_i)|^4 + |z_2(v_i)|^2$, $D = \mathrm{lcm}(S_0, S_1, S_2, S_3)$, and $P(\lambda) = \sum_i \lambda_i \mathcal{H}(F_2(v_i))/S_i$. The exact scalar polynomial

$$G_\sigma(\lambda) = D\,\langle P(\lambda), \mathcal{H}(F_2(q(\lambda)))\rangle \tag{82}$$

can be homogenized to total barycentric degree five by multiplication of its terms with suitable powers of $\sum_i \lambda_i = 1$. The accompanying rational-integer verifier constructs this homogeneous polynomial exactly from the *integer-scaled* vertex representation $V_i = N v_i$, $F_{2,N}(V) = (z_1(V)^2, N z_2(V))$, which equals $N^2 F_2(v)$ and hence defines the same normalized unit-cube field. It multiplies the homogenized polynomial by $(\sum_i \lambda_i)^{d-5}$ for $d \in \{5, 8, 10\}$, and checks *every* degree-d monomial coefficient strictly positive. Such a coefficient check implies $G_\sigma(\lambda) > 0$ on the entire closed simplex; no floating-point sampling or numerical root search is used.

For the fixed-domain field, *all* $36 \cdot 5^3 = 6000$ oriented tetrahedra pass already at homogeneous degree $d = 5$. Separately, the native-cube normalization $F_2(z_1, z_2) = (z_1^2, z_2)$ on $\partial[-5, 5]^4$ passes in 5856 cells at $d = 5$, another 96 at $d = 8$, and the remaining 48 at $d = 10$. These are two analytically distinct but degree-two families. No cell is left unresolved in either mode. Consequently $\langle P(q), n_2(q)\rangle > 0$ at every point of ∂Q, so P never vanishes and the straight-line path $(1-t)P + tn_2$ is nowhere zero. Normalization gives the desired homotopy. The exact rational inverse-image algorithm independently verifies closed, oriented preimage polygons and gives linking -2 in three rationally tilted projections. An orientation-reversing domain reflection gives $+2$. This establishes the signed conclusion without using the anticipated integer as an input to the linker. □

Corollary 4 (A conservative universal resolution threshold for the polynomial family)**.** *For any integer $k \geq 1$, the vertex-sampled barycentric PL map associated with Eq. (79) is homotopic to the analytic map on ∂Q whenever*

$$\boxed{N > 4\sqrt{3}\, k\, 2^{(k-1)/2}.} \tag{83}$$

For $k = 2$ this yields the conservative sufficient threshold $N \geq 20$, while the exact Bernstein witness of Theorem 19 improves the particular validated instance *to $N = 5$. This result does not imply*

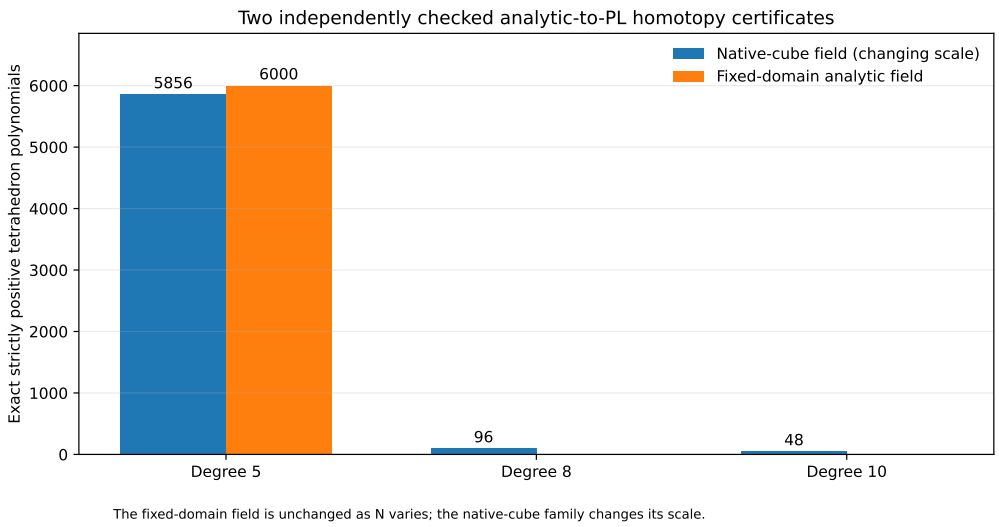


FIG. 17. Complete finite algebraic proof of the analytic-to-PL homotopy for the charge-two benchmark. Bars count tetrahedra certified at the earliest displayed homogeneous Bernstein degree for two analytically distinct, explicitly labelled normalization conventions. Every monomial positivity check uses integer arithmetic.

that the numerical algorithm succeeds at every smaller resolution.

Proof. On each cubical boundary facet at least one of the four real coordinates has absolute value one. Hence $\left\|(z_1^k, z_2)\right\|_2 \geq 1$. Along that facet the differential of F_k has operator norm at most $k2^{(k-1)/2}$, since $|z_1| \leq \sqrt{2}$. Normalization to the unit sphere has differential norm at most $1/\left\|F_k\right\| \leq 1$, and the standard Hopf map from the unit three-sphere to the unit two-sphere has derivative norm at most 2. Thus n_k is Lipschitz on each facet with $L_k \leq 2k2^{(k-1)/2}$. Every Freudenthal tetrahedron in a boundary facet has Euclidean diameter at most $2\sqrt{3}/N$. On any cell, $\|P(q) - n_k(q)\| \leq L_k(2\sqrt{3}/N) < 1$, since P is a convex combination of the vertex images. This excludes a zero of P and gives a nonvanishing straight-line homotopy. At no stage is a fixed global Lipschitz estimate asserted across arbitrary paths passing through different facets; the cellwise bound suffices. □

A. Failure modes and independence checks

The independent Python verifier checks all 6000 cell polynomials in each of the two normalization modes, evaluates the polynomial identity at 63 predetermined rational barycentric points per mode against an independently implemented direct substitution, verifies a reflection-reversed positive-charge benchmark, and repeats the negative-charge linking on $N = 7$ and $N = 9$ meshes. As a

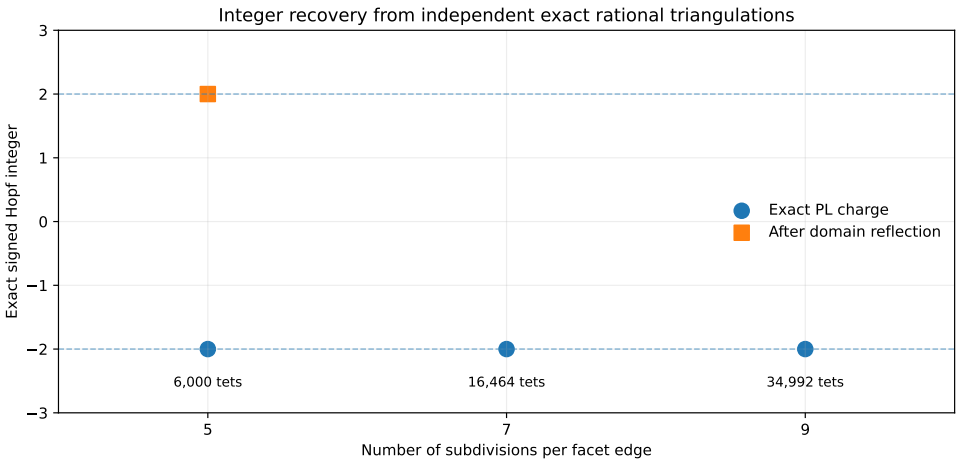


FIG. 18. Exact charge-two linking from independently constructed rational boundary-fourcube meshes. On $N = 5$ the analytic transfer is additionally proved by a complete homogeneous-polynomial positivity witness. Repeated meshes employ the same fixed-domain analytic field. Their PL integer checks test refinement stability; full analytic transfer requires a positive witness or the conservative bound of Corollary 4.

separate blind-format control, the complete $N = 5$ rational vertex data are exported to a standalone JSON file and the pre-existing `certified_hopf_linking_v8.py` is invoked without an analytic reference field, returning $H = -2$ with closed oriented inverse-image cycles. The $N = 4$ native-cube charge-two input is explicitly rejected because the PL interpolant has a zero in a tetrahedron. Thus insufficient spatial resolution is not silently repaired by integer rounding. Earlier unverified $k = 3, 4$ numerical prototypes exposed discretization/regularity issues. The subsequently audited rational-chart results are given in Section XX; numerical integer agreement alone is never equated with a local straight-line homotopy witness.

XX. RATIONAL-CHART CERTIFICATION OF CHARGES THREE AND FOUR

A. A singular-point-free chart for global inverse-image curves

The v8 linker projected a cubical boundary facet out of the domain; this restricted inverse-image cycles for larger charges. We instead puncture the boundary at the *single point* $q_+ = (N, 0, 0, 0)$ in the coordinates $q = (w, \boldsymbol{u}) \in \partial[-N, N]^4$. Define the rational map

$$\boxed{\Psi_N(w, \boldsymbol{u}) = \frac{\boldsymbol{u}}{\|\boldsymbol{u}\|^2 + 2N(N - w)} \in \mathbb{Q}^3} \tag{84}$$

for rational input points away from q_+. Its denominator is positive off q_+. For every direction $\boldsymbol{u} = t\boldsymbol{v}$, $\|\boldsymbol{v}\| = 1$, the cubical surface consists successively of a front-cap segment $w = N$, at most

one side-wall segment with the largest coordinate $|u_j| = N$, and a back-cap segment $w = -N$. Along the front cap the radial coordinate is $1/t$, strictly decreasing with t. Along the side wall, as w decreases, it is strictly decreasing because t is fixed. Along the back cap it is $t/(t^2 + 4N^2)$, strictly increasing in t on $0 \le t \le \sqrt{3}N < 2N$, hence decreases as one travels from the side wall to the back pole. These continuous pieces join without a jump and cover $(0, \infty)$ exactly once. Thus Ψ_N is a homeomorphism $\partial[-N, N]^4 \setminus \{q_+\} \to \mathbb{R}^3$; its inverse preserves each outward ray and its global orientation is calibrated against the charge-one benchmark. This is a *rational radial chart*, not an assertion that the cube is round in its ambient metric.

For two regular rational target rays $p, q \in \mathbb{Q}^3$ the algorithm identifies the oriented preimage inside each affine simplex using signed 3×3 determinants and the strict *positive ray* condition. It then checks that each oriented preimage is a disjoint union of closed cycles. Each segment $s(t) = (1-t)a + tb$ on the boundary is sent to the rational curve $\Psi_N(s(t))$; this curve is in general *not* a straight segment. Therefore replacing it with a chord without an explicit perturbation bound would be invalid.

Theorem 20 (Exact rational linking under certified chord approximation)**.** *Suppose an oriented triangulation of the cubical three-sphere passes the manifold, orientation and nonzero-field checks, and p, q are regular values yielding disjoint, closed polygonal preimages avoiding the chart pole. Subdivide the preimages into finitely many affine parameter segments s_i, t_j; let C_i, D_j be the straight chords joining the endpoints of $\Psi_N \circ s_i$ and $\Psi_N \circ t_j$. Let $e_i, f_j \ge 0$ be rigorous uniform bounds for the respective pointwise deviations from those chords. If*

$$\boxed{\operatorname{dist}(C_i, D_j) > e_i + f_j \quad \textit{for every } (i, j),} \tag{85}$$

then the linking number of the original oriented preimages equals that of the rational chord polygons in $\mathbb{R}^3$, which can be calculated by exact rational oriented-crossing signs. Any violated strict predicate forces an explicit ABSTAIN *outcome.*

Proof. Linearly interpolate each smooth image segment to its chord, fixing the segment endpoints. By the uniform error bounds the moving point of each interpolating segment lies within e_i or f_j, respectively, of its chord at the same parameter. The strict distance hypothesis guarantees that images of the *two distinct components* cannot intersect at any intermediate time. Linking of disjoint oriented closed maps is invariant under this homotopy. Self-intersections of either individual intermediate curve are immaterial for the mutual Gauss linking degree, provided the two entire maps remain disjoint. A generic rational linear projection determines crossings from the signed determinant predicates; nontransverse coincidences are rejected. □

In the implementation the chord error is bounded by the standard interpolation inequality $\|g - \ell\|_\infty \le \|g''\|_\infty/8$ on each *rescaled subinterval* and a rational upper bound for the second derivative of $\Psi_N \circ s$. Bounding boxes with directed integer rounding reject obviously separated pairs; uncertain pairs require an exact rational constrained segment-distance calculation. Both branches are conservative, not Monte Carlo. Three rationally perturbed projection directions must return the same integer.

B. Reproduced integer results for the polynomial family

Let $F_k(z_1, z_2) = (z_1^k, z_2)$ on the fixed domain $\partial[-1, 1]^4$, $k \in \mathbb{N}$, and form the two-sphere map $n_k = \mathcal{H}(F_k)/\|F_k\|^2$ in the orientation adopted above. The polynomial map $F_k : \mathbb{C}^2 \to \mathbb{C}^2$ has an isolated zero at the origin with *positive complex local multiplicity* k. Its normalized boundary map has degree k, so composition with the Hopf fibration has Hopf invariant $-k$ in the cubical outward-orientation convention of the implementation, and a domain reflection changes its sign. These are established consequences of degree theory, not a novel classification of $\pi_3(S^2)$.

The executable `verify_v10_higher_charge.py` independently obtains the following integers from rational PL vertex data. Each test uses regular rational target directions, strict nonzero tests, closed oriented inverse images, chord separation and three exact rational projections:

Family	Subdivisions per facet	Tetrahedra	Inferred H	Chord pairs certified
F_1	4	3072	-1	19600
F_2	5	6000	-2	94864
F_3	5	6000	-3	176336
F_3 reflected	5	6000	$+3$	176336
F_4	5	6000	-4	705276
F_4 reflected	5	6000	$+4$	705276

Additional $N = 7$ checker runs reproduced $H = -3$ and $H = -4$ on 16464 tetrahedra. Two independent JSON tables with 1040 rational three-component sample vectors were then passed to the extractor *without* the analytic generating formula; the returned integers were -3 and -4. Expected charges are used solely as *post-inference test assertions*. These are blind-input-format checks, not blinded laboratory trials. The supplied rational chart is valid for all regular preimage cycles avoiding its single pole, while external arbitrary inputs can legitimately trigger abstention.

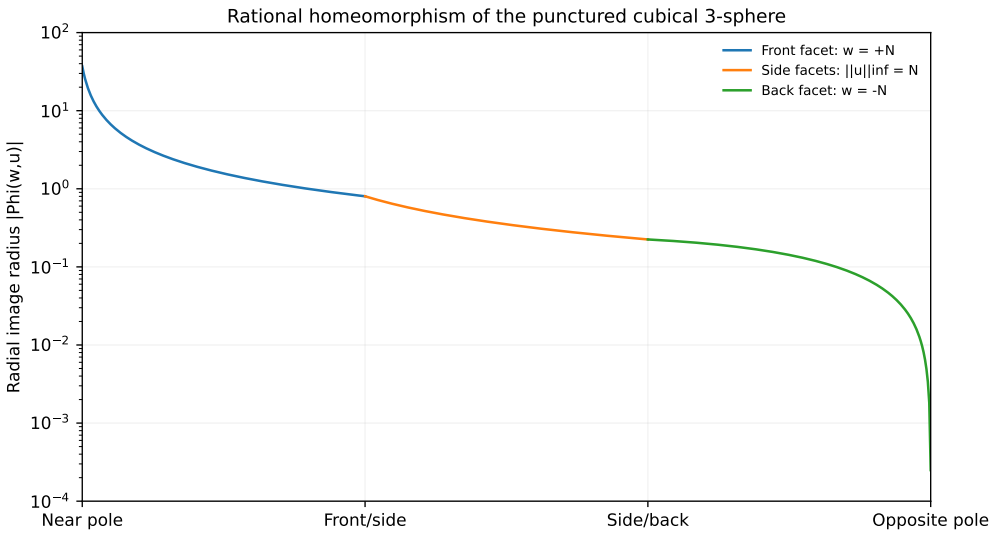


FIG. 19. A rational global chart punctures the boundary of the fourcube at one point. Unlike excluding a cubical facet, it accommodates regular inverse-image loops traversing multiple facets. The geometry of the resulting curved segments is controlled by strict rational chord-error and separation bounds.

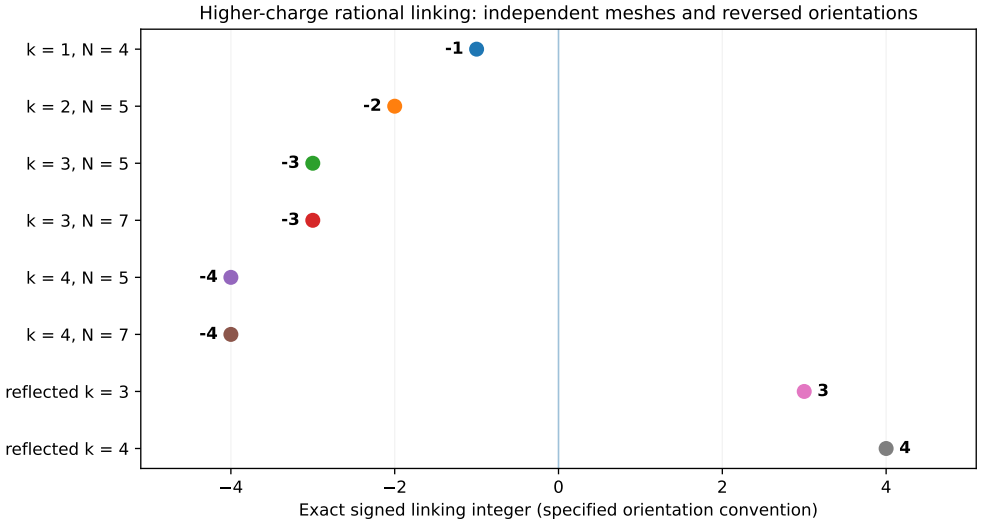


FIG. 20. Reproducible exact rational PL charges ± 3 and ± 4 at $N = 5$, including orientation reversal. The plotted counts describe certified chord pairs; they are not quantum measurement budgets.

C. Analytic-to-discrete transfer: what is and is not proved

Proposition 5 (Class-level analytic–PL transfer for the tested charge family)**.** *On an oriented cubical three-sphere, if the degree of F_k is independently established as k and an independently validated PL map P has a certified Hopf invariant $-k$ in the same orientation convention, then the*

sphere-valued normalization $P/\|P\|$ is homotopic to the analytic map n_k. The reflection version has the opposite integer.

Proof. The classification $[S^3, S^2] \cong \mathbb{Z}$ by Hopf invariant implies two continuous maps are homotopic if and only if their Hopf integers coincide. The analytic value follows from the local holomorphic degree, while the PL value follows from the independently certified rational-crossing calculation. □

This proof establishes *existence* of a homotopy in the tested analytic family. It does not construct the straight-line homotopy at each point and therefore cannot replace a local Bernstein or Lipschitz positivity certificate when the analytic degree is *unknown*. For unknown laboratory fields, a calibrated measurement/error and regularity bound remains essential before attributing a PL integer to the underlying physical system.

D. Falsification and correction of a proposed differential constant

A proposed bound $\|dn_2\|_{\mathrm{op}} = 4/\sqrt{3}$ for the degree-two field on the entire cube boundary is false for the stated *ambient Euclidean differential norm*: at $|z_1|^2 = 1/2$ and $|z_2| = 1$, direct substitution into the horizontal-Wronskian formula yields $12/5 > 4/\sqrt{3}$. The proposed polynomial identity also fails at $t = 1/2$, where its left-hand side is -2.

Proposition 6 (A corrected sufficient degree-two sampling bound)**.** *For the analytic field $n_2 = \mathcal{H}(z_1^2, z_2)/(|z_1|^4+|z_2|^2)$, the full ambient differential obeys $\|dn_2\|_{\mathrm{op}} \le 5/2$ at every point of $\partial[-1,1]^4$. Its barycentrically interpolated vertex field on the standard Freudenthal tetrahedra is therefore homotopic to the analytic field whenever $N \ge 9$.*

Proof. The horizontal complex differential identity, a consequence of the Hopf projection, is

$$\|dn_2(z)\|_{\mathrm{op}}^2 = \frac{4|z_1|^2(4|z_2|^2 + |z_1|^2)}{\left(|z_1|^4 + |z_2|^2\right)^2}. \tag{86}$$

On the region $|z_1| \ge 1$ the bound $\|dn_2\| \le 4/\sqrt{3} < 5/2$ follows by setting $s = |z_2|^2/|z_1|^4$ and maximizing $4(4s + 1)/(1 + s)^2$. On the remaining cubical boundary region $|z_1| \le 1$ we have $|z_2| \ge 1$. Monotonicity in $|z_2|^2 \ge 1$ reduces the maximum to $|z_2|^2 = 1$. Setting $t = |z_1|^2 \in [0, 1]$, it is sufficient to prove

$$P(t) = 25t^4 + 34t^2 - 64t + 25 > 0. \tag{87}$$

For the three rational intervals $[0, 1/2]$, $[1/2, 3/4]$, $[3/4, 1]$ all five exact degree-four Bernstein coefficients are strictly positive, as independently output by the verifier. This yields $\|dn_2\|^2 < 25/4$ on

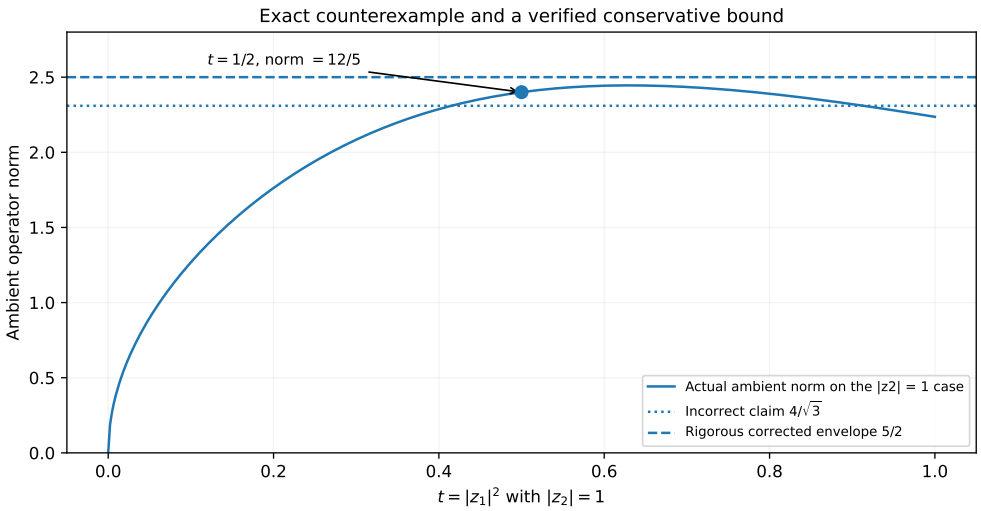


FIG. 21. The claimed $4/\sqrt{3}$ ambient bound is contradicted by a rational counterexample; a rigorously checked $5/2$ bound recovers the degree-two $N \geq 9$ sufficient transfer threshold.

this region. Every tetrahedron has Euclidean diameter at most $2\sqrt{3}/N$ and lies in one face; hence a convex combination of vertex samples differs from the analytic field by at most $(5/2)(2\sqrt{3}/N) < 1$ at $N \geq 9$. The straight-line interpolant cannot vanish, and normalization gives the homotopy. □

The last bound is independent of the rational-chart method and is a conservative *fixed-domain* estimate; in particular it does not imply a universal resolution threshold linear in k for arbitrary high-degree polynomial maps. Numerical evidence about balancing $\alpha_k z_1^k$ is not a proof of a uniform bound. In fact, the proposed $k = 4$, $\alpha = 1/4$, $N = 5$ sample is rejected by the exact zero-in-simplex test. This negative control demonstrates why purported "automatic fixes" to discretization cannot replace strict checks.

XXI. LINEAR-RESOLUTION CERTIFICATION FOR ARBITRARY POSITIVE CHARGE

The preceding conservative resolution condition, Eq. (83), grows exponentially with k because it bounds the differential of the *unnormalized* complex map F_k before applying the Hopf projection. The horizontal complex derivative instead cancels precisely the radial component responsible for this overestimate. This leads to a bound linear in the charge k for *every* positive integer k, not merely the previously verified small-charge samples.

Theorem 21 (Linear, order-sharp charge-dependent regularity)**.** *For every integer $k \geq 1$, let n_k be the smooth rational Bloch field on $X = \partial[-1,1]^4$ defined in Eq. (79). Its full ambient Euclidean differential on every open cubical facet satisfies*

$$\boxed{\sqrt{k^2+1} \;\leq\; \sup_{q\in X} \|Dn_k(q)\|_{\mathrm{op}} \;\leq\; k+2.} \tag{88}$$

The supremum in the lower bound may be interpreted using tangent directions inside the intersection of adjacent facets; its value is also the limit of facet-tangent derivatives at interior points. The bounds show that the differential scale is $\Theta(k)$ for this particular family.

Proof. Set $a = |z_1|$, $b = |z_2|$, $U = a^k$. A standard computation for the Hopf projection (equivalently, the Fubini–Study horizontal Wronskian) gives the *exact full ambient* differential norm

$$\|Dn_k\|_{\mathrm{op}} = \frac{2\sqrt{k^2a^{2k-2}b^2 + a^{2k}}}{a^{2k}+b^2}, \qquad (a,b) \neq (0,0), \tag{89}$$

with the continuous derivative interpretation at $a = 0$ and $k = 1$. On X, at least one of a, b is at least one, as a real cubical coordinate equals ± 1. By $\sqrt{A^2+B^2} \leq A + B$ for nonnegative A, B,

$$\|Dn_k\| \leq T_1 + T_2, \quad T_1 = \frac{2ka^{k-1}b}{a^{2k}+b^2}, \quad T_2 = \frac{2a^k}{a^{2k}+b^2}. \tag{90}$$

If $a \geq 1$, use $2Ub \leq U^2 + b^2$ to obtain $T_1 \leq k/a \leq k$ and $T_2 \leq 2/U \leq 2$. If instead $a \leq 1$ and $b \geq 1$, then $U \leq 1$ and $b \mapsto b/(U^2+b^2)$ decreases on $[1,\infty)$. Therefore

$$T_1 \leq \frac{2ka^{k-1}}{1+a^{2k}} \leq k+1. \tag{91}$$

For $k = 1$, this follows from $2/(1+a^2) \leq 2$. For $k \geq 2$, the interior maximum occurs at $a^{2k} = (k-1)/(k+1)$, where the last expression is $(k+1)a^{k-1} \leq k+1$. Also $T_2 \leq 2U/(1+U^2) \leq 1$. Both cases give $\|Dn_k\| \leq k+2$.

For the converse, take the edge-intersection point $(z_1, z_2) = (1,1)$ on the cube boundary, and vary the two free imaginary coordinates $x = \Im z_1$, $z = \Im z_2$ while keeping the two real coordinates equal to 1. At $(x,z) = (0,0)$ the Bloch vector is $(1,0,0)$ and its infinitesimal transverse component satisfies $dn_{k,y} = k\,dx - dz$. The operator norm on these allowed two directions is exactly $\sqrt{k^2+1}$. Continuity along either incident open facet gives the same limiting lower bound. Thus no family-wide differential ceiling can be $o(k)$. □

Corollary 5 (Linear mesh threshold and certified analytic-to-PL transfer)**.** *On the standard boundary-of-fourcube Freudenthal triangulation with N equal edge subdivisions, form the barycentric interpolant P_N of the exact vertex values $n_k(v)$. The strict, explicit sufficient condition*

$$\boxed{N > 2\sqrt{3}\,(k+2)} \tag{92}$$

implies $P_N(q) \neq 0$ *for every point of* X*, and* $P_N/\|P_N\| \simeq n_k$ *through the* explicit normalized straight-line homotopy*. Consequently the exact integer Hopf invariant of the normalized PL field is the analytic integer* $(-k$ *in our cubical orientation convention). The number of tetrahedra is* $48N^3 = O(k^3)$*, and the number of distinct vertices is* $(N+1)^4 - (N-1)^4 = 8N^3 + 8N$*.*

Proof. Every Freudenthal tetrahedron lies in a single cubical facet and has Euclidean diameter at most $2\sqrt{3}/N$. Within it, $P_N(q) = \sum_i \lambda_i n_k(v_i)$ with $\lambda_i \geq 0$ and $\sum_i \lambda_i = 1$. By Theorem 21 and convexity,

$$\|P_N(q) - n_k(q)\| \leq \sum_i \lambda_i (k+2)\|v_i - q\| \leq \frac{2\sqrt{3}(k+2)}{N} < 1. \tag{93}$$

Hence $P_N(q) \cdot n_k(q) > 0$ and in particular $P_N(q) \neq 0$. The straight line $(1-t)n_k + tP_N$ stays in the open hemisphere centered on $n_k(q)$ for each q, so its normalization gives an explicit global homotopy. The complex local degree of F_k supplies the analytic integer. There are eight cubical three-facets and six Freudenthal tetrahedra in each of N^3 unit cells on a facet, yielding $48N^3$ cells; the difference of all lattice vertices and interior lattice vertices yields $8N^3 + 8N$ unique boundary vertices. □

Corollary 6 (Conditional Pauli-shot transfer with an explicit resource bound)**.** *Assume the unknown physical field is precisely* n_k*, or more generally a field with a separately established facetwise Lipschitz ceiling* $k+2$ *on the same cubical domain. Measure* m *independent, calibrated* ± 1 *outcomes for* each *of the three Pauli components at each of the* $V = 8N^3 + 8N$ *vertices. Denote their empirical three-vectors by* b_v*, and use their barycentric interpolant* B_N*. Set* $\mu = 1 - 2\sqrt{3}(k+2)/N > 0$*. If*

$$\boxed{m > \frac{6}{\mu^2} \log\left(\frac{6V}{\delta}\right), \qquad 0 < \delta < 1,} \tag{94}$$

then with probability at least $1 - \delta$ *the everywhere-defined normalized field* $B_N/\|B_N\|$ *is homotopic to the physical field, and therefore has exactly the correct Hopf integer. An independent* sound *exact integer extractor for the measured rational vertex means may then report that integer when all its strict regularity predicates pass; the inequality does not assert that a particular finite numerical linker always terminates or accepts.*

Proof. For a fixed vertex and Pauli component Hoeffding's inequality yields $\Pr(|\widehat{b} - n_{k,a}(v)| > t) \leq 2\exp(-mt^2/2)$. Let $\eta = \sqrt{6\log(6V/\delta)/m} < \mu$. Apply a union bound over the $3V$ components with $t = \eta/\sqrt{3}$, giving failure probability at most $6V\exp(-m\eta^2/6) = \delta$. On the complementary

event $\max_v \|b_v - n_k(v)\| \le \eta < \mu$. The same barycentric error decomposition as in Eq. (93) then gives the genuinely strict bound $\|B_N(q) - n_k(q)\| \le \eta + 2\sqrt{3}(k+2)/N < 1$ at every point. This supplies the normalized homotopy. Neither a spectral gap alone nor a synthetic statistical model validates the required physical regularity and calibration hypotheses for real data. □

Remark 5 (Relation to the previous small-charge and fundamental limits)**.** *The new all-k sufficient bound replaces the* exponentially growing estimate *of Corollary 4; it does not claim that the smaller exact charge-two witness at $N = 5$ or the validated $k = 2$ bound $N \ge 9$ is superseded for that special case. The upper bound $O(k^3)$ is a sufficient uniform triangulation size, not an algorithmic or measurement lower bound for this restricted algebraic family. Linear optimality in Theorem 21 concerns the* differential regularity growth*; the general hidden-Hopf-bubble lower bound concerns a much larger Lipschitz class. The two claims must not be conflated. The two-band operator $H_k = n_k \cdot \boldsymbol{\sigma}$ has eigenvalues ± 1 and constant gap 2 for every k, illustrating that a fixed gap alone does not bound the topological integer. This is a synthetic construction, not a realization of a four-dimensional Yang–Mills mass gap or a three-dimensional Coulomb inverse theorem.*

k	Previous sufficient N	Linear sufficient N	$48N^3$ (new)	$\sqrt{k^2+1}$ lower slope
1	7	11	63888	$\sqrt{2}$
2	20	14	131712	$\sqrt{5}$
3	42	18	279936	$\sqrt{10}$
4	79	21	444528	$\sqrt{17}$
5	139	25	750000	$\sqrt{26}$
10	1568	42	3556224	$\sqrt{101}$

All mesh numbers are independently rounded with exact integer-square comparisons in the supplementary verifier. The earlier sufficient $k = 1$ bound is better; for $k \ge 2$ the new sufficient N values improve the previous coarse general estimate. The $k = 2$ special $N \ge 9$ condition remains stronger than either coarse general formula.

XXII. BIDEGREE HOPF TOMOGRAPHY: NEAR-SHARP METRIC CONTROL AND EXACT TWO-LOOP IDENTIFICATION

The general geometric $O(L^4)$ distortion estimate for $S^3 \to S^2$ and constructions achieving its order are classical: our polynomial subfamily below neither attains nor improves that universal exponent [9, 10]. What is new *within the present certification protocol* is the explicit interaction

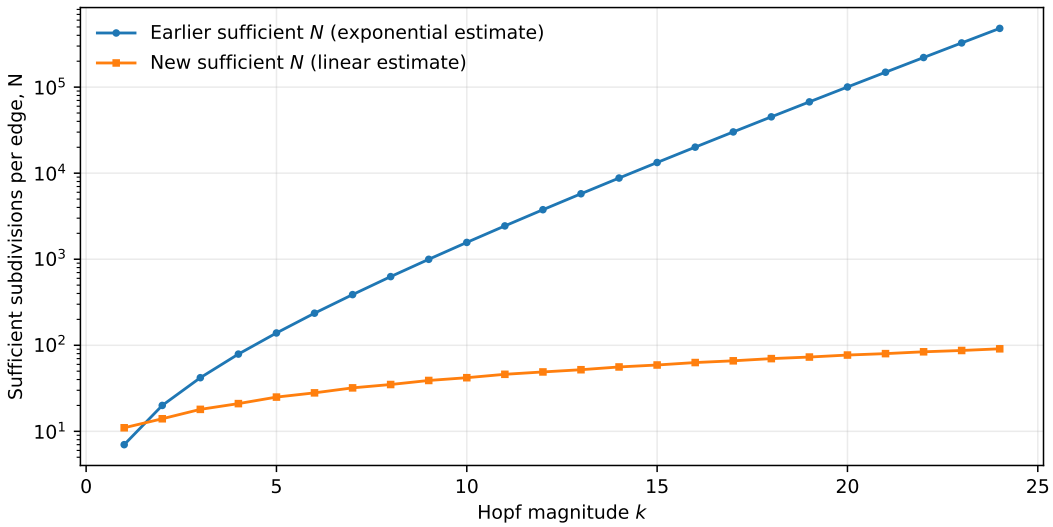


FIG. 22. Sufficient subdivisions versus topological charge on a logarithmic ordinate. The earlier exponentially growing *sufficient bound* is compared against the new linear sufficient bound. Neither curve represents a proved minimal interpolation resolution, laboratory shot count, or computational complexity lower bound.

of a bidegree parameterization, a rationally checkable cell-resolution guarantee, and a finite-shot two-loop estimator with a matching aliasing obstruction. The statements below do not assert first discovery of the topological map or the winding-number principle.

Let $X = \partial[-1,1]^4 \subset \mathbb{C}^2$, oriented as in the preceding integer-extraction sections, and let $p, q \geq 1$ be integers. Define

$$F_{p,q}(z_1, z_2) = (z_1^p, z_2^q), \qquad n_{p,q} = \frac{(2\,\mathrm{Re}(F_1\overline{F_2}),\ 2\,\mathrm{Im}(F_1\overline{F_2}),\ |F_1|^2 - |F_2|^2)}{|F_1|^2 + |F_2|^2}. \tag{95}$$

The denominator never vanishes on X. Here $\|Dn\|_{\mathrm{amb}}$ denotes the operator norm of the full Euclidean differential in a neighborhood of each open cubical facet, so the bound also controls facet-tangent derivatives and simplex chord estimates.

Theorem 22 (Certified bidegree Hopf class and additive-one metric bracket)**.** *With the cubical orientation fixed in the preceding reconstruction sections,*

$$\boxed{H(n_{p,q}) = -pq.} \tag{96}$$

Writing $a = |z_1|$, $b = |z_2|$, its exact ambient derivative norm is

$$\|Dn_{p,q}\|_{\mathrm{amb}} = \frac{2\sqrt{p^2 a^{2p-2} b^{2q} + q^2 a^{2p} b^{2q-2}}}{a^{2p} + b^{2q}}, \tag{97}$$

with the continuous interpretation at zero bases. Put

$$L_{p,q}^2 = \max\{p^2 + (q+1)^2, (p+1)^2 + q^2\}. \tag{98}$$

Then the following globally valid facetwise *inequalities hold:*

$$\boxed{\sqrt{p^2+q^2} \;\le\; \sup_X \|Dn_{p,q}\|_{\mathrm{amb}} \;\le\; L_{p,q} \;\le \sqrt{p^2+q^2}+1.} \tag{99}$$

The lower bound is the limit of allowed facet-tangent derivatives at the intersection point $(z_1, z_2) = (1, 1)$.

Proof. Normalize $F_{p,q}$ to a map $X \simeq S^3 \to S^3$; it has degree pq because its Hopf-coordinate angular variables are sent to $(p\theta_1, q\theta_2)$ while the radial coordinate is monotonically reparameterized. Composition with the Hopf projection multiplies its invariant by that degree, and the existing cubical orientation introduces the minus sign of Eq. (96). For the metric assertion, differentiate the complex projective Hopf coordinate: its horizontal Wronskian is $F_2\,dF_1 - F_1\,dF_2$. The map to the unit Bloch sphere has infinitesimal metric $2|F_2\,dF_1 - F_1\,dF_2|/(|F_1|^2+|F_2|^2)$, proving Eq. (97) by orthogonality of the two complex input variables. Define

$$A = \frac{2p\,a^{p-1}b^q}{a^{2p}+b^{2q}}, \qquad B = \frac{2q\,a^p b^{q-1}}{a^{2p}+b^{2q}}; \qquad \|Dn_{p,q}\|_{\mathrm{amb}} = \sqrt{A^2+B^2}. \tag{100}$$

On X either $a \ge 1$ or $b \ge 1$. If $a \ge 1$, the arithmetic-geometric inequality $2a^pb^q \le a^{2p}+b^{2q}$ gives $A \le p/a \le p$. For the other term, if $b \ge 1$ then $B \le q$; if $b \le 1$ and $a \ge 1$, monotonicity in $a^p \ge 1 \ge b^q$ yields $B \le 2qb^{q-1}/(1+b^{2q}) \le q+1$. The final inequality follows by differentiating the rational expression: for $q \ge 2$ its maximum over $b \in [0,1]$ occurs at $b^{2q} = (q-1)/(q+1)$ with value $(q+1)b^{q-1} \le q+1$; for $q = 1$ it is at most 2. Thus $\|Dn\|^2 \le p^2+(q+1)^2$ in this case. Exchanging the coordinates proves $\|Dn\|^2 \le (p+1)^2+q^2$ when $b \ge 1$. At $(1,1)$ the two independent permitted imaginary tangent variations induce differential Bloch phase $p\,d\,\mathrm{Im}\,z_1 - q\,d\,\mathrm{Im}\,z_2$, whose Euclidean norm is $\sqrt{p^2+q^2}$. Finally $L_{p,q}^2 = p^2+q^2+2\max(p,q)+1 \le (\sqrt{p^2+q^2}+1)^2$. □

Corollary 7 (Bidegree analytic-to-PL transfer and square-charge cost)**.** *Use the same* N*-subdivision Freudenthal cubical boundary mesh as before. If*

$$\boxed{N^2 > 12\max\{p^2+(q+1)^2, (p+1)^2+q^2\},} \tag{101}$$

then the barycentric interpolant of exact vertex Bloch vectors is nowhere zero and its normalization has Hopf integer $-pq$*. Its mesh contains* $48N^3$ *tetrahedra and* $8N^3+8N$ *distinct vertices. For balanced* $p = q = r$*, the charge magnitude is* r^2 *while* $N = O(r)$ *and the sufficient triangulation budget is* $O(|H|^{3/2})$*. This does not state a minimax or output-sensitive complexity lower bound.*

Proof. Every tetrahedron has diameter at most $2\sqrt{3}/N$ and the field has derivative ceiling $L_{p,q}$. The simplex interpolation discrepancy is at most $2\sqrt{3}L_{p,q}/N < 1$; therefore the normalized straight-line homotopy is nowhere singular. Count simplices and vertices as in Corollary 5. □

A. Two closed equatorial loops replace the full three-dimensional grid under a structural promise

Define two circles inside the cubical boundary,

$$C_1(\theta) = (e^{i\theta}, 1), \qquad C_2(\theta) = (1, e^{i\theta}), \qquad 0 \le \theta < 2\pi. \tag{102}$$

For the complex transverse Bloch observable $u = n_x + in_y$, exact substitution into Eq. (95) gives

$$u(C_1(\theta)) = e^{ip\theta}, \qquad u(C_2(\theta)) = e^{-iq\theta}, \qquad \boxed{H(n_{p,q}) = \mathrm{wind}(u \circ C_1)\ \mathrm{wind}(u \circ C_2).} \tag{103}$$

This two-loop identity is *not* valid for arbitrary maps $X \to S^2$; it is a consequence of the specified globally known polynomial form.

Theorem 23 (Exact finite-shot two-loop integer certificate)**.** *Assume a priori that the unknown field belongs* exactly *to the family* $\{n_{p,q} : 1 \le p, q \le R\}$*, with known integer* $R \ge 1$*. Fix an integer* $M > 2R$ *and sample the two loops at* $\theta_j = 2\pi j/M$*,* $j = 0, \ldots, M-1$*. At each site measure the two Pauli axes* x, y *with* m independent calibrated ± 1 *outcomes per axis. Let* $\widehat{u}_{\ell,j}$ *be the empirical rational complex means and let* W_ℓ *be the integer winding of the resulting closed polygon around the origin, determined by an exact rational ray-crossing predicate. For* $0 < \delta < 1$*, if*

$$\boxed{m > \frac{4}{\cos^2(\pi R/M)} \log\!\left(\frac{8M}{\delta}\right),} \tag{104}$$

then, with probability at least $1 - \delta$*,* both *polygonal windings are unambiguously defined and*

$$\boxed{(W_1, W_2) = (p, -q), \qquad H = W_1 W_2 = -pq.} \tag{105}$$

There are $2M$ *loop-site instances, and* $4Mm$ *individual Pauli outcomes. Taking* $M = 4R$ *gives* $m = O(\log(R/\delta))$ *and a sufficient total budget* $O(R \log(R/\delta))$*. For the balanced subfamily* $p = q = R$*, this is* $O(\sqrt{|H|} \log |H|)$ only because the global polynomial family is promised in advance.

Proof. At any of the $4M$ measured scalar means, Hoeffding's bound is $2e^{-mt^2/2}$. A union bound implies that every empirical complex-vertex error is at most η except with probability at most $8M \exp(-m\eta^2/4)$. Under Eq. (104) take $\eta = \sqrt{4\log(8M/\delta)/m} < \cos(\pi R/M)$. For a pure winding $e^{is\theta}$ with $|s| \le R < M/2$, each straight chord between neighboring ideal vertices subtends angle at most $2\pi R/M < \pi$, so the ideal polygon has norm at least $\cos(\pi R/M) > 0$ everywhere and winding s. The linear interpolation of measured endpoints is at every point within distance η of that ideal polygon. Consequently the interpolating straight-line homotopy avoids zero. Both measured

polygons have exactly the ideal windings simultaneously. All their coordinates are rational sample means, so the winding integer can be computed with strict rational orientation predicates without floating-point rounding. The probability concerns only the declared independence and calibration model. □

Proposition 7 (Exact phase aliasing and the essential structural promise)**.** *If only the two sampled loops of Eq.* (102) *are observed on an M-point uniform grid, fields $n_{p,q}$ and $n_{p+M,q}$ generate* identical *noiseless data at every measured site, yet their Hopf charges differ by Mq. Therefore no estimator from these loop-grid observations alone can identify arbitrary, unbounded integer bidegrees, even when the two-band Hamiltonian has constant gap 2. Moreover, if the global polynomial-family assumption is dropped, observing the full* continuous *two loops does not certify a general $S^3 \to S^2$ homotopy class: a localized Hopf bubble can be inserted in an open ball disjoint from both loops, leaving their observations unchanged while shifting the global Hopf charge by ± 1.*

Proof. For the first assertion, at each θ_j one has $e^{i(p+M)\theta_j} = e^{ip\theta_j}$, while the second loop is independent of p. Every $n_{p,q} \cdot \boldsymbol{\sigma}$ has eigenvalues ± 1. For the second assertion, the union of the two circles is a proper closed subset of the three-sphere; select a disjoint three-ball. On a smaller ball the smooth field can be contracted to a constant via a boundary-preserving local homotopy. Inserting a compactly supported representative of the generator of $\pi_3(S^2) \cong \mathbb{Z}$ changes its global Hopf class by one while agreeing outside that ball, as in the localized construction of Theorem 3. No data on either loop can detect this modification. □

Remark 6 (Model identification is not unrestricted topology certification)**.** *The loop theorem identifies the bidegree within a known synthetic parametric model; it must* never *be used as an experimental certificate without an independent global bound showing that the physical field is homotopic to a fitted member. Replacing the structural promise by finite-sample goodness of fit is logically invalid because of Proposition 7. The full-mesh certificate of Corollary 7, or an independent regularity/tail certificate from earlier sections, is then still required. The new reconstruction method does not settle the completeness of an LTI-SDP spectral-gap hierarchy, the interacting 3D Hohenberg–Kohn inverse problem, or any ProofAtlas global conjecture.*

B. Reproduction and relation to existing topology algorithms

The companion `verify_v12_bidegree_loops.py` tests 32,400 exact rational evaluations of Eq. (97) across 144 positive exponent pairs, 246 pairs of rational polygon-winding controls, a

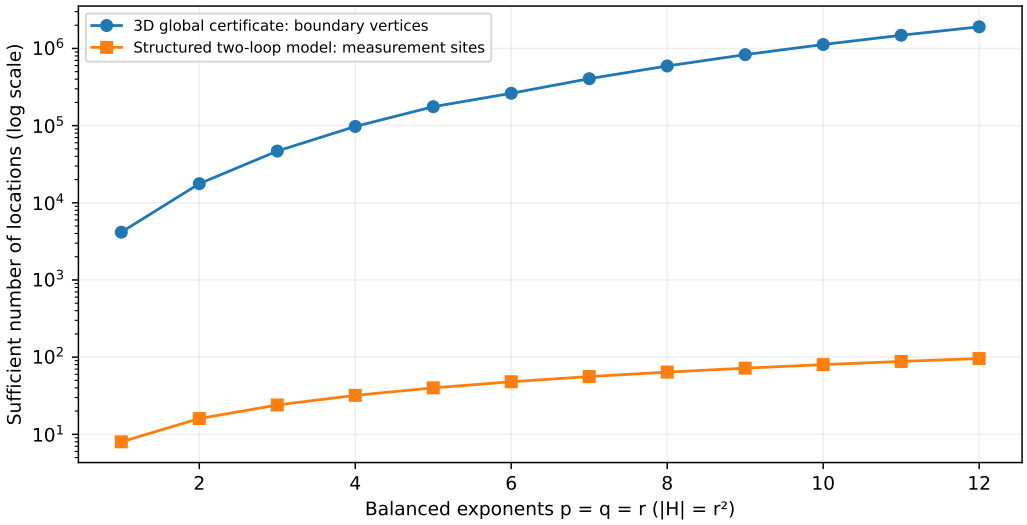


FIG. 23. Two different *sufficient* measurement-location budgets for the structured balanced field $n_{r,r}$: $2M = 8r$ sites for the two-loop winding protocol with a priori model assumption ($M = 4r$), versus the $8N^3 + 8N$ vertices of the global analytic-to-PL grid ($N^2 > 12L_{r,r}^2$). Different information guarantees make these curves incomparable as unrestricted minimax bounds.

deliberate M-aliasing counterexample, and 100 fixed-seed synthetic Pauli replications for $R = 6$, $M = 24$, and $\delta = 0.01$. Exact arithmetic certifies the tested rational predicates; finite regression tests are *not* a proof of the continuum inequalities or a formal verifier of the full extraction algorithm. The universal quartic Hopf distortion bound predates this manuscript [9, 10]; for arbitrary simplicial input, independent cochain-based Hopf computation is developed by Musin and Shamazov [12]. The contribution here is a self-contained conditional reconstruction protocol with explicit error budgets and an equally explicit impossibility boundary.

XXIII. CONCLUSION

For a piecewise-linear Bloch field with independently justified spatial regularity and readout calibration, its integer Hopf invariant can be certified through exact rational linking and a nonvanishing homotopy margin. The resulting framework combines finite-shot concentration, Fourier error envelopes, sharp anisotropic simplex robustness, structured two-loop recovery, and pointwise affine or directional-only calibration. Its additional minimax linear-inverse construction identifies the exact worst-case error constant for a fixed redundant matrix, without forcing inconsistent noisy measurements to satisfy artificially exact equations. The analytic premises and physical observation model are essential: finite unstructured measurements alone do not determine the topology

of an arbitrary continuum field. The present numerical examples are synthetic, not independent experimental evidence.

SUPPLEMENTARY MATHEMATICAL DERIVATIONS

Appendix A: Problem, scope, and prior work

The Hopf invariant of $n : \mathbb{S}^3 \to \mathbb{S}^2$ is the linking number of inverse images of two regular target values, after fixing orientations. This classical construction underlies topological textures and some two-band insulator models. Quantum-simulator and electrical-circuit implementations of Hopf-insulator physics already exist [29, 30]. Recent discussions emphasize sampling obstructions and discrete algorithms [31, 32]. Our aim is narrower than a new topological classification: to make the entire *integer extraction and finite vertex-error tolerance* checkable on explicitly given rational test fields.

The domain of our exact test is $\partial[-1,1]^4 \cong \mathbb{S}^3$, *not* a three-dimensional Brillouin torus $\mathbb{T}^3$. No result here solves the nonzero-weak-Chern-class classification problem on $\mathbb{T}^3$. A physical continuum field and its sampled interpolant are distinct objects unless an independently justified regularity or analytic-to-discrete homotopy argument links them. We expressly distinguish these claims below.

Appendix B: Exact input geometry and nonvanishing tests

Take the oriented boundary $X = \partial[-1,1]^4$ with a consistent Freudenthal tetrahedral subdivision of its eight cubic facets. The mesh parameter N counts subdivisions of each facet edge. At $N = 5$, the mesh has $M = 6^4 - 4^4 = 1040$ distinct rational vertices and $8 \cdot 6 \cdot 5^3 = 6000$ oriented tetrahedra. The checker verifies shared-face orientation, adjacency, closed vertex links, and connectedness; $X \cong \mathbb{S}^3$ follows from the convexity of the fourcube. These machine checks do not independently formalize the theorem that the cubical boundary is a sphere.

Let $b_i \in \mathbb{Q}^3$ denote vectors supplied at mesh vertices v_i. On each tetrahedron $\sigma = [v_0, v_1, v_2, v_3]$, use barycentric coordinates $\lambda_j \geq 0$, $\sum_j \lambda_j = 1$ and define

$$b_{\mathrm{PL}}(x) = \sum_{j=0}^{3} \lambda_j b_{v_j}, \qquad n_{\mathrm{PL}}(x) = \frac{b_{\mathrm{PL}}(x)}{\|b_{\mathrm{PL}}(x)\|_2}. \tag{B1}$$

The implementation rejects any tetrahedron for which the origin belongs to the convex hull of its four images. This is checked with exact rational arithmetic; absence of vertex zeros alone would be insufficient.

Appendix C: Rational inverse-image and linking algorithm

For a face $[i, j, k]$ of a tetrahedron and a rational unit target direction p (represented by an integer vector whose squared Euclidean norm is a positive perfect square), write

$$D = \det(b_i, b_j, b_k), \quad w_i = \det(p, b_j, b_k), \quad w_j = \det(b_i, p, b_k), \quad w_k = \det(b_i, b_j, p). \tag{C1}$$

When $D \neq 0$, the ray from the origin through p intersects the *interior* of the affine face image if and only if

$$\boxed{Dw_i > 0, \quad Dw_j > 0, \quad Dw_k > 0.} \tag{C2}$$

The barycentric coordinates are $\lambda_a = w_a/(w_i + w_j + w_k)$. The factor D in Eq. (C2) is necessary: merely requiring the three w to have the same sign can mistake $-p$ for p. Degenerate faces and edge hits are rejected or handled by explicit rank tests. Each regular tetrahedron contributes either zero or two oppositely oriented crossings of its boundary; the resulting inverse-image arcs must join into closed oriented cycles.

For higher-charge models, inverse-image curves can traverse all cubical facets. The checker uses the rational chart (in coordinates scaled so the fourcube has edge $[-N, N]$)

$$\Phi(w, \boldsymbol{u}) = \frac{\boldsymbol{u}}{\boldsymbol{u} \cdot \boldsymbol{u} + 2N(N - w)}, \qquad (w, \boldsymbol{u}) \in \partial[-N, N]^4 \setminus \{(N, 0, 0, 0)\}. \tag{C3}$$

It verifies that the pole does not lie on either inverse image. Images of affine inverse-image segments under (C3) are generally curved. They are replaced by rational chords only after bounding the second derivative along each arc and checking that the entire pair of link components remains disjoint under the resulting tube homotopies. Separation is checked using outward integer bounds, with exact constrained segment-distance calculations in ambiguous cases. A noncertifiable pair produces `ABSTAIN`, not an extrapolated linking number.

Theorem 24 (Exact rational linking under the checked hypotheses)**.** *Suppose the input triangulation is an oriented three-sphere; Eq.* (B1) *is everywhere nonzero; two rational target directions are regular; their preimages form two closed disjoint oriented cycles; and rational chord approximants preserve disjointness through a certified homotopy. For every rational generic planar projection, the exact crossing signs satisfy*

$$H(n_{\mathrm{PL}}) = \mathrm{Lk}(L_p, L_q) = \frac{1}{2} \sum_{c \in \mathrm{Cross}(L_p, L_q)} \mathrm{sgn}(c) \in \mathbb{Z}. \tag{C4}$$

The checker can compute the right side without floating-point rounding. It does not thereby determine the class of an unknown continuum field.

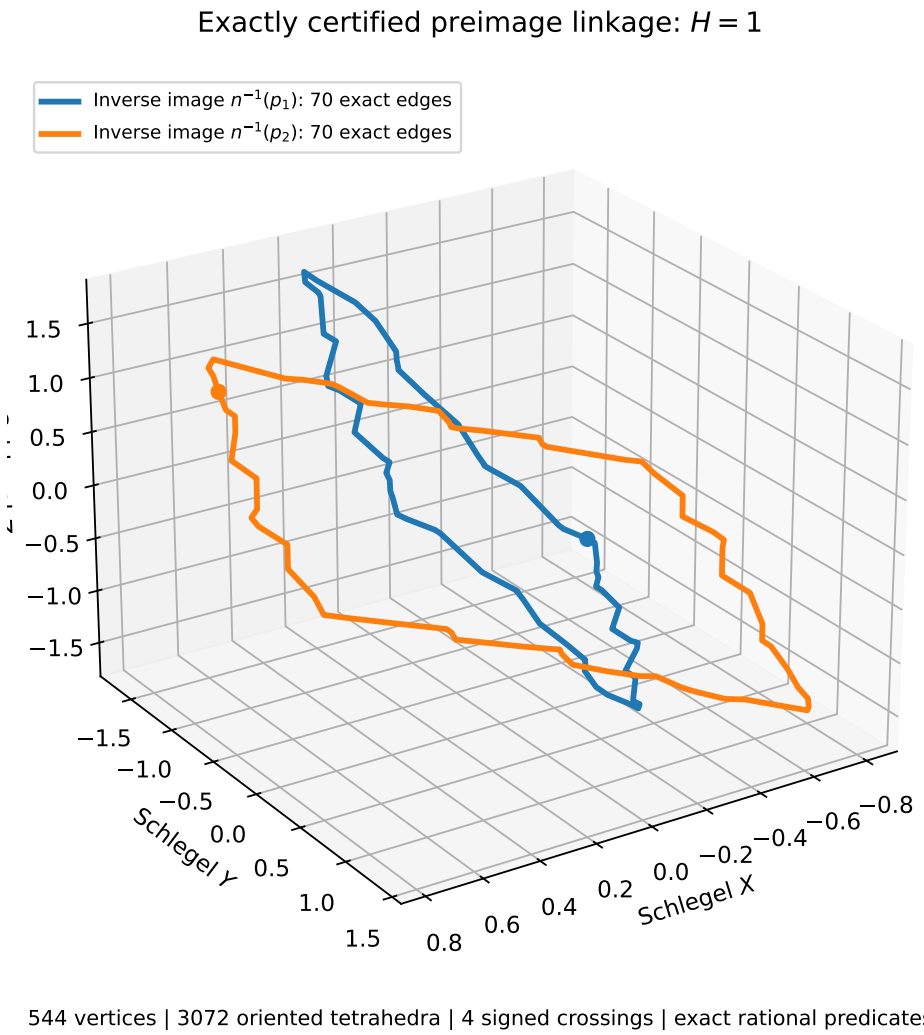


FIG. 24. Example inverse-image curves and exact crossing calculations from the inherited audited implementation. These are synthetic rational fields; the graphic is not an experimental measurement.

Proof. Nonvanishing gives a continuous sphere-valued map. Under the stated regularity assumptions its distinct inverse images are disjoint closed oriented one-manifolds. The Pontryagin–Hopf construction identifies their linking with H. Homotopy of the curves through disjoint pairs preserves linking. For a generic projection of two oriented closed polygonal cycles the linking number is half the signed intercomponent crossing sum. Each sign is determined by exact rational orientation and over/under predicates; only the hypotheses of the theorem, not the floating-point proximity of an integer, permit the conclusion. □

Appendix D: New result: finite rational vertex-error certificate

The preceding algorithm certifies the topology of the exact input data. The next theorem gives a practical way to decide whether arbitrary uncertainty in *every* vertex value could change that answer, without enumerating the uncertain maps. Let $\|v\|_1 = \sum_a |v_a|$ and $\|v\|_\infty = \max_a |v_a|$.

Theorem 25 (Cellwise separating-plane robustness)**.** *For every tetrahedron σ choose any nonzero rational vector $u_\sigma \in \mathbb{Q}^3$ and set*

$$\rho_\sigma = \frac{\min_{i \in \mathrm{Vert}(\sigma)} u_\sigma \cdot b_i}{\|u_\sigma\|_1}, \qquad \rho = \min_\sigma \rho_\sigma. \tag{D1}$$

Suppose all *these quantities are strictly positive. Let the observed vertex vectors be replaced by arbitrary $\widetilde{b}_i$ satisfying $\max_i \left\|\widetilde{b}_i - b_i\right\|_\infty \le \epsilon < \rho$. Then the two normalized piecewise-linear maps are homotopic and have the same exact Hopf integer:*

$$\boxed{\epsilon < \rho \quad \Longrightarrow \quad H(\widetilde{n}_{\mathrm{PL}}) = H(n_{\mathrm{PL}}).} \tag{D2}$$

No independence assumption, probability distribution, or reference analytic field is required for this deterministic statement.

Proof. For i a vertex of σ, Holder's inequality gives $u_\sigma \cdot \widetilde{b}_i \ge u_\sigma \cdot b_i - \|u_\sigma\|_1 \epsilon > 0$. The same strict inequality holds for the linear interpolation $(1-t)b_i + t\widetilde{b}_i$ at every $t \in [0,1]$. Barycentric combinations preserve the inequality on the entire tetrahedron. Thus the affine field never vanishes at any stage of the global vertexwise linear homotopy. Normalizing this field produces a continuous homotopy between the normalized PL maps. Hopf invariance follows. □

Corollary 8 (Polyhedral stability chamber)**.** *Fix the finite orientation-compatible mesh and the vectors u_σ in Theorem 25. In the space $\mathbb{R}^{3M}$ of all vector-valued vertex datasets define*

$$\mathcal{C}_u = \left\{ (c_i)_{i=1}^M : \; u_\sigma \cdot c_i > 0 \text{ for every } \sigma \text{ and } i \in \mathrm{Vert}(\sigma) \right\}. \tag{D3}$$

This is an open convex polyhedral cone. Any two data vectors in $\mathcal{C}_u$ define homotopic normalized PL maps, even if their componentwise distance is arbitrarily large. Thus a single finite rational certificate defines a whole explicit domain of constant topological phase in data space.

Proof. Each condition is a strict homogeneous linear inequality, so their intersection is an open convex cone. The straight-line path between two points of that cone remains within it, and the proof of Theorem 25 applies verbatim with a common positive normal on each simplex. □

Corollary 9 (A finite proof object)**.** *For a rational dataset it is sufficient to provide one triple of integers $u_\sigma \neq 0$ for each tetrahedron and a positive rational number ρ such that $u_\sigma \cdot b_i \geq \rho \|u_\sigma\|_1$ at every incident vertex. An independent checker needs only rational dot products, absolute values and comparisons, not a solver or a floating-point optimization routine.*

For each of four canonical models in Sec. F, we searched the *fixed small candidate set* $\{-1,0,1\}^3 \setminus \{0\}$ for a witness vector per cell, saved all 6,000 triples and rechecked all $4 \times 6000 \times 4 = 96{,}000$ strict vertex inequalities exactly. The resulting global values are exact rational sufficient margins, *not* assertions of the largest possible perturbations. Because the witness construction restricts directions to 26 choices, the radii could in principle be improved by richer exact optimization.

Appendix E: From uncertainty margins to Pauli-shot budgets

If b_i are the calibrated Bloch expectations and each of their three components is estimated from N_{shot} independent binary ± 1 outcomes, Hoeffding's inequality followed by a union bound gives, for M measured sites,

$$\mathbb{P}\left(\max_{i,a} |\widehat{b}_{i,a} - b_{i,a}| \geq \epsilon\right) \leq 6M \exp(-N_{\text{shot}}\epsilon^2/2). \tag{E1}$$

A synthetic family with validated deterministic vertex margin ρ is consequently guaranteed to retain the same *piecewise-linear* class with probability at least $1-\delta$ whenever

$$\boxed{N_{\text{shot}} > \frac{2}{\rho^2} \log \frac{6M}{\delta}.} \tag{E2}$$

For the integer shot counts in Table II, the verifier goes further: it checks the probability inequality with *exact rational arithmetic*, using the finite positive Taylor lower bound

$$\sum_{j=0}^{70} \frac{(N_{\text{shot}}\rho^2/2)^j}{j!} > \frac{6M}{\delta}. \tag{E3}$$

Since the positive Taylor partial sum lies below the full exponential, Eq. (E3) proves the required upper bound on failure probability. This finite rational check validates the proposed sufficient shot counts without relying on a floating-point logarithm. Their optimality is not claimed.

For real measurements, one should instead recompute a margin $\widehat{\rho}$ *from the observed rational means* and issue an integer certificate only when the independently justified simultaneous error envelope is smaller than $\widehat{\rho}$. Because Eq. (E1) controls all components simultaneously, this data-dependent abstention preserves frequentist coverage. It does not by itself identify the topology of

TABLE II. New finite rational perturbation witnesses for the fixed $N = 5$ mesh ($M = 1040, 6000$ tetrahedra). The last two columns are *sufficient theoretical* Pauli budgets at $\delta = 0.01$ for ideal independent measurements, not observed experimental resources.

$\lvert H\rvert$	Exact ρ	Decimal ρ	Shots/axis/site	Total binary outcomes
1	4/9	0.444444	136	424,320
2	32/95	0.336842	236	736,320
3	17173/83331	0.206082	629	1,962,480
4	54457/1742586	0.031251	27,328	85,263,360

the underlying continuous momentum-space field: a separately verified regularity, tail, or analytic-to-PL transfer condition is still necessary. There were no physical Pauli-shot experiments in this study.

Appendix F: Exact polynomial benchmark and reproducible results

On the cubical three-sphere identify (w, x, y, z) with $(z_1, z_2) = (w + ix, y + iz)$. Consider the polynomial family

$$F_k(z_1, z_2) = (z_1^k, z_2), \qquad n_k = \frac{\mathcal{H}(F_k)}{|z_1|^{2k} + |z_2|^2}, \quad \mathcal{H}(a, b) = \big(2\Re(a\bar{b}), 2\Im(a\bar{b}), |a|^2 - |b|^2\big). \tag{F1}$$

F_k has isolated complex local degree k at the origin, and its normalized boundary map has degree k. Thus the analytic Hopf class has magnitude k; the sign is determined by our explicit domain and target orientation conventions. We calculate the *discrete* class independently from rational field values at all mesh vertices and from Eq. (C4). Both classes coincide on the test family at $N = 5$ for $k = 1, 2, 3, 4$, yielding a homotopy by the classification $[\mathbb{S}^3, \mathbb{S}^2] \cong \mathbb{Z}$. For $k = 2$ we also have a constructive, local analytic-to-PL homotopy certificate by exact polynomial positivity (below). For $k = 3, 4$ equality of known classes is a special-property argument and should not be mistaken for a local sampling theorem valid for arbitrary unknown fields.

1. Independent local analytic transfer for charge two

For the $k = 2$ family, define on each simplex

$$P(\lambda) = \sum_i \lambda_i n_2(v_i), \qquad G_\sigma(\lambda) = D_\sigma\Big\langle P(\lambda), \mathcal{H}\Big(F_2\big(\sum_i \lambda_i v_i\big)\Big)\Big\rangle, \tag{F2}$$

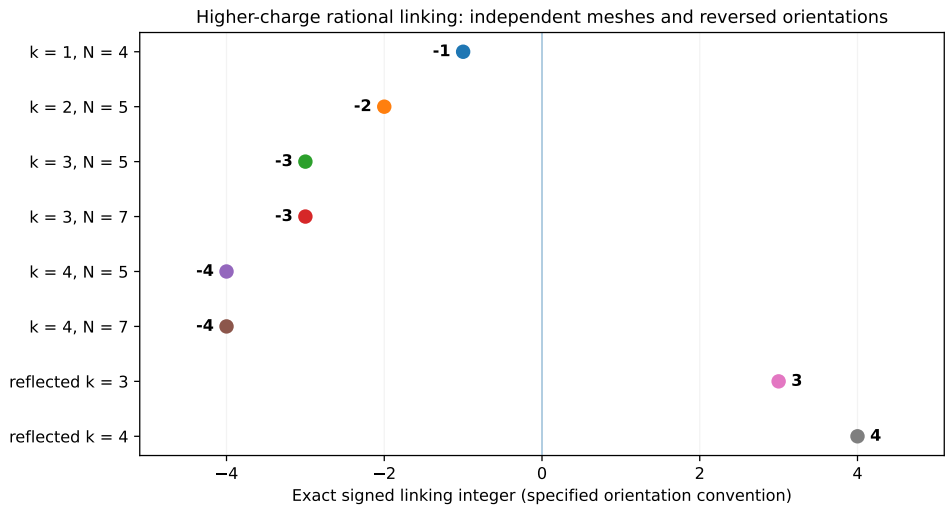


FIG. 25. Exact integer checks for the higher-charge synthetic models of the predecessor, including orientation reversals. The reproduced integer concerns the exact PL map.

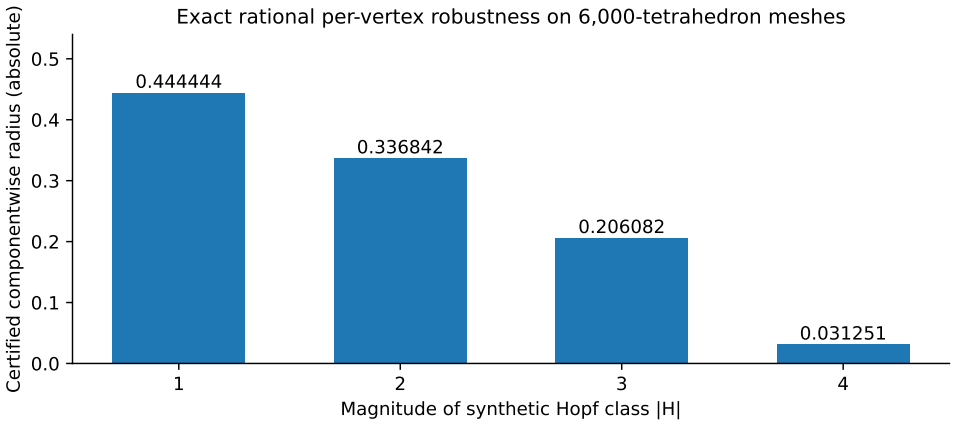


FIG. 26. New exact rational sufficient perturbation radii for the four canonical synthetic classes. The marked tolerances are provable lower bounds on robustness, not optimized distances to the set of topology-changing maps.

with integer clearing denominator $D_\sigma > 0$. After homogenizing this polynomial to degree five in barycentric coordinates, an exact positive-coefficient check succeeds on *every* one of the 6000 tetrahedra. Positivity of the homogeneous Bernstein coefficients on the standard simplex implies $G_\sigma > 0$, so the normalized analytic and PL fields are pointwise non-antipodal, and the straight-line homotopy is nonvanishing. The certificate is reproduced by `bernstein_transfer_v9.py` and `verify_analytic_transfer_v9.py`, with exact integer arithmetic. This separate transfer is stronger than merely observing equal final integers for two maps.

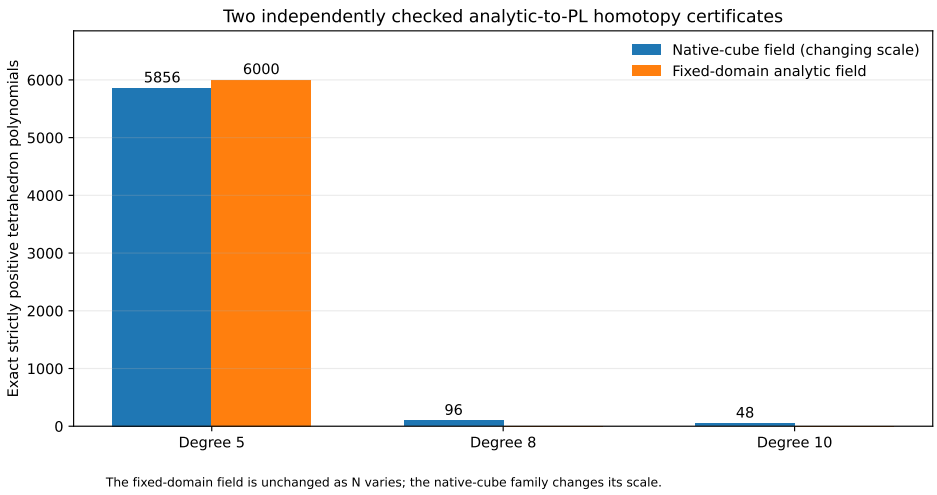


FIG. 27. Exact polynomial positivity checks for the analytic-to-PL transfer of the degree-two field, distinguishing two normalization conventions. The fixed-domain test requires no degree elevation beyond five.

Appendix G: Limitations and publication-level claims

Three levels of evidence must not be conflated. *First*, exact rational preimage linking assigns a true integer to the specified rational PL map, under the checker's listed hypotheses. *Second*, the new rational separator certificate proves perturbation robustness of that PL integer, with an explicit finite-shot translation when the Pauli measurement model is calibrated. *Third*, the integer belongs to an unknown continuous physical field only after proving a field-to-PL homotopy by external regularity, spectral locality, or a comparable calibrated constraint. Our synthetic degree-family classification does not supply such a condition for arbitrary experiments.

The exact checker was not ported to Lean, Coq, or another proof assistant; its arithmetic primitives and mesh construction have not been independently peer reviewed. Existing results on Hopf invariants, linked preimages, and simplicial computations are extensive [28, 32]. The separation-witness method is an elementary convexity argument; its presentation here should be assessed as a concrete verified proof object and resource benchmark, not declared a novel universal theorem or a major open-problem solution. Tests at $N = 5$ and selected $N = 7$ meshes show useful behavior but are not proofs of optimal runtime, arbitrary-charge termination, or universal sampling convergence. This work does not analyze laboratory data or determine any cosmological topology.

A consequential next step is to obtain a three-dimensional laboratory dataset with axis-resolved repetition counts, systematic calibration bounds and independently validated spatial regularity;

then run the complete certificate without knowing the topological label. For mathematical verification, the highest-value follow-up is to machine-check the small rational separator and orientation kernels against a formal specification. None of the named open problems in the external catalogues is claimed resolved by the present experiment.

Appendix H: Central problem and the publication boundary

Consider a finite, oriented simplicial three-manifold X with vertex set $\mathcal{V}$ and tetrahedral set $\mathcal{T}$. Let $b : \mathcal{V} \to \mathbb{R}^3$ be rational. For x in a tetrahedron $\sigma = [v_0, v_1, v_2, v_3]$, define the barycentric field

$$B_b(x) = \sum_{j=0}^{3} \lambda_j(x) b_{v_j}, \qquad n_b(x) = B_b(x) / \left\| B_b(x) \right\|_2 . \tag{H1}$$

When B_b is nowhere zero, $n_b : X \to \mathbb{S}^2$ is continuous. On $X \cong \mathbb{S}^3$, it has an integer Hopf invariant. Our previous working rational linker computes this number for admissible inputs, but does not by itself quantify the *largest tolerable adversarial vertex error*. This paper isolates and solves that finite-dimensional robustness problem. Inverse-image linking is classical, and independent cochain algorithms already exist [33, 34]; convex duality is classical as well. We therefore do not claim to have invented the Hopf invariant or linear programming.

We focus on one mathematical question: an exactly auditable stability radius for a normalized simplicial field, rather than a broad survey of topological phases.

Appendix I: Sharp pointwise robustness theorem

Write $K_\sigma = \operatorname{conv}\{b_v : v \in \sigma\}$, and define

$$\rho(b) := \min_{\sigma \in \mathcal{T}} \min_{z \in K_\sigma} \left\| z \right\|_\infty . \tag{I1}$$

The perturbation class is $\left\| \tilde{b}_v - b_v \right\|_\infty \le \varepsilon$ for *every* vertex; it permits arbitrary spatial correlations between errors.

Theorem 26 (Sharp robust nonvanishing and homotopy radius)**.** *Assume $\rho(b) > 0$. If $0 \le \varepsilon < \rho(b)$, then for every such vertex perturbation $\tilde{b}$, both normalized fields $n_b, n_{\tilde{b}}$ are defined on all of X and are explicitly homotopic through*

$$n_t(x) = \frac{(1-t)B_b(x) + tB_{\tilde{b}}(x)}{\left\| (1-t)B_b(x) + tB_{\tilde{b}}(x) \right\|_2}, \quad 0 \le t \le 1. \tag{I2}$$

At $\varepsilon = \rho(b)$ there exists a perturbation in this class for which the PL field vanishes at some point. Thus $\rho(b)$ is the exact universal radius for nonvanishing under arbitrary bounded vertex errors and for this uniform straight-line homotopy certificate. *It need not be the infimum of errors among only those* nonvanishing *perturbations that change the Hopf class.*

Proof. Choose $x \in \sigma$ with barycentric weights λ_j. By convexity and the triangle inequality,

$$\left\|B_{\tilde{b}}(x) - B_b(x)\right\|_\infty \le \sum_j \lambda_j \left\|\tilde{b}_{v_j} - b_{v_j}\right\|_\infty \le \varepsilon. \tag{I3}$$

Every $B_b(x)$ has infinity norm at least $\rho(b)$, and the entire unnormalized segment in Eq. (I2) therefore has norm at least $\rho(b)-\varepsilon > 0$. For sharpness choose a simplex σ_* and weights λ_j^* attaining the compact minimum, and set $z_* = \sum_j \lambda_j^* b_{v_j}$. Shift all four of those vertices by $-z_*$ (other vertices unchanged). Each coordinate shift has absolute value at most $\rho(b)$, and the perturbed PL field vanishes at the barycentric point with coefficients λ_j^*. □

Theorem 27 (Exact finite-dimensional convex dual)**.** *For every finite nonempty rational vertex set $\{b_v : v \in \sigma\}$,*

$$\boxed{\min_{\lambda \ge 0,\ \sum \lambda = 1} \left\|\sum_{v\in\sigma} \lambda_v b_v\right\|_\infty = \max_{\|u\|_1 \le 1} \min_{v\in\sigma} u \cdot b_v.} \tag{I4}$$

The optimum admits rational primal and dual witnesses. To prove $\rho(b) \ge r \in \mathbb{Q}_{>0}$ it is sufficient to provide, for each tetrahedron, a nonzero integer vector $w_\sigma \in \mathbb{Z}^3$ such that

$$\boxed{w_\sigma \cdot b_v \ge r \left\|w_\sigma\right\|_1 \quad \text{for all } v \in \sigma.} \tag{I5}$$

Conversely a single rational convex combination with infinity norm r proves $\rho(b) \le r$. Coincident lower and upper witnesses prove equality exactly.

Proof. The dual representation $\|z\|_\infty = \max_{\|u\|_1\le 1} u \cdot z$ holds in $\mathbb{R}^3$. Both optimization sets are convex compact polyhedra and the objective is bilinear, so the elementary finite-dimensional minimax theorem interchanges minimum and maximum. Minimizing a linear form over a convex hull equals its minimum over vertices, proving Eq. (I4). Rational input produces rational extreme-point optimal witnesses by linear programming duality. Multiplication by a common denominator yields integer w_σ. Division by $\|w_\sigma\|_1$ gives an admissible dual vector proving Eq. (I5). □

Appendix J: Four exactly attained radii on the cubical three-sphere

The benchmark $X = \partial[-1,1]^4$ is divided into $N = 5$ subdivisions per cubical edge and triangulated by Freudenthal tetrahedra. There are precisely $8N^3 + 8N = 1040$ vertices and $48N^3 = 6000$

tetrahedra. Writing $z_1 = w + ix$ and $z_2 = y + iz$ on the boundary, define

$$F_k(z_1, z_2) = (z_1^k, z_2), \tag{J1}$$

$$b_k = \frac{\left(|z_1|^{2k} - |z_2|^2,\ 2\Re(z_1^k\overline{z_2}),\ 2\Im(z_1^k\overline{z_2})\right)}{|z_1|^{2k} + |z_2|^2}. \tag{J2}$$

A consistent sign convention for the last coordinate is fixed in the included integer verifier. Each vertex is a triple of exact rationals. The reference Hopf charges of these test maps are $-k$ in the orientation convention of the inherited linker; the new robustness checker does not use that information or an expected Hopf charge. Analytic degree and the earlier exact linker are independent mathematical inputs to identifying this family as Hopf benchmarks, not inputs to Eq. (I1).

TABLE III. Exact radii ρ_k, attaining cell indices, and independently certified 99% sufficient fixed-sample budgets. Each cell index is zero-based in the stated Freudenthal enumeration. All numbers were verified using Python `Fraction`; floating-point LP solutions were used only to search for proof witnesses.

$\lvert H\rvert$	Exact ρ_k	Cell	m	Total outcomes
1	17/35	4006	114	355,680
2	5521/12731	396	142	443,040
3	69351883/206498067	4596	237	739,440
4	14330719/71515656	3466	665	2,074,800

For each k, the archived witness file contains precisely 6000 signed integer triples w_σ. The checker reconstructs *all* 1040 exact rational vertex fields from the polynomial formula, then checks Eq. (I5) at four vertices per cell, a total of $4{\cdot}6000{\cdot}4 = 96{,}000$ inequalities. Separate primal witnesses establish equality for a selected minimizing simplex in each charge family; the certified optimal values appear in Table III. Every asserted equality is cross-multiplied into an integer identity. In particular, for $k = 4$, the attaining simplex has vertices $(511, 509, 547, 795)$ and rational convex image

$$z_* = (-r, r, -r), \qquad r = \frac{14330719}{71515656}. \tag{J3}$$

The associated exact ℓ_1-unit dual vector (numerators and denominators given in the machine data) yields the same value r on the three active vertices. Therefore the global minimum is *equal*, not merely close, to the rational r.

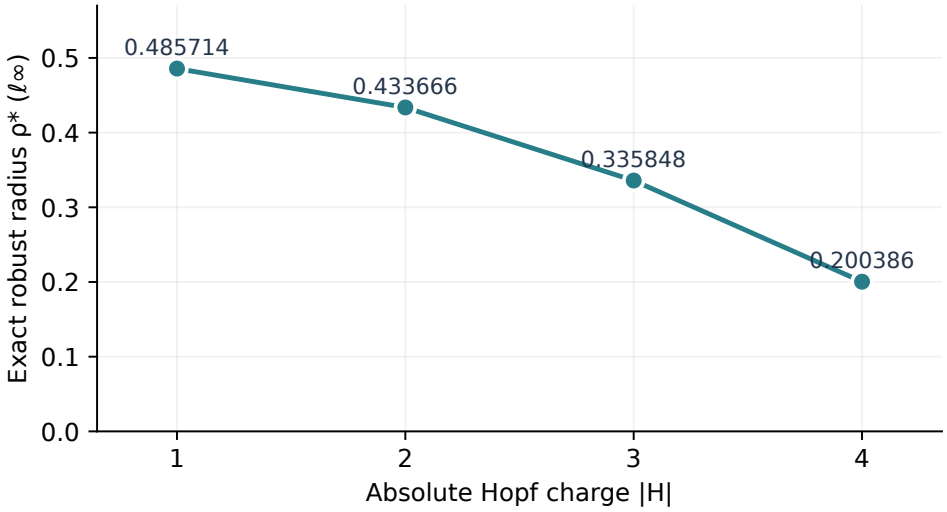


FIG. 28. The four exact rational uniform adversarial radii in the ℓ_∞ norm. The line joining the discrete charges is purely a visual guide, not a continuous physical law.

Appendix K: A reference-free certificate for measured data

A useful consequence of Theorems 26–27 does *not* require pre-selecting b_k. Suppose the measured vertex means $\hat{b}_i$ are rational (as they are for exact recorded integer shot counts). Compute $\rho(\hat{b})$ and the Hopf invariant $H(n_{\hat{b}})$ using a separate sound reference-free integer extractor. If calibrated confidence intervals guarantee $\left\|\hat{b}_i - b_i\right\|_\infty < \varepsilon$ simultaneously for all vertices, and

$$\varepsilon < \rho(\hat{b}), \tag{K1}$$

then the *unknown true vertex-interpolated field* n_b has exactly the measured PL integer, without a reference Hamiltonian or a pre-specified Hopf charge. Strict acceptance or abstention is essential at equality. This criterion is a deterministic consequence of the observed input and its trusted error intervals; the exact-radius computation itself cannot certify the physical calibration.

The extension from the *true vertex-interpolated field* to a genuinely continuous physical field $n(x)$ is an additional and logically independent task. For example, if $\sup_x \|n(x) - B_b(x)\|_\infty \le \tau$, then $\varepsilon + \tau < \rho(\hat{b})$ is a sufficient physical transfer condition. A generic sufficient spatial estimate is $\tau \le Lh$ for verified Lipschitz ceiling L and simplex diameter h. This estimate is not supplied by a spectral gap alone and must not be inferred from the integer result. Model-specific polynomial positivity certificates may give substantially sharper transfer bounds.

Appendix L: Conditional finite-shot theorem and exact resource audit

Corollary 10 (Distribution-free bounded Pauli-shot certificate)**.** *Let the means of the calibrated, independent* ± 1 *Pauli outcomes at* V *vertices be the components of a fixed field* b*. At each vertex and axis perform* m *independent shots. Under no assumptions on the dependence of* different vertices *beyond valid per-axis fixed-shot Hoeffding estimates, if*

$$\boxed{m > \frac{2}{\rho(b)^2} \log \frac{6V}{\delta}, \qquad 0 < \delta < 1,} \tag{L1}$$

then, with probability at least $1 - \delta$*, the empirical PL field is nonzero and homotopic to* n_b*. Conditional on an independently sound integer extractor returning a valid certificate, its integer is therefore correct for the true PL field.*

Proof. For one component $\Pr(|\hat{b}_{ia} - b_{ia}| \geq t) \leq 2\exp(-mt^2/2)$. A union bound over $3V$ components gives $\Pr(\max_{i,a} |\hat{b}_{ia} - b_{ia}| \geq \rho(b)) \leq 6V \exp[-m\rho(b)^2/2] < \delta$. On the complementary event Theorem 26 applies. □

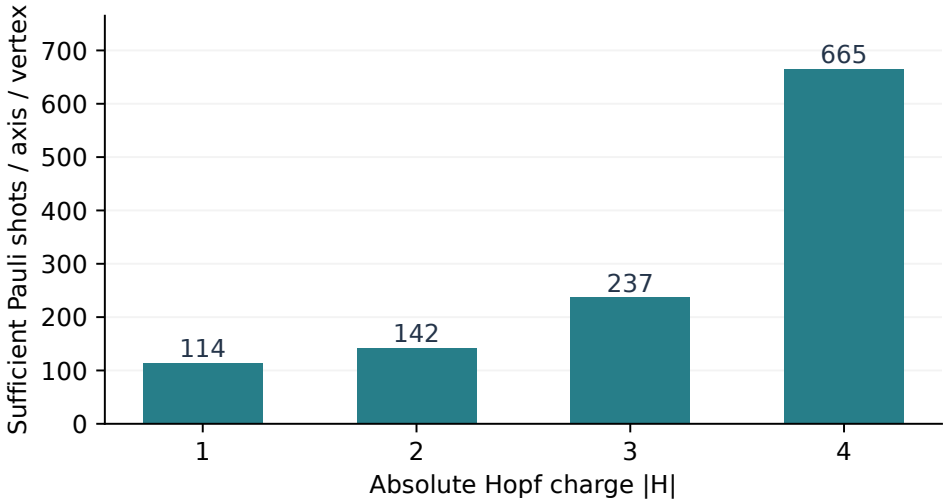


FIG. 29. Proved sufficient shot budgets for the *synthetic benchmark* at $\delta = 0.01$. No actual laboratory Pauli measurements are used. There are $3 \times 1040 \times m$ binary outcomes per experiment.

For every integer budget in Table III, the verifier proves the exact inequality $\exp(m\rho_k^2/2) > 600V$ by lower-bounding the exponential with a *finite positive rational Taylor sum.* This removes even floating-point evaluation of logarithms or exponentials from the acceptance predicate. The resulting budgets are sufficient, not minimax-optimal, not expected early-stopping times, and not evidence of hardware speedup.

Appendix M: Algorithm, proof objects, and falsification

The archived data contain four JSON files `EXACT_OPTIMUM_DUALS_Hk_N5.json`, one per charge family. Each stores 6000 signed integer separating normals, an attaining simplex, its four exact convex coefficients, and an exact dual normal at that simplex. The pure-standard-library program `verify_optimal_v12.py` reconstructs the mesh and fields independently, verifies all 96,000 inequalities and all eight matching primal/dual claims, checks fixed-shot probabilities with exact arithmetic, and performs a deliberately corrupted-witness negative control. Two earlier externally serialized rational vertex fields for $k = 3, 4$ were cross-checked against regenerated vertex values by a separate checker. The original numerical LP search is disclosed but is neither imported nor trusted by the proof checker.

The scope of the theorem is an elementary but *sharp* finite-dimensional robustness statement; the scientific claim beyond standard convex duality is the deployment as an independently auditable, fail-closed topology-tomography certificate and the four rational instances. This does not solve the general Hohenberg–Kohn inverse stability problem, prove completeness of a thermodynamic spectral-gap SDP hierarchy, or establish physically realized Hopf phases. Prior art on exact Hopf extraction [34] and classical finite-dimensional convex geometry must be acknowledged. The software is checked by independent arithmetic tests but has *not* been fully formalized in Lean or peer reviewed; assertions that it is unprecedented would exceed the evidence.

Appendix N: Research implications and next falsifiable steps

The same primal–dual architecture applies to finite-simplicial fields on arbitrary oriented compact three-manifolds and to any target modeled by normalized nonzero vectors, although the invariant being certified may no longer be an absolute Hopf integer. It also extends to mixed *anisotropic* independent error bounds, for which weighted ℓ_∞ distances and their dual weighted ℓ_1 support constraints replace Eq. (I4). Those statements do not resolve nonlinear continuum inverse problems, but suggest a practical three-part validation agenda: an independently implemented cochain Hopf extractor; machine-checked finite convex-dual certificates (Lean or an equally trusted checker); and application to real, three-dimensional qubit tomography with external calibration and derivative bounds. The maintained physics problem list explicitly identifies local spectral-gap SDP completeness (#058) and 3D interacting density–potential inverse stability (#066) as distinct open problems [36]. Their formulations may motivate techniques but are not consequences of the

present result.

Appendix O: Conclusion

A topological integer can be stable under arbitrary bounded perturbations of finitely sampled rational data even when the error pattern is fully adversarial. The sharp threshold *for nonvanishing and straight-line PL homotopy* is an exactly solvable convex optimization quantity. Four nontrivial cubical Hopf benchmarks admit finite primal–dual rational proof objects attaining that optimum. This enables a scientifically useful distinction between having a reproducible *integer from synthetic vertices* and possessing a genuine *experimentally validated topological measurement*. The latter remains a separate frontier.

Appendix P: Scope: what the data can and cannot establish

A sample-derived topological integer is not automatically the integer of the unknown continuous field. A sound chain has three separate links: a rational triangulation and nonzero PL map; an exact integer extractor for that map; and a *separate* measurement or regularity certificate relating it to the physical field. Inverse-image linking goes back to classical topology [38]; independent cochain extraction has also been studied [40]. Finite spatial sampling without regularity permits compactly supported hidden Hopf charge [41]. These distinctions determine our claim boundaries.

Let X be a finite oriented simplicial three-manifold with vertices $\mathcal{V}$ and tetrahedra $\mathcal{T}$. Every rational vertex field $b_i \in \mathbb{Q}^3$ defines, in tetrahedron σ, the affine map

$$B_b(x) = \sum_{i\in\sigma} \lambda_i(x) b_i, \qquad n_b(x) = \frac{B_b(x)}{\|B_b(x)\|_2} \tag{P1}$$

when B_b never vanishes. For $X \cong \mathbb{S}^3$, the map $n_b : X \to \mathbb{S}^2$ has an integer Hopf invariant. A physically unknown field must be independently shown homotopic to n_b before this integer has experimental meaning.

Appendix Q: Exact integer recovery from rational simplices

The inherited integer engine triangulates $X = \partial[-1,1]^4$, validates the oriented boundary complex, and takes inverse images of two rational regular target points. On a triangle with vertex images b_0, b_1, b_2, the exact Cramer determinants test whether a *positive*, rather than merely a signed, ray meets the image simplex. Its edge/face crossings are stitched into closed, consistently

oriented inverse-image curves. A rational chart from $\mathbb{S}^3$ with one excluded point to $\mathbb{R}^3$ is used, and exact crossing-sign arithmetic computes their linking number. Replacing curved chart arcs by rational chords requires a separate certified separation check, not only numerical agreement of projected crossings.

The earlier implementation produces $H \in \{-4,-3,-2,-1,0,1,2,3,4\}$ in designated rational synthetic test families and explicitly abstains on degenerate inputs. Its successful unit and higher-charge regressions are preserved as executable dependencies in the accompanying archive. Repeating three projections checks consistency but is *not* a formal proof of software correctness. Our paper's new exact geometric radii, by contrast, depend only on finite convex geometry and can be verified independently of the linker. Neither task assumes access to the correct integer during prediction.

Appendix R: Sharp isotropic robustness, with attained proofs

Let $K_\sigma = \operatorname{conv}\{b_i : i \in \sigma\}$ and define

$$\rho(b) = \min_{\sigma \in \mathcal{T}} \min_{z \in K_\sigma} \|z\|_\infty . \tag{R1}$$

This quantity is associated with the normalized PL map, not an arbitrary unresolved continuum texture.

Theorem 28 (Sharp nonvanishing and straight-line certificate)**.** *If $\rho(b) > 0$, every field $\tilde{b}$ satisfying $\left\|\tilde{b}_i - b_i\right\|_\infty < \rho(b)$ for all i has a nowhere-zero PL interpolant and is homotopic to b by the pointwise normalized straight line. At equality there exists an allowed perturbation causing a zero. Hence $\rho(b)$ is the exact maximal* universal nonvanishing/linear-homotopy radius*; it is not proved to be the minimal perturbation that changes the Hopf integer while leaving both endpoint fields nonsingular.*

Proof. For $x \in \sigma$, convexity gives $\left\|B_{\tilde{b}}(x) - B_b(x)\right\|_\infty \le \sum_i \lambda_i \left\|\tilde{b}_i - b_i\right\|_\infty < \rho(b)$. The entire straight-line segment stays at least $\rho(b) - \varepsilon > 0$ away from zero in the infinity norm. For sharpness, let $z_* \in K_{\sigma_*}$ attain the minimum and shift all four image vertices of σ_* by $-z_*$. The perturbed affine image then vanishes at the attaining barycentric point, with coordinatewise shifts bounded by $\rho(b)$. □

Theorem 29 (Exact primal–dual certificate)**.** *For each simplex,*

$$\min_{\lambda_i \ge 0,\, \sum \lambda_i = 1} \left\|\sum \lambda_i b_i\right\|_\infty = \max_{\|u\|_1 \le 1} \min_i u \cdot b_i. \tag{R2}$$

For rational inputs, rational primal and dual optimal witnesses exist. For any candidate $r > 0$, one integer normal $w_\sigma \neq 0$ satisfying $w_\sigma \cdot b_i \geq r \|w_\sigma\|_1$ for all four vertices of every simplex proves $\rho(b) \geq r$. A single attaining rational barycentric combination with infinity norm r proves equality.

Proof. Use the dual norm identity $\|z\|_\infty = \max_{\|u\|_1 \leq 1} u \cdot z$. Both optimization sets are compact convex polytopes; finite-dimensional minimax interchanges the extrema, and the minimum of a linear form over a convex hull occurs at a vertex. Rational linear programs possess rational extreme-point witnesses. Clearing denominators makes each nonzero normal integral. □

The benchmark triangulates $\partial[-1,1]^4$ with five subdivisions per cubical edge, yielding $V = 8N^3 + 8N = 1040$ vertices and $T = 48N^3 = 6000$ oriented tetrahedra. For $z_1 = w + ix$ and $z_2 = y + iz$, the rational field of integer charge magnitude k is the Hopf projection of (z_1^k, z_2). Exact arithmetic independently regenerates every sample. The four sharp isotropic radii, already present in the earlier compact v12, are

$$\rho_k \in \left\{ \frac{17}{35}, \frac{5521}{12731}, \frac{69351883}{206498067}, \frac{14330719}{71515656} \right\}_{k=1}^{4}. \tag{R3}$$

These are rational *benchmarks*; their non-monotone variation is not a physical scaling law. The archived checker verifies $4 \cdot 6000 \cdot 4 = 96,000$ isotropic dual inequalities, an attaining primal mixture for each charge, and a deliberately corrupted-witness rejection case.

Appendix S: Anisotropic robustness: a new exact benchmark

Often different components admit different calibrated errors. Let $D = \mathrm{diag}(d_x, d_y, d_z)$ with $d_a > 0$, and permit arbitrary correlated perturbations $|\tilde{b}_{ia} - b_{ia}| \leq t d_a$. Define

$$\rho_D(b) := \min_{\sigma \in \mathcal{T}} \min_{z \in K_\sigma} \left\| D^{-1} z \right\|_\infty. \tag{S1}$$

Theorem 30 (Sharp weighted adversarial certificate)**.** *The largest uniform scale of this anisotropic error box guaranteeing nonvanishing and the straight-line PL homotopy is $\rho_D(b)$, with a singular perturbation possible at equality. Its exact dual is*

$$\boxed{\rho_D(b) = \min_{\sigma} \max_{\sum_a d_a |u_a| \leq 1} \min_{i \in \sigma} u \cdot b_i.} \tag{S2}$$

For rational b, D, the optimal radius and an attaining primal and dual admit rational certificates. A simplex-wise integer normal w_σ proves $\rho_D(b) \geq r$ whenever $w_\sigma \cdot b_i \geq r \sum_a d_a |w_{\sigma a}|$ for every incident vertex.

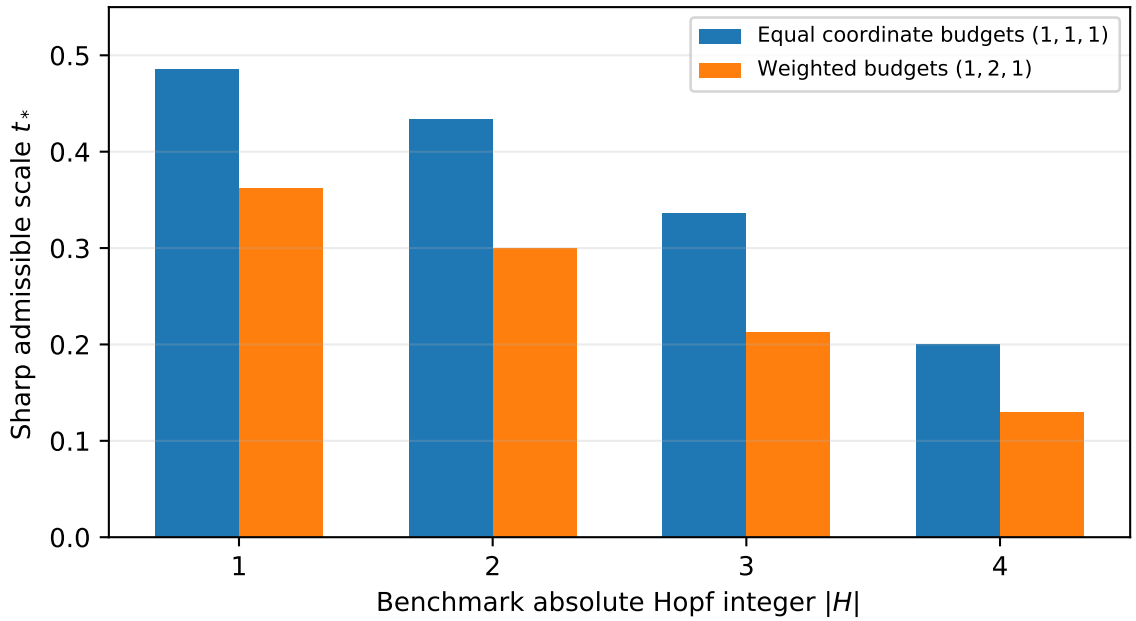


FIG. 30. Exactly certified perturbation scales for the same four rational PL benchmark fields, under equal axis scales and $(1, 2, 1)$ axis scales. The plotted bars are not interpolated physical data.

Proof. Apply Theorems 28 and 29 to $b'_i = D^{-1}b_i$. The perturbation condition and the constraint on the new dual variable $u = D^{-1}u'$ become

$$\left\| D^{-1}(\tilde{b}_i - b_i) \right\|_\infty \le t, \qquad \|Du\|_1 \le 1.$$

Exact primal attainment follows by moving the attaining simplex's vertices by $-z_*$, within the scaled box. All rationality statements follow from the finite rational LP. □

We choose the concrete *anisotropic audit pattern* $(d_x, d_y, d_z) = (1, 2, 1)$, without pretending that this ratio characterizes any particular device. In addition to the four known isotropic cases, independently reconstructed 24,000 integer normals and four exact attaining convex combinations prove the following *new* sharp values:

$$\begin{aligned} \rho_D(b_1) &= \frac{17}{47}, & \rho_D(b_2) &= \frac{15067}{50343}, \\ \rho_D(b_3) &= \frac{361487}{1701213}, & \rho_D(b_4) &= \frac{14330719}{110685656}. \end{aligned} \tag{S3}$$

Every one of the 96,000 weighted dual inequalities and the eight primal–dual equality checks is replayed by a pure standard-library rational checker; floating-point linear programming was used solely for finding candidates, not for acceptance.

Appendix T: Adaptive confidence, selected certificates, and an audited Pauli example

For unknown rational means b_i and observed empirical rational vectors $\widehat{b}_i$, one may choose a different integer normal w_σ *after* seeing the data. Suppose simultaneous coordinatewise confidence radii ε_{ia} have been established. The finite, checkable acceptance predicate is

$$\boxed{w_\sigma \cdot \widehat{b}_i > \sum_a |w_{\sigma a}|\varepsilon_{ia} \quad \text{for every } \sigma \text{ and } i \in \sigma.} \tag{T1}$$

Theorem 31 (Data-dependent anisotropic, optional-stopping-safe certificate)**.** *Consider V predetermined measurement sites. At each site and each of three Pauli axes, let the calibrated outcomes $X_{ia,t} \in \{-1, +1\}$ be independent across the time index t with fixed mean b_{ia}; dependence between different sites or axes is permitted. Define the confidence sequence*

$$\varepsilon(t) = \sqrt{\frac{2}{t}\log\left(\frac{12Vt^2}{\delta}\right)}, \quad 0 < \delta < 1. \tag{T2}$$

With probability at least $1 - \delta$, $|\widehat{b}_{ia}(t) - b_{ia}| \le \varepsilon(t)$ for all *sites, axes and integer times $t \ge 1$ simultaneously. Consequently any observation-dependent time choices t_{ia} and any rational separating normals computed from the same data may be inserted in Eq. (T1), using error ceilings not smaller than $\varepsilon(t_{ia})$. Acceptance certifies that the unknown true vertex-interpolated PL field and the observed PL field are both nonvanishing and linearly homotopic. On $X \cong S^3$ their exact integer Hopf invariants agree.*

Proof. For each fixed (i, a, t), Hoeffding gives $\Pr(|\widehat{b}_{ia}(t) - b_{ia}| > e) \le 2\exp(-te^2/2)$. Choosing Eq. (T2) bounds this probability by $\delta/(6Vt^2)$. Sum over all $3V$ axes/sites and all positive integer times. The elementary bound $\sum_{t\ge1} t^{-2} < 2$ makes the resulting total strictly smaller than δ. On this single simultaneous event, the data-dependent normals and time selections do not change coverage. Inequality (T1) then yields $w_\sigma \cdot b_i > 0$ for all vertices of every simplex, both at the observed and true endpoints and throughout their interpolation. This excludes zero in each simplex by convexity and proves homotopy. □

Corollary 11 (Finite rational confidence predicate)**.** *For rational $e > 0$, positive integer t, and rational δ, the inequality $e > \varepsilon(t)$ in Eq. (T2) is implied by the* finite rational test

$$\sum_{j=0}^{J} \frac{(te^2/2)^j}{j!} > \frac{12Vt^2}{\delta}, \qquad J \in \mathbb{N}. \tag{T3}$$

Thus the accepting branch of an optional-stopping-safe confidence certificate need not trust a floating-point logarithm, exponential, or inverse optimizer.

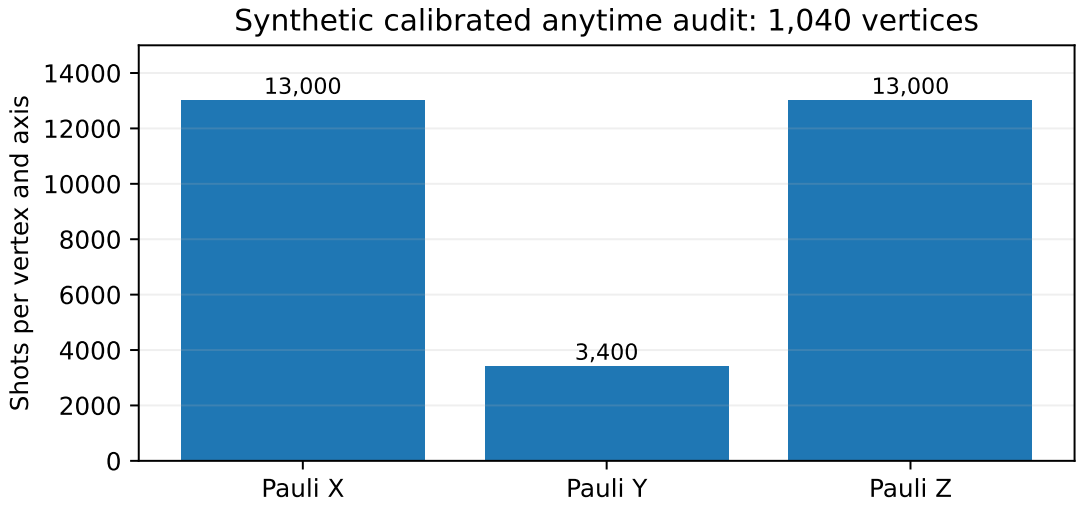


FIG. 31. Synthetic Pauli-count audit with different axis-specific shot allocations, total 30,576,000 simulated binary outcomes. This is *not* a demonstrated improvement over an optimized experimental schedule.

Proof. Every term in the exponential series is nonnegative for $te^2/2 > 0$, so its finite partial sum is a strict lower bound on $\exp(te^2/2)$. Equation (T3) implies $te^2/2 > \log(12Vt^2/\delta)$, equivalently $e > \varepsilon(t)$. □

For an explicitly labeled *synthetic* record with $V = 1040$, $\delta = 0.01$, and an inherited integer-normal family for the charge-four field, we simulated only binomial outcome *counts*, not millions of explicit per-shot records. We chose $t_{ix} = t_{iz} = 13,000$, $t_{iy} = 3,400$, $e_x = e_z = 2/25$, and $e_y = 4/25$. The standard-library checker reads the independent count JSON and verifies the strict predicate at all 24,000 simplex-vertex incidences, the rational Taylor bound for each axis, and a deliberately corrupted-normal rejection control. It uses no knowledge of the generating field when checking the observed data, while the associated field model and calibration premises remain part of this synthetic test.

Appendix U: Structured bidegree maps and two-loop recovery

The accumulated v12 archive contained an independent exact-family theorem not present in the compact four-page v12. We preserve it here because it illustrates the remarkable difference between unrestricted topological inference and model-conditioned parameter identification.

On the oriented cubical sphere $X = \partial[-1,1]^4 \subset \mathbb{C}^2$, put

$$F_{p,q}(z_1, z_2) = (z_1^p, z_2^q),$$
$$n_{p,q} = S^{-1} \begin{pmatrix} 2\,\mathrm{Re}(F_1\overline{F_2}) \\ 2\,\mathrm{Im}(F_1\overline{F_2}) \\ |F_1|^2 - |F_2|^2 \end{pmatrix}, \quad S = |F_1|^2 + |F_2|^2. \tag{U1}$$

for positive integers p, q. The denominator never vanishes on X. In the orientation used by the supplied cubical algorithm, $H(n_{p,q}) = -pq$.

Proposition 8 (Bidegree differential and uniform transfer)**.** *The ambient differential norm and a global valid facewise ceiling are*

$$\|Dn_{p,q}\|_{\mathrm{amb}} = \frac{2\sqrt{p^2a^{2p-2}b^{2q} + q^2a^{2p}b^{2q-2}}}{a^{2p} + b^{2q}}, \tag{U2}$$
$$L_{p,q}^2 = \max\{p^2 + (q+1)^2, (p+1)^2 + q^2\},$$

where $a = |z_1|$, $b = |z_2|$. *One has* $\sqrt{p^2+q^2} \le \sup_X \|Dn_{p,q}\|_{\mathrm{amb}} \le L_{p,q} \le \sqrt{p^2+q^2} + 1$. *Thus a cubical Freudenthal triangulation with* $N^2 > 12L_{p,q}^2$ *yields a normalized PL interpolant homotopic to* $n_{p,q}$. *For balanced* $p = q = r$, *its sufficient tetrahedral budget is* $O(r^3) = O(|H|^{3/2})$, not *an output-optimal lower bound.*

Proof. The horizontal differential of the Hopf projection is $2|F_2dF_1 - F_1dF_2|/(|F_1|^2+|F_2|^2)$, giving Eq. (U2). Since X is a cubical boundary, $a \ge 1$ or $b \ge 1$. Writing the square root as $\sqrt{A^2+B^2}$ with $A = 2pa^{p-1}b^q/(a^{2p}+b^{2q})$ and $B = 2qa^pb^{q-1}/(a^{2p}+b^{2q})$, when $a \ge 1$ we have $A \le p$, $B \le q+1$ (maximize $2qb^{q-1}/(1+b^{2q})$ for $b \le 1$); interchange a, p with b, q for the other case. At $(z_1, z_2) = (1,1)$, the two free imaginary directions yield differential norm $\sqrt{p^2+q^2}$. The facewise simplex diameter is at most $2\sqrt{3}/N$; interpolation error < 1 then gives a normalized straight-line homotopy, as in Theorem 28. The degree-pq map followed by the standard Hopf projection has $H = -pq$ in the chosen domain orientation. □

Let $C_1(\theta) = (e^{i\theta}, 1)$, $C_2(\theta) = (1, e^{i\theta})$. The transverse Bloch observable $u = n_x + in_y$ on these *physical circles within the cubical boundary* obeys $u(C_1) = e^{ip\theta}$, $u(C_2) = e^{-iq\theta}$.

Theorem 32 (Conditional two-loop finite-shot certificate)**.** *Suppose the unknown global field is promised to lie* exactly *in the bidegree family* (U1) *with* $1 \le p, q \le R$. *Measure Pauli* x, y *at* $M > 2R$ *equally spaced locations on each loop, with* m *independent calibrated* ± 1 *outcomes per*

component and site. Let W_1, W_2 be the exact rational-polygon winding numbers of the two empirical mean sequences. If

$$m > 4\cos^{-2}(\pi R/M)\log(8M/\delta), \tag{U3}$$

then with probability at least $1-\delta$ the polygons avoid zero and $(W_1, W_2) = (p, -q)$; hence $H = W_1W_2 = -pq$. At $M = 4R$ the sufficient binary-outcome count is $O(R\log(R/\delta))$. This is family-conditioned parameter inference*, not unrestricted reconstruction from arbitrary $S^3 \to S^2$ fields.*

Proof. The chord of the ideal winding polygon between two successive points is at least $\cos(\pi R/M) > 0$ from the origin because each angular step is less than π. For each of the $4M$ real means Hoeffding gives $2e^{-m\eta^2/4}$ with an ℓ_2 complex vertex-error target η. The union bound gives failure $\leq 8Me^{-m\eta^2/4}$. Choose $\eta = \sqrt{4\log(8M/\delta)/m} < \cos(\pi R/M)$. Every point of the empirical polygon then stays within η of the ideal polygon, so their straight-line interpolation avoids zero and preserves winding. Exact rational ray crossings compute W_1, W_2 without a floating-point integer-rounding rule. □

Proposition 9 (Aliasing and hidden charge)**.** *On M uniform samples of C_1, C_2, degrees p and $p + M$ are observationally identical even with noiseless values, but their Hopf integers differ by Mq. Without the global bidegree-family promise, even* continuous *observations of both entire loops cannot determine the global Hopf integer: insert an arbitrary compactly supported Hopf bubble in a three-ball disjoint from the loops. A constant spectral gap of the normalized two-band Hamiltonian alone does not remove either obstruction.*

Appendix V: Retained results, verification scope, and scientific priorities

This canonical manuscript retains the mathematically relevant core of both uploaded v12 branches: (i) rational integer linking and explicit abstentions; (ii) exact isotropic PL robustness and convex duality; (iii) exact anisotropic radii and data-adaptive confidence; (iv) the bidegree metric and two-loop results; and (v) the impossibility boundary without structural priors. The longer inherited results on Fourier interpolation, spatial covariance, spectral-tail envelopes, Pauli confidence allocation, lower bounds for hidden bubbles, and positive bosonic BdG spectral stability remain logged in `THEOREM_LEDGER.md`, with provenance to the original uploaded cumulative manuscript. They are *not* deleted from the research record, but they are not asserted as new central results of this focused article.

The independent exact finite checker is deliberately much smaller than a proof of the full rational crossing engine. Its accepting computations consist of rational arithmetic over explicit, externally inspectable witnesses; no LP solver is trusted at verification time. The original bidegree verifier independently tests 32,400 rational differential instances, 246 pairwise exact polygon winding controls, deliberate phase aliasing and 100 simulated shot replications; finite tests do not prove global statements. In addition, the main rational PL linker passes its preserved synthetic charge and abstention regressions. No Lean formalization or external laboratory data are included. For actual experiments one must independently bound calibration bias, drift, source regularity and model mismatch, and prove transfer from the true vertex-interpolated field to the continuous physical map.

Current mathematics catalogues motivate, but are *not* solved by, this work. Physics Open Problems #058 asks for convergence or a counterexample to a local spectral-gap SDP hierarchy, and #066 asks for a quantitative three-dimensional Hohenberg–Kohn inverse theorem; neither follows from our finite-dimensional PL certificate [44]. ProofAtlas describes its ranking as model-assessed importance rather than expert validation [45]. OpenAI Math expressly warns that some unformalized mathematical manuscripts may contain errors [46]. Near-term credible milestones are an independent cochain-based implementation, a Lean-verified core rational checker, and a public raw experimental tomography benchmark with a trusted calibration and regularity budget.

Appendix W: Conclusion

Exact Hopf integers, their sharp adversarial PL stability radii, and finite-shot identification become one auditable inference workflow only when each conditional arrow is explicitly checked. The new anisotropic four-family exact certificates and the data-dependent optional-stopping predicate strengthen the previous isotropic archive without importing historical filler or claiming priority for classical convex duality and linking topology. The two-loop bidegree result demonstrates the resource reduction possible with a true structural model promise; its aliasing theorem shows why that reduction is not universal. The primary remaining obstacles are formal verification of the whole topological integer checker, independent algorithmic replication, and scientifically calibrated measurements of genuinely unknown three-dimensional fields.

DATA AND CODE AVAILABILITY

The synthetic rational vertex fields, all isotropic and anisotropic primal–dual proof objects, synthetic Pauli count records, pure-standard-library proof checkers, and reproduction instructions accompany the v13 canonical archive. A permanent public repository release or DOI has not yet been assigned; the cited external scientific sources are independent of these new test artifacts. No laboratory data are reported.

Appendix X: Exact affine calibration phases and a spatial-gauge impossibility theorem

Scope and notation. Let $X = \mathbb{S}^3$ with its prescribed orientation and $n : X \to \mathbb{S}^2$ continuous. The experimentally reconstructed idealized Bloch direction may be distorted by a *single spatially constant* real 3×3 matrix A and a constant offset c:

$$\widetilde{n}(x) = \frac{An(x) + c}{\|An(x) + c\|_2}, \qquad \det A \neq 0, \tag{X1}$$

provided the numerator does not vanish. This is a *model of calibrated readout* and does not assert that an arbitrary real affine map is a completely positive physical qubit channel. We distinguish the topology of a reconstructed vector field from the physics of a laboratory Hamiltonian.

Theorem 33 (Exact global affine Hopf phase classification)**.** *For an invertible A, set $a = A^{-1}c$ and $\eta = \|a\|_2$. For the direction map $F_{A,c} : \mathbb{S}^2 \to \mathbb{S}^2$, $F_{A,c}(u) = (Au + c)/\|Au + c\|_2$, the following classification holds:*

1. *If $\eta < 1$, $F_{A,c}$ is defined everywhere and $\deg F_{A,c} = \operatorname{sgn}(\det A)$. For every $n : \mathbb{S}^3 \to \mathbb{S}^2$,* including reflected readout axes*,*

$$\boxed{H(F_{A,c} \circ n) = H(n).} \tag{X2}$$

2. *If $\eta > 1$, $F_{A,c}$ is defined everywhere, is null-homotopic and $H(F_{A,c} \circ n) = 0$.*

3. *If $\eta = 1$, the numerator vanishes at the unique direction $u_* = -A^{-1}c \in \mathbb{S}^2$. If $H(n) \neq 0$ then n is surjective, so at some point of X the transformed field vanishes: the normalized reconstruction is undefined.*

If A is singular, whenever the normalized affine map is everywhere defined its image avoids at least one point of $\mathbb{S}^2$, hence $H(F_{A,c} \circ n) = 0$.

Proof. For $\eta < 1$, $Au + tc = A(u + ta)$ cannot vanish for any $u \in \mathbb{S}^2$ and $0 \leq t \leq 1$, since vanishing would imply $\|u\|_2 = t\eta < 1$. This homotopes $F_{A,c}$ to $u \mapsto Au/\|Au\|_2$, whose degree is $\operatorname{sgn} \det A$ by the polar decomposition of A. The classical postcomposition law of the Hopf invariant is $H(f \circ n) = (\deg f)^2 H(n)$; this follows, for example, from naturality of the square of a degree-two cohomology generator of the mapping cone, or the signed preimage-linking formula. Since $(\pm 1)^2 = 1$, Eq. (X2) follows. If $\eta > 1$, the straight homotopy $F_t(u) = (tAu + c)/\|tAu + c\|_2$ from the constant direction $c/\|c\|_2$ to $F_{A,c}(u)$ does not vanish: $tAu + c = 0$ would require $\|a\|_2 = t \leq 1$. The critical case follows immediately from $u_* = -a$ and surjectivity of any n with $H(n) \neq 0$, since

a nonsurjective map into $\mathbb{S}^2$ is null-homotopic. If $\operatorname{rank} A < 3$, the numerator lies in a proper affine subspace of $\mathbb{R}^3$; if that subspace misses the origin its normalized image lies in an open hemisphere, and if it contains the origin the image lies in a proper great subsphere. In either case it avoids a point and is null-homotopic. □

Corollary 12 (Exact polynomial decision predicate and gap bound)**.** *For rational nonsingular A, c the topological regime is decided without division or floating-point arithmetic by the sign of the rational polynomial*

$$\boxed{\mathcal{D}(A,c) = \det(A)^2 - \|\operatorname{adj}(A)c\|_2^2.} \tag{X3}$$

Positive, negative, and zero $\mathcal{D}$ are equivalent to $\eta < 1$, $\eta > 1$, and $\eta = 1$ respectively. The reconstructed two-band Hamiltonian $\widetilde{\mathcal{H}}(x) = (An(x) + c) \cdot \boldsymbol{\sigma}$ has spectral gap satisfying

$$\boxed{\Delta_{\text{gap}} = 2 \min_x \|An(x) + c\|_2 \;\geq\; 2\sigma_{\min}(A)\,|1 - \eta|.} \tag{X4}$$

If $H(n) \neq 0$, the image of n equals all of $\mathbb{S}^2$, so the bound is independent of the detailed domain texture. A positive lower bound γ on $\sigma_{\min}(A)$ can itself be certified by exact principal-minor inequalities for $A^{\mathsf{T}}A - \gamma^2 I$ (strict inequalities certify $\sigma_{\min} > \gamma$).

Proof. Because $A^{-1} = \operatorname{adj}(A)/\det(A)$, the sign of $1 - \|A^{-1}c\|_2^2$ is precisely the sign of (X3) for invertible A. Since $An + c = A(n + a)$ and $\|Av\| \geq \sigma_{\min}(A)\|v\|$, the reverse triangle inequality gives $\|An + c\| \geq \sigma_{\min}(A)|\|a\| - \|n\|| = \sigma_{\min}(A)|\eta - 1|$. The two eigenvalues are $\pm\|An + c\|$. □

Corollary 13 (Uniform calibration uncertainty and a finite witness)**.** *Suppose $\eta_0 \neq 1$ for invertible nominal (A_0, c_0) and*

$$\|\Delta A\|_{\text{op}} + \|\Delta c\|_2 < \sigma_{\min}(A_0)|1 - \|A_0^{-1}c_0\|_2|. \tag{X5}$$

Then the two reconstructed fields corresponding to (A_0, c_0) and $(A_0 + \Delta A, c_0 + \Delta c)$ are everywhere defined and homotopic by their normalized coefficient-vector straight line. No stability of the inverse matrix along that line is required. In the rational example $A_0 = \operatorname{diag}(2, 3, 5)$, $c_0 = (1, 0, 0)$, the right-hand side is 1. If every entry of ΔA is at most $1/20$ in magnitude and every component of Δc at most $1/10$, then $\|\Delta A\|_{\text{op}} \leq \|\Delta A\|_F \leq 3/20$, $\|\Delta c\|_2 \leq 3/10$, and their sum is at most $9/20 < 1$. Thus the certificate accepts an entire explicit rational error box.

Proof. For any unit vector u and $t \in [0, 1]$, the perturbation $t(\Delta A\, u + \Delta c)$ has norm at most the left side of (X5). The bound (X4) for (A_0, c_0) excludes zero throughout the normalized straight line. □

Remark 7 (Varying Bloch-vector magnitude)**.** *If the uncalibrated coefficient field is* $b(x) = r(x)n(x)$ *with* $0 < r_{\min} \le r(x) \le r_{\max}$*, the same argument gives* $H((Ab+c)/\|Ab+c\|) = H(n)$ *whenever* $\eta < r_{\min}$*, and zero whenever* $\eta > r_{\max}$*, provided the corresponding fields are defined. For* $r_{\min} \le \eta \le r_{\max}$ *these interval bounds alone do not determine the topology. This is a sufficiency result, not an assertion of a universally sharp transition for variable* $r(x)$*.*

Theorem 34 (Spatially varying readout rotations destroy unconditional identifiability)**.** *Allow an arbitrary continuous unknown map* $R : \mathbb{S}^3 \to SO(3)$ *and suppose a reported unit field is* $m(x) = R(x)n(x)$*. Even infinitely many noiseless samples of* m*, including complete knowledge of* m *itself, do not identify* $H(n)$ *without a constraint on* R*: the same* m *has representations with a constant true field of Hopf invariant zero and with a true field of arbitrary prescribed integer Hopf invariant. Explicitly, for a unit quaternion* $q \in \mathbb{S}^3$*, let*

$$R_k(q)v = q^k v q^{-k}, \qquad m_k(q) = R_k(q)\,\mathbf{e}_1 \quad (k \in \mathbb{Z}), \tag{X6}$$

where imaginary unit quaternions are identified with vectors in $\mathbb{R}^3$*. Then* m_k *is a smooth unit vector field with* $H(m_k) = kH(m_1)$*,* $|H(m_1)| = 1$*, while the underlying constant field* $n(q) = \mathbf{e}_1$ *has* $H(n) = 0$*. The* same observed m_k *could instead originate from the true field* $n = m_k$ *and identity readout rotation. Accordingly an unrestricted spatial calibration gauge is a structural obstruction, not statistical noise that can be cured by increasing the shot count.*

Proof. Quaternion conjugation maps $\mathbb{S}^3$ continuously into $SO(3)$ and restricts to a proper rotation of $\mathbb{S}^2$. The map $q \mapsto q\mathbf{e}_1 q^{-1}$ is the Hopf fibration. The power map $q \mapsto q^k$ induces multiplication by k on $\pi_3(\mathbb{S}^3) \cong \mathbb{Z}$, since pointwise multiplication of maps into the topological group $SU(2)$ is the additive group law on π_3. Hence its degree is k and precomposition multiplies the Hopf invariant by k. The two incompatible true/calibration decompositions yield identical observed fields pointwise, proving non-identifiability. □

Remark 8 (Topological and physical boundaries)**.** *On* $X = \mathbb{T}^3$ *the same* target *degree classification holds for the primary weak Chern vector, which changes by* $\deg F_{A,c}$*; an absolute Hopf integer exists in the ordinary way when those weak classes vanish. Nonzero weak classes require their secondary Pontryagin conventions and are not assigned a spurious absolute invariant here. For the* $\mathbb{S}^3$ *results, a nontrivial spatially varying* R *need not be deformable to a constant rotation. If an* independently proved *null-homotopy* $R_t(x)$ *to the identity exists, it does give a homotopy of* $R(x)n(x)$ *to* $n(x)$*. Instrument calibration metadata and regularity are therefore identifiable physical assumptions rather than cosmetic details.*

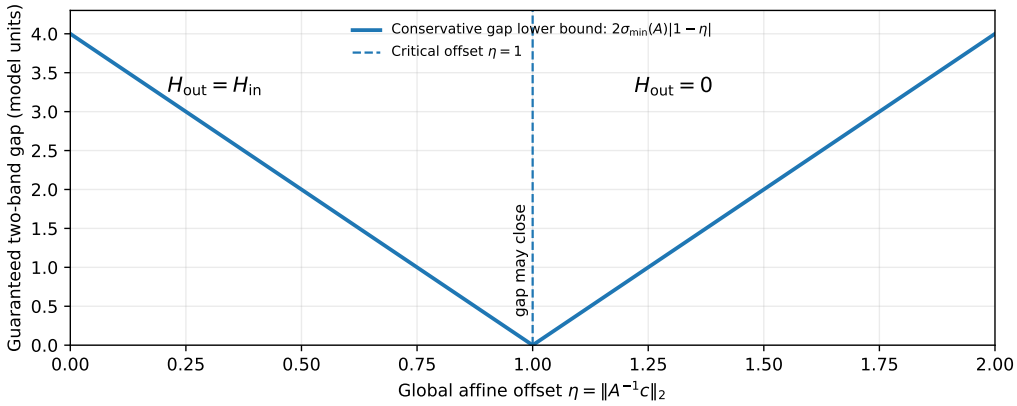


FIG. 32. A global affine readout of a surjective unit Bloch field has a sharp algebraic phase boundary at $\eta = \|A^{-1}c\|_2 = 1$. In the stable region H is preserved even for $\det A < 0$; beyond the boundary the normalized output is null-homotopic. The gap bar shows the conservative bound $2\sigma_{\min}(A)|1-\eta|$, not a measured gap. A spatially varying unknown rotation defeats any unconditional inference from the output alone.

a. Exact tests, attribution, and next experiment. The executable standard-library module `verifier/verify_affine_gauge_v15.py` independently checks the polynomial discriminant against exact inversion on 245 rational matrices/offsets; evaluates quaternion conjugations at 405 rational group points for $k = -4, \ldots, 4$; tests a critical singular response and a rational uncertainty box; and emits `V15_AFFINE_GAUGE_RESULTS.json`. These finite tests do not prove the degree-composition and non-identifiability theorems; the continuous proofs are above. The post-composition law and quaternionic Hopf fibration are classical; gauge freedom in self-consistent tomography is likewise established in prior work [48–50]. The contribution claimed here is an *auditable calibration-specific combination* and its exact experimental decision rule, with no unreviewed claim of first discovery. A decisive laboratory test would independently calibrate a fixed affine response (A, c) over the sampled parameter manifold, determine whether the response really is spatially constant, and compare the recovered class and gap to an independently calibrated reference reconstruction. It must not mistake a software calibration transformation for a new physical phase transition.

Appendix Y: Spatially varying readout gauges: exact charge addition and reference-state cancellation

Standing hypotheses. Let $X = \mathbb{S}^3$ carry a fixed orientation and let $n : X \to \mathbb{S}^2$ be continuous. We call $H(n) \in \mathbb{Z}$ the Hopf invariant normalized so that $h(q) = q\mathbf{i}q^{-1}$ on unit quaternions has $H(h) = 1$. This appendix resolves the topological ambiguity in Theorem 34 *conditionally on independent calibration information.* It does not reconstruct an arbitrary unknown gauge from a single measured texture and does not extend an absolute Hopf integer to $\mathbb{T}^3$ with nonvanishing weak Chern classes.

Theorem 35 (Exact spatial-frame charge addition)**.** *Let $R : X \to SO(3)$ and $m(x) = R(x)n(x)$. Every R has a continuous lift $g : X \to SU(2) \cong \mathbb{S}^3$ under the standard twofold cover* $\mathrm{Ad} : SU(2) \to SO(3)$*. Define*

$$\nu(R) := \deg(g) \in \mathbb{Z}. \tag{Y1}$$

This is independent of the choice of lift. With the stated orientation convention,

$$\boxed{H(m) = H(n) + \nu(R).} \tag{Y2}$$

In particular, if $e \in \mathbb{S}^2$ is any fixed axis and $r(x) = R(x)e$, then

$$\boxed{H(n) = H(m) - H(r).} \tag{Y3}$$

Thus the winding correction can be determined from a separately measured constant-input calibration field*, without explicitly recovering g or a pointwise 3×3 rotation matrix, provided the experiment uses the same gauge R for both fields.*

Proof. Since $X = \mathbb{S}^3$ is simply connected, every R lifts through the covering $SU(2) \to SO(3)$; two lifts differ by the constant central factor -1, whose left multiplication preserves the degree of $SU(2)$. The Hopf bundle $S^1 \to S^3 \xrightarrow{h} S^2$ admits a lift $p : X \to S^3$ of n, because its pullback Chern class belongs to $H^2(S^3; \mathbb{Z}) = 0$. Thus $n(x) = p(x)\mathbf{i}p(x)^{-1}$, up to a constant orientation-preserving rotation of the target, and $H(n) = \deg(p)$. Conjugation by $g(x)$ gives $m = h(gp)$. Pointwise multiplication $\mu : SU(2) \times SU(2) \to SU(2)$ is additive on $\pi_3(SU(2))$, hence $\deg(gp) = \deg(g) + \deg(p)$. The constant e likewise has a constant quaternion lift, so $H(Re) = \deg(g)$. This proves both identities. The degree/hopf-winding identification and the group law are classical [51, 56]. □

Theorem 36 (Reference correction under a nonconstant affine readout)**.** *Suppose* $A : X \to GL(3,\mathbb{R})$ *and* $c : X \to \mathbb{R}^3$ *are continuous; their determinant sign is constant since* X *is connected. Assume the independently certified* pointwise low-offset condition

$$\eta(x) := \|A(x)^{-1}c(x)\|_2 < 1 \qquad \textit{for every } x \in X. \tag{Y4}$$

For any fixed $e \in \mathbb{S}^2$ *put*

$$m(x) = \frac{A(x)n(x) + c(x)}{\|A(x)n(x) + c(x)\|_2}, \qquad r(x) = \frac{A(x)e + c(x)}{\|A(x)e + c(x)\|_2}. \tag{Y5}$$

Then both fields exist everywhere and the exact two-readout subtraction law *holds:*

$$\boxed{H(n) = H(m) - H(r).} \tag{Y6}$$

The same statement covers orientation-reversing linear readout matrices, singular-value anisotropy, and arbitrary spatial variation of A*; it does* not *require* A *to be approximately constant.*

Proof. The normalization of $A(n+tc)$ written correctly as $An + tc = A(n + tA^{-1}c)$ never vanishes for $t \in [0,1]$ by (Y4); hence the offset can be removed by homotopy for both input fields. Use the continuous polar decomposition $A(x) = O(x)P(x)$, with $O(x) \in O(3)$ and $P(x)$ positive definite. Since $(1-t)P + tI$ is positive definite, $An/\|An\|$ is homotopic to On. Choose a fixed orthogonal matrix J with $\det J = \det O \in \{\pm 1\}$ and $R(x) := O(x)J^{\mathsf{T}} \in SO(3)$. Then $On = R(Jn)$ and $Oe = R(Je)$. Postcomposition with a fixed reflection J preserves the Hopf integer because $H(Jn) = \deg(J)^2 H(n) = H(n)$; and Je is a constant reference. Apply Theorem 35 to these two fields and subtract. No spatial $SU(2)$ gauge fixing is required by the experimental correction formula. □

Corollary 14 (High-offset phase with spatially varying bias)**.** *If instead* $\eta(x) > 1$ *everywhere, then for any* n *the output* m *of* (Y5) *is homotopic to* $\widehat{c}(x) = c(x)/\|c(x)\|_2$ *and therefore*

$$\boxed{H(m) = H(\widehat{c}).} \tag{Y7}$$

Unlike the constant *offset case of Theorem 33, a varying high-amplitude offset can itself carry any Hopf integer. For example* $A = I$*,* $n = \mathbf{i}$*,* $c(q) = 3q\mathbf{i}q^{-1}$ *produces* $H(m) = 1$ *from a topologically trivial input, without closing the readout vector gap.*

Proof. Because $\|A^{-1}c\| > 1$, the family $tAn + c = A(tn + A^{-1}c)$ does not vanish for $t \in [0,1]$. Its normalized version contracts the output to $\widehat{c}$. In the example, the c-field has norm 3 whereas the constant contribution has norm 1, so the homotopy never encounters zero. □

Theorem 37 (Sharp universal frame-distance threshold and finite-sampling obstruction)**.** *Let* $R_0 : X \to SO(3)$ *be an independently reconstructed continuous calibration frame and* R *the true frame. If*

$$\boxed{\sup_{x\in X} \|R(x)R_0(x)^{\mathsf{T}} - I\|_{\mathrm{op}} < 2,} \tag{Y8}$$

then R *and* R_0 *are homotopic through rotations and* $\nu(R) = \nu(R_0)$*. The strict constant* 2 *is* universally sharp*: there are nonzero-winding frames at distance exactly* 2 *from the identity, whereas no nonzero-winding frame can satisfy the strict inequality for a constant reference frame. A direct covering-net condition is*

$$\max_i \|R(x_i) - R_*\|_{\mathrm{op}} + L_R h < 2, \tag{Y9}$$

where (x_i) *is an* h*-net and* L_R *is an* independently established *Lipschitz ceiling. Under* (Y9)*,* $\nu(R) = 0$*. In contrast, for* every finite *subset* $S \subset X$ *there exist smooth frames* R_0, R_1 *that agree pointwise on* S *but have* $\nu(R_1) - \nu(R_0) = 1$*. Therefore even perfect finite calibration samples cannot alone certify the winding class.*

Proof. Every rotation Q with $\|Q - I\|_{\mathrm{op}} < 2$ has angle strictly less than π, and hence a unique continuous principal logarithm in $\mathfrak{so}(3)$. Compactness of X makes this uniform. For $Q(x) = R(x)R_0(x)^{\mathsf{T}}$, $R_t(x) = \exp(t \log Q(x))R_0(x)$ is the required homotopy. The complement of the half-turn locus in $SO(3)$ is an open 3-ball, so nonzero-winding R must reach that locus relative to every constant reference. The explicit frame $R(q)v = qvq^{-1}$ has $\nu(R) = 1$ and achieves distance 2 at pure imaginary quaternions. The net inequality follows from a triangle inequality and the Lipschitz ceiling. For the no-go, choose a closed ball $B \subset X \setminus S$; take a smooth based map $g : X \to SU(2)$ of degree one which equals 1 outside B (a smoothed collapse/degree-one map of $B/\partial B \cong S^3$). The frames $R_0 = I$ and $R_1 = \mathrm{Ad}_g$ coincide on S, but $\nu(R_1) = 1$. □

Remark 9 (Statistical and physical certification)**.** *Applying* (Y6) *to finite-shot data requires* two *separately justified continuum topological certificates, one for the unknown-state readout* m *and one for the constant-input control* r*. If their false-certification risks are* δ_m *and* δ_r*, the joint false conclusion has risk at most* $\delta_m + \delta_r$ *by the union bound; independent noise between scans is* not *required for this union step. Any uncertainty in* (Y4) *or in the spatial calibration assumptions needs its own coverage budget. If the same underlying device does not act identically on the two scans, subtraction is not valid. Exact integer linking tests for piecewise-linear reconstructions are available in the previous appendices, but the continuum transfer conditions remain mandatory. On*

a physical Brillouin torus, nonzero primary weak Chern classes obstruct an unrestricted absolute Hopf integer and are outside the scope of this theorem.

a. Exact reproducible tests and novelty boundaries. The standard-library v16 verifier checks 5,184 quaternionic identities of the form $q^j(q^k\mathbf{i}q^{-k})q^{-j} = q^{j+k}\mathbf{i}q^{-(j+k)}$ for 64 exactly rational unit quaternions, all $j, k \in [-4, 4]$; 576 rational spatial-frame and low-offset response checks; a rational polynomial discriminant; and an exactly attained half-turn frame distance. A separate end-to-end test reconstructs two *external* 1,040-vertex rational fields on a 6,000-tetrahedron sphere with the inherited exact linking code. It returns $H(r) = 1$ and $H(m) = 2$ in three independent rational projections each; a strict curve-separation check accepts the former at three and the latter at ten chord subdivisions (1,296,000 cross-component pairs for the latter). The derived integer $H(m)-H(r) = 1$ is a completed discrete two-readout demonstration. A coarser three-subdivision observed-field attempt abstains; this negative result is retained in the audit rather than silently discarded. These tests are synthetic and do not establish the continuum topological theorem, which is proved above, nor do they validate a laboratory calibration. The $SU(2)$ winding/Hopf correspondence and the gauge ambiguity of self-consistent tomography are in established literature [51–53]. The contribution claimed here is the explicit conditional *two-readout* correction theorem, the nonconstant-affine extension and its exact failure criteria as one auditable synthesis. Its independent novelty and generality relative to the complete literature have not been certified.

Appendix Z: Four-reference affine readout inversion and optimal tetrahedral calibration

Experimental scope and relation to Appendix E. The previous two-scan theorem corrects an arbitrary spatial affine readout using a constant-state control only when the separately justified low-offset condition holds. This section gives an alternative when the offset may be large: four calibrated, affinely independent input Bloch vectors identify all twelve pointwise affine parameters from *raw three-component expectation values*. This identification does not apply if a detector reports only normalized directions, discards amplitude, drifts between probes, or is nonlinear in the input state. Those are separate, testable physical assumptions; we provide an exact model-violation control and an explicit impossibility example. The geometry of symmetric tetrahedral states and their value for qubit tomography is established in prior literature [54, 55]; our statement specializes this design to exact rational PL Hopf certificates rather than claiming discovery of tetrahedral state tomography.

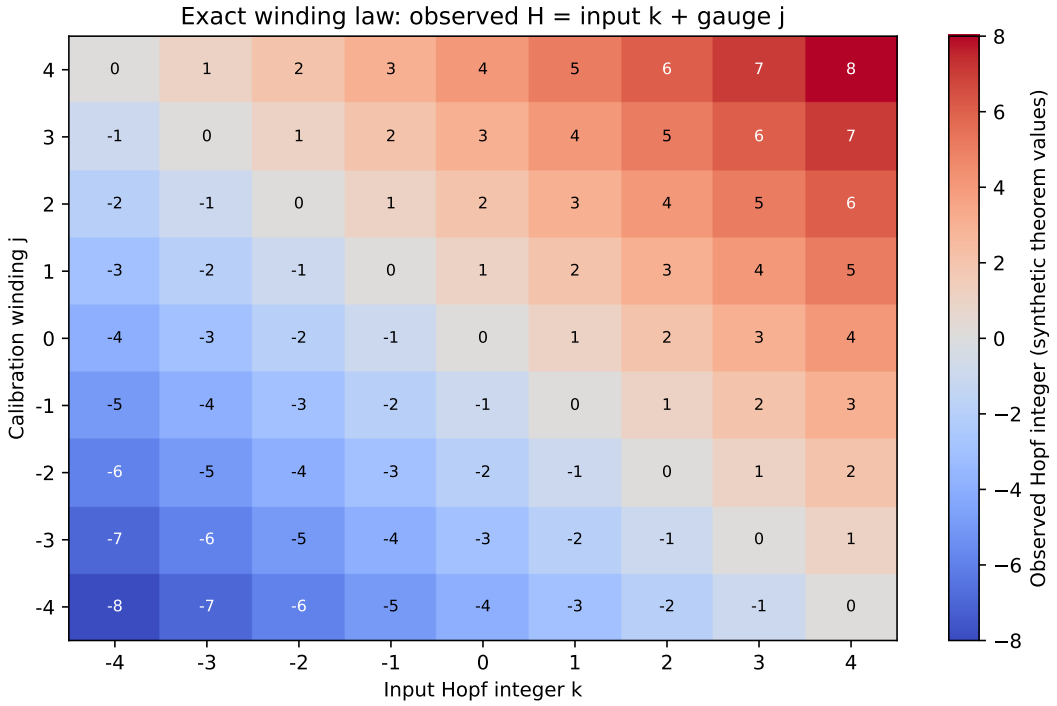


FIG. 33. Exact additive gauge-charge model in the degree-k and degree-j quaternion family. Each cell is the theoretical observed integer $H(m) = k + j$; independent constant-input calibration gives $H(r) = j$, and subtraction recovers the k of the input. The grid visualizes a mathematical identity, not laboratory measurements.

Theorem 38 (Exact pointwise affine inversion and four-reference necessity)**.** *Let* $r_1, \dots, r_4 \in \mathbb{R}^3$ *be affinely independent known Bloch vectors* ($\|r_j\|_2 \le 1$) *and* $Z = [(r_j^{\mathsf{T}}, 1)^{\mathsf{T}}]_{j=1}^4 \in \mathbb{R}^{4\times 4}$. *At every site* x *suppose the* unnormalized *detector response is* $y_x(n) = A_x n + c_x$, *with* $A_x \in GL(3, \mathbb{R})$ *and* $c_x \in \mathbb{R}^3$, *and that this same response is used for calibration and target data. From four probe outputs* $Y_x = [y_x(r_1) \ \cdots \ y_x(r_4)] \in \mathbb{R}^{3\times 4}$ *one reconstructs uniquely*

$$\boxed{[A_x \ c_x] = Y_x Z^{-1}, \qquad n_x = A_x^{-1}(y_x(n_x) - c_x).} \tag{Z1}$$

No low-offset condition is needed. For fewer than four affinely independent reference states, the full arbitrary affine response (A, c) *is not uniquely identifiable, even if* A *is required to be invertible.*

Proof. The calibration equations are the matrix identity $Y_x = [A_x \ c_x]Z$. Invertibility of Z proves the first two identities. For any $s \le 3$ probes their augmented vectors $z_j = (r_j, 1)$ span a proper subspace of $\mathbb{R}^4$. Let $\ell \ne 0$ annihilate their span, and $v \in \mathbb{R}^3 \setminus \{0\}$. The distinct affine laws $[A \ c]$ and $[A \ c] + tv\ell^{\mathsf{T}}$ agree on all probes. For sufficiently small nonzero t the new linear block remains invertible by continuity of the determinant. This is an identifiability lower bound for *the complete*

affine response, not a claim that four probes are always information-theoretically necessary to identify only a Hopf integer under extra physical priors. □

Theorem 39 (Sharp D- and E-optimal pure-state reference geometry)**.** *Among all four pure qubit probe vectors $r_j \in \mathbb{S}^2$, the largest possible calibration determinant is*

$$\boxed{|\det Z| \le \frac{16}{3\sqrt{3}},} \tag{Z2}$$

with equality exactly when $r_i \cdot r_j = -1/3$ for $i \neq j$, up to a common orthogonal transformation and permutation. The same tetrahedral configuration maximizes $\sigma_{\min}(Z)$, attaining $2/\sqrt{3}$, and minimizes the spectral condition number $\kappa_2(Z)$, attaining $\sqrt{3}$. Thus these are global *determinant, smallest-singular-value and conditioning optima within four pure Bloch references, not merely locally optimized numerical proposals.*

Proof. Set $\bar{r} = \frac{1}{4}\sum_j r_j$ and $C = \sum_j (r_j - \bar{r})(r_j - \bar{r})^{\mathsf{T}}$. The Schur complement of the lower-right entry of ZZ^{T} gives $\det(ZZ^{\mathsf{T}}) = 4 \det C$. Since $\operatorname{tr} C = 4 - 4\|\bar{r}\|_2^2 \le 4$, the arithmetic-geometric mean inequality on the three eigenvalues gives $\det C \le (4/3)^3$. Equality forces $\bar{r} = 0$ and $C = (4/3)I_3$, which for four unit vectors implies their Gram matrix equals $(4/3)(I_4 - \mathbf{1}\mathbf{1}^{\mathsf{T}}/4)$, hence the regular tetrahedron. Moreover $\lambda_{\min}(ZZ^{\mathsf{T}}) \le \lambda_{\min}(\sum_j r_j r_j^{\mathsf{T}}) \le \operatorname{tr}(\sum_j r_j r_j^{\mathsf{T}})/3 = 4/3$. Since the constant-output direction implies $\sigma_{\max}(Z) \ge 2$, necessarily $\kappa_2(Z) \ge 2/(2/\sqrt{3}) = \sqrt{3}$. For a tetrahedron the singular values of Z are $2, 2/\sqrt{3}, 2/\sqrt{3}, 2/\sqrt{3}$, attaining all bounds simultaneously. □

Theorem 40 (Sharp amplification of reference noise)**.** *For the regular unit tetrahedron r_j, the unique barycentric coefficients of a physically allowed target $n \in \mathbb{B}^3$ are*

$$\lambda_j(n) = \frac{1 + 3r_j \cdot n}{4}, \qquad n = \sum_{j=1}^{4} \lambda_j(n) r_j, \quad \sum_j \lambda_j(n) = 1. \tag{Z3}$$

The exact *worst-case calibration-noise amplification is*

$$\boxed{\sup_{\|n\|_2 \le 1} \sum_{j=1}^{4} |\lambda_j(n)| = 2.} \tag{Z4}$$

For perturbed calibration outputs $\widehat{y}_j = y(r_j) + e_j$ and target output $\widehat{y} = y(n) + e$, let $(\widehat{A}, \widehat{c})$ be the affine fit to the perturbed probes; suppose $\widehat{A}$ is nonsingular. If all errors obey $\|e\|_\infty \le \varepsilon_t$ and $\|e_j\|_\infty \le \varepsilon_c$, then the recovered $\widehat{n} = \widehat{A}^{-1}(\widehat{y} - \widehat{c})$ satisfies the exact deterministic inequality

$$\boxed{\|\widehat{n} - n\|_\infty \le \|\widehat{A}^{-1}\|_{\infty\to\infty}(\varepsilon_t + 2\varepsilon_c).} \tag{Z5}$$

The factor 2 is sharp for arbitrary independent bounded reference-output perturbations. For comparison, four references at $0, e_1, e_2, e_3$ *have* $|\det Z| = 1$, $\kappa_2(Z) = \sqrt{(5+\sqrt{21})/(5-\sqrt{21})}$, *and worst-case interpolation amplification* $1+2\sqrt{3}$. *The determinant gain* $16/(3\sqrt{3})$ *and error-factor reduction* $(1+2\sqrt{3})/2$ *compare different calibration designs under the* same componentwise error assumptions*; they do not imply the same ratio of physical experimental shot costs without accounting for probe preparation and count allocation.*

Proof. Affine interpolation gives $\widehat{A}n + \widehat{c} = \sum_j \lambda_j(n)\widehat{y}_j$ and $An + c = \sum_j \lambda_j(n) y(r_j)$. Subtraction yields the exact error equation

$$\widehat{A}(\widehat{n} - n) = e - \sum_j \lambda_j(n) e_j. \tag{Z6}$$

The error bound follows by the induced matrix norm. To prove sharpness of the scalar amplification, use $\sum |\lambda_j| = \max_{\epsilon_j \in \{-1,1\}} \sum_j \epsilon_j \lambda_j$ and maximize the linear expression over the unit ball. Grouping the 16 sign choices by the number of negative signs, their maxima are $1, 2, \sqrt{3}, 1, -1$. The maximum is 2, attained at $n = -r_j$, where $\lambda_j = -1/2$ and the other three barycentric coefficients equal $1/2$. Independently choosing the signs of e_j realizes the factor. For the axis-plus-center reference design, $\lambda = (1 - n_x - n_y - n_z, n_x, n_y, n_z)$ and the analogous sign maximization gives $1 + 2\sqrt{3}$. □

Corollary 15 (Sharp heterogeneous reference-error budget via sixteen algebraic witnesses)**.** *If the four pure tetrahedral calibration outputs have possibly unequal componentwise error ceilings* $|e_{j,a}| \le \varepsilon_{j,a}$, *then for each measurement coordinate a the sharp amplification against an unknown physical target* $n \in \mathbb{B}^3$ *is the finite, algebraically evaluable expression*

$$\boxed{L_a = \frac{1}{4} \max_{s \in \{-1,1\}^4} \left[\sum_{j=1}^{4} s_j \varepsilon_{j,a} + 3 \left\| \sum_{j=1}^{4} s_j \varepsilon_{j,a} r_j \right\|_2 \right].} \tag{Z7}$$

For target-axis error ceilings $|e_a| \le \varepsilon_{t,a}$, *every reconstructed component satisfies*

$$|\widehat{n}_b - n_b| \le \sum_{a=1}^{3} |(\widehat{A}^{-1})_{ba}| \, (\varepsilon_{t,a} + L_a). \tag{Z8}$$

The maximum in Eq. (Z7) uses just sixteen sign choices and square roots of explicit algebraic numbers; it requires no continuous optimization or statistical independence of reference errors. With equal reference ceilings, $L_a = 2\varepsilon_c$; *if only one tetrahedral calibration preparation has error ceiling* ε *and the other three are exact, then* $L_a = \varepsilon$, *an exact factor-two improvement over the equal-ceiling fallback. Anisotropic reference and target shot allocations may therefore be certified* after acquisition *using the simultaneous anytime event of Corollary 18.*

Proof. Insert the affine barycentric coefficients from Eq. (Z3) and use

$$\sum_j \varepsilon_{j,a}|\lambda_j(n)| = \max_s \sum_j s_j \varepsilon_{j,a}\lambda_j(n).$$

Because the maximum is finite, exchange it with the maximum over $\|n\|_2 \le 1$. Each fixed-sign expression is affine in n and is maximized by aligning n with its vector coefficient, giving Eq. (Z7). Apply the triangle inequality separately to the three rows of the exact identity (Z6). When only reference j is uncertain, $\sup_{\|n\|\le 1} |\lambda_j(n)| = 1$, whereas with all four equal ceilings the sharp amplification is 2 by Theorem 40. □

Corollary 16 (Exact algebraic tomography and topology certification without a reference Hopf charge)**.** *On an oriented rational triangulated three-sphere, let $\widehat{b}_i \in \mathbb{Q}^3$ be vertex estimates obtained by pointwise affine inversion from rational raw counts and rational reference vectors. Define their PL nonvanishing radius*

$$\rho_\infty(\widehat{b}) = \min_\sigma \min_{\lambda\in\Delta^3} \left\| \sum_{i\in\sigma} \lambda_i \widehat{b}_i \right\|_\infty . \tag{Z9}$$

If $\rho_\infty(\widehat{b}) > 0$ and every site's independently justified affine-calibration error bound from Eq. (Z5) is strictly smaller than $\rho_\infty(\widehat{b})$, then the PL field of the true vertex data is everywhere nonzero and homotopic to the recovered PL field. In particular, the certified integer Hopf invariants coincide. Rational witnesses from Appendices A–C can supply lower bounds *for ρ_∞ without running a floating-point optimizer. An independently validated spatial regularity or spectral-tail bound is still required to identify the continuous physical field with its sampled PL representative.*

Proof. On each tetrahedron the difference between the true and reconstructed PL vector fields is a barycentric combination of vertex errors; its coordinate-infinity norm is bounded by the maximum vertex error. If this is smaller than the global positive nonvanishing radius, the segment joining these fields cannot reach zero. Normalize the segment to obtain a continuous homotopy. This argument is deterministic and permits choosing witnesses and $\widehat{A}^{-1}$ *after viewing the data*, provided the error bounds hold on a simultaneous confidence event. □

Corollary 17 (A rational control with a held-out fifth preparation)**.** *For a centered tetrahedron the additional calibration preparation $r_0 = 0$ predicts the exact affine identity*

$$\boxed{y(0) = \frac{1}{4}\sum_{j=1}^{4} y(r_j).} \tag{Z10}$$

Thus, with componentwise calibration errors at most ε_c and control-output errors at most ε_0, any component of the measured discrepancy exceeding $\varepsilon_c+\varepsilon_0$ falsifies *the affine readout model. This fifth-state test is a necessary check, not a complete proof of affineness: the explicit nonzero polynomial response deviation*

$$D(n) = \epsilon v \prod_{j=0}^{4} \|n - r_j\|_2^2 \tag{Z11}$$

vanishes at all five *controls but changes generic target-state readings. Finite control scans therefore require a separately justified bound on model nonlinearity for universal claims.*

Corollary 18 (Anytime-valid calibration with independent Pauli shots)**.** *Assume stationary conditionally unbiased independent ±1 shot streams for three axes, four references and one target at each of M spatial sites. Put $K = 15M$. For the mean of any one stream after $t \geq 1$ observations let*

$$\varepsilon_t = \sqrt{\frac{2}{t}\log\Big(\frac{K\pi^2 t^2}{3\delta}\Big)}. \tag{Z12}$$

With probability at least $1-\delta$, all *such means simultaneously lie within their respective radii of their true means for* every *$t \geq 1$. Consequently sitewise values $\varepsilon_c = \max_{j,a} \varepsilon_{t_{ja}}$ and $\varepsilon_t^{\mathrm{tar}} = \max_a \varepsilon_{t_{0a}}$ may be substituted into Eq. (Z5) under arbitrary data-dependent stopping. With a held-out fifth reference use $K = 18M$ instead. These are conservative Hoeffding bounds, not an information-theoretically optimal measurement protocol; time drift, correlated apparatus bias, invalid reference preparation or a nonlinear detector invalidate the stipulated shot model.*

Proof. For a single bounded ±1 stream Hoeffding's lemma and Chernoff's bound give $\mathbb{P}(|\widehat{\mu}_t - \mu| > a) \leq 2\exp(-ta^2/2)$. With Eq. (Z12), the right side equals $6\delta/(K\pi^2t^2)$. The union over K streams and all positive integers t has probability at most δ because $\sum_{t\geq1} t^{-2} = \pi^2/6$. The event is uniform over times and sites, so data-driven stopping, calibration-matrix fitting and cell-separator selection are allowed without post-selection leakage from this coverage event. □

a. Machine-checkable example and exact comparisons. The independent standard-library program `verifier/verify_tetra_affine_v17.py` checks 4,160 exact site-by-site inversion identities using all four previously certified rational $k = 1, 2, 3, 4$ texture families. The full-site synthetic example employs a rational *mixed-state* tetrahedron with vertices $(\pm1/3, \pm1/3, \pm1/3)$ in the tetrahedral sign pattern, because this makes every probe, matrix, offset and determinant rational; its radius is $1/\sqrt{3}$, not the pure optimal radius. Its determinant is exactly $16/27$. The mixed-state

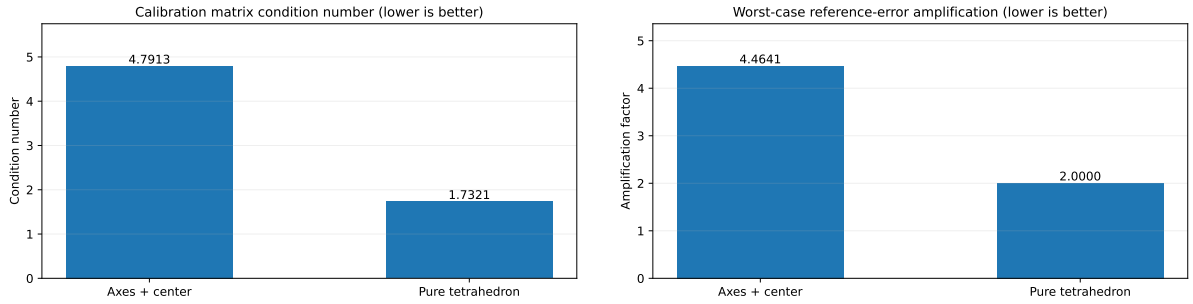


FIG. 34. Exact design benchmarks for two explicitly specified four-preparation calibration schemes. The pure regular tetrahedron has globally optimal conditioning and a sharp factor-two interpolation error. Axis-plus-center is a comparator with a mixed central probe; the graphs do not account for experimental state-preparation costs.

reference has interpolation factor $(1+3\sqrt{3})/2 < 31/10$, *not* the pure-state factor 2. An independent perturbed run applies exact 10^{-4} componentwise changes on the 1,040 sites of the charge-four case, verifies the calibration-error inequality with the conservative $31/10$ constant and checks all 6,000 tetrahedral separating inequalities for the *observed* reconstructed vectors. Its computed lower topological margin is positive and exceeds its maximum affine-error bound. The fifth held-out reference rejects the explicit quadratic detector deviation $n_x^2/30$ with exact discrepancy $1/270 > 2 \times 10^{-4}$, while the five-control polynomial of Eq. (Z11) confirms the unavoidable finite-calibration no-go. No noise model was estimated from laboratory hardware; all readout data and errors are synthetic exact fractions, and the program does not establish historical priority or replace an independent formal proof of the inherited integer-linking kernel.

b. Comparison of complementary readout strategies. There are three distinct, *noninterchangeable* operating regimes. With a separately certified small affine offset, the v16 *two-scan* charge-subtraction identity avoids pointwise matrix inversion and can accommodate spatially varying gauge winding. Without a small-offset guarantee but with a justified invertible affine response and access to raw three-axis expectation vectors, the v17 *four-reference plus target* design identifies arbitrary pointwise $A(x), c(x)$ and permits sharp conditional error propagation. One additional centered reference supplies a model-falsification control but cannot certify global linearity by itself. None is a solution of the unrelated LTI-SDP hierarchy completeness problem or the continuum three-dimensional Hohenberg–Kohn inverse theorem. These ideas provide finite proof-object and stability design patterns rather than any claimed settlement of those open problems.

Appendix AA: Directional-only affine readouts: an exact hierarchy of minimal reference sets

Motivation and relation to Appendices E–F. Appendix F inverts a full three-component affine detector response using four known probes, assuming that the *amplitude* of each response is retained. Many readouts instead report only the unit direction $d = y/\|y\|$ of a three-vector $y = An + c$. This is a different inverse problem: the four-state formula $[A\ c] = YZ^{-1}$ cannot be used because the four unknown response amplitudes are missing. We determine the exact reference counts needed to identify the affine parameters up to one common positive scale, state the additional condition needed to recover the *input state*, and give an exact-rational error certificate. This is a specialization of classical projective calibration/observability, not a claim to have invented projective geometry or ordinary four-state qubit process tomography [54, 57].

Theorem 41 (Minimal directional calibration ladder with explicit rational pure-state witnesses)**.** *Let $r_1, \dots, r_m \in \mathbb{S}^2$ be known pure Bloch vectors and suppose that the device returns only oriented rays*

$$d_j = \frac{Ar_j + c}{\|Ar_j + c\|_2} \in \mathbb{S}^2, \quad Ar_j + c \neq 0. \tag{AA1}$$

All following results concern generic parameters satisfying the stated model, not arbitrary or ill-conditioned probe choices:

Assumed response model	*Free parameters / scale*	*Minimal m*	*Exact maximal rank*
$A = A^{\mathsf{T}} \succ 0,\ c = 0$	$6 - 1 = 5$	3	5
$A \in GL(3),\ c = 0$	$9 - 1 = 8$	4	8
$A = A^{\mathsf{T}} \succ 0,\ c \in \mathbb{R}^3$	$9 - 1 = 8$	4	8
$A \in GL(3),\ c \in \mathbb{R}^3$	$12 - 1 = 11$	6	11

At this minimal count, generically the response parameters are uniquely identifiable up to a common positive *scale. The displayed ranks have explicit exact-rational certificates, including pure states with coordinate denominators at most three. With one fewer probe, no method can generically identify the full stated model even locally under its open positivity and invertibility conditions.*

Proof. Write $w_j = Ar_j + c$ and use any nonzero representative of the oriented ray d_j. The equation $d_j \times w_j = 0$ gives two independent equations

$$d_{j,p}(Ar_j + c)_k - d_{j,k}(Ar_j + c)_p = 0, \qquad k \in \{1, 2, 3\} \setminus \{p\}, \quad d_{j,p} \neq 0. \tag{AA2}$$

These equations are homogeneous and *linear* in the 6, 9, 9 or 12 coefficients of the stated response model. Each ray gives at most two linearly independent scalar constraints. Since a common nonzero scale is invisible, full projective identifiability requires $2m \geq D-1$, where $D \in \{6,9,9,12\}$ is the model dimension. Thus m is at least $3,4,4,6$, respectively. If $2m < D-1$, the nullspace has dimension at least two, and small perturbations in a second null direction preserve strict positive ray orientations and the open SPD/invertibility constraints.

To prove the converse generically, take the rational unit vectors

$$\begin{aligned} r_1 &= (-2,-2,-1)/3, \quad r_2 = (-1,0,0), \\ r_3 &= (-2,2,-1)/3, \quad r_4 = (0,-1,0), \\ r_5 &= (0,0,1), \quad r_6 = (0,1,0). \end{aligned} \tag{AA3}$$

For the symmetric $c = 0$ model use the three vectors $(1,0,0),(0,1,0),r_1$; for the general linear model add $(0,0,1)$; for symmetric A with translation use r_1, r_2, r_3, r_4; for arbitrary A, c use all six in Eq. (AA3). At $A = I, c = 0$, form the rational coefficient matrices from Eq. (AA2), selecting the first nonzero component of each d_j as the pivot. Their sizes and explicit nonzero maximal minors are respectively

model	matrix size	nonzero minor
$A = A^{\mathsf{T}},\ c = 0$	6×6	$(5 \times 5):\ 8/81$
A arbitrary, $\ c = 0$	8×9	$(8 \times 8):\ 8/81$
$A = A^{\mathsf{T}},\ c$ arbitrary	8×9	$(8 \times 8):\ 640/6561$
A, c arbitrary	12×12	$(11 \times 11):\ 1024/6561$

. (AA4)

For each model the true parameter vector is in the nullspace, so its rank is at most $D-1$; the nonzero minor forces equality. The minor stays nonzero on an open, Zariski-dense set of response parameters since it is a nonzero polynomial in the entries of A, c when using the unnormalized representatives $Ar_j + c$. Row scaling by the observed unit-vector normalizations does not change rank. The one-dimensional nullspace fixes the channel up to a real scalar. The sign of the scalar is fixed by requiring $d_j \cdot (Ar_j + c) > 0$ for all reference rays. This proves generic uniqueness modulo a *positive* scale. The model with $A = A^{\mathsf{T}} \succ 0$ uses the open SPD subset containing $A = I$. □

Theorem 42 (Sharp one-ray inversion threshold and a two-to-one obstruction)**.** *Assume the full affine channel has been identified up to positive scale, with $A \in GL(3,\mathbb{R})$. Write $u = A^{-1}d$ and $v = A^{-1}c$. If*

$$\boxed{\eta^2 := \|v\|_2^2 < 1,} \tag{AA5}$$

then every oriented output ray $d \in \mathbb{S}^2$ uniquely determines the pure input state $n \in \mathbb{S}^2$ through

$$\boxed{t = \frac{u \cdot v + \sqrt{(u \cdot v)^2 + \|u\|^2(1 - \|v\|^2)}}{\|u\|^2} > 0, \qquad n = tu - v.} \tag{AA6}$$

This result is unchanged if (A, c) is rescaled by the same positive factor. At $\eta > 1$ the input-to-direction map need not be injective, even with perfectly known A, c. *At $\eta = 1$ the normalized response is undefined for at least one pure input. The strict condition* (AA5) *coincides with the affine topological stability regime of Appendices D–E.*

Proof. The output ray condition means $An + c = td$ with $t > 0$. Therefore $n = tu - v$ and

$$\|u\|^2 t^2 - 2(u \cdot v)t + (\|v\|^2 - 1) = 0.$$

If $\|v\| < 1$, the constant term is negative and the leading term positive, so precisely one root is positive and it is Eq. (AA6). Conversely, $A = I, c = (0, 0, 2)$ maps both $n = (0, 0, 1)$ and $n = (0, 0, -1)$ into the *same* positive ray $(0, 0, 1)$, with amplitudes 3 and 1, respectively. At $A = I, c = (0, 0, 1)$ the input $n = (0, 0, -1)$ produces the zero response. The same-scale invariance follows by replacing (A, c) with (sA, sc) and t with st. □

Proposition 10 (Exact-rational noise-margin certificate for directional inversion)**.** *Let M_0 be an exactly rational, consistent bearing matrix of size $2m \times D$ and rank $D - 1$. Choose $D - 1$ rows and $D - 1$ columns giving an invertible rational square submatrix P_0, and fix the remaining coefficient as a scale gauge. The selected equations have the form $P_0 z_0 = -q_0$. Let $P_1 z_1 = -q_1$ be the corresponding equations for perturbed bearing representatives, with unchanged pivot choices. Put*

$$C = \|P_0^{-1}\|_\infty, \quad e = \|P_1 - P_0\|_\infty, \quad f = \|q_1 - q_0\|_\infty. \tag{AA7}$$

If $Ce < 1$, then P_1 is invertible and the parameter error is bounded by

$$\boxed{\|z_1 - z_0\|_\infty \le \frac{C(f + e\|z_0\|_\infty)}{1 - Ce}.} \tag{AA8}$$

Every quantity in this certificate is rational. With noisy data, the unused bearing equations generally have a nonzero residual; it must be checked against the stated observation-error model before treating the projective calibration as a physically supported estimate.

Proof. Subtract $P_1 z_1 = -q_1$ and $P_0 z_0 = -q_0$ to obtain $P_0(z_1 - z_0) = -(q_1 - q_0) - (P_1 - P_0)z_1$. By the triangle inequality, $(1 - Ce)\|z_1 - z_0\|_\infty \le C(f + e\|z_0\|_\infty)$. The strict Neumann condition also guarantees invertibility of P_1. □

Corollary 19 (Amplitude-free topological reconstruction, conditional on readout physics)**.** *On a triangulated* $\mathbb{S}^3$*, suppose at every vertex the instrument is an invertible pointwise affine map in the* raw *Bloch coordinates, followed only by positive normalization of its output. If six sufficiently generic pure references have measured oriented rays and the corresponding exact bearing matrix has rank* 11*, the channel is identified up to positive scale at that vertex. If Eq.* (AA5) *is separately certified and the input target is pure, the target Bloch direction is uniquely recovered by Eq.* (AA6)*. If all reconstructed fields satisfy an independently verified nonvanishing and uncertainty margin as in Appendices A–C, the discrete integer Hopf class can be certified without ever measuring the detector's output amplitude. Moving from the discrete model to an unknown continuum texture still requires a verified spatial regularity or spectral-tail bound. With non-negligible observation errors, exact rank is replaced by an error-aware fit and interval/spectral certificate, not by forcing a noisy matrix to have rank* 11*.*

Remark 10 (Free topological protection with positive-definite spatial gain)**.** *When* $A(x) = A(x)^{\mathsf{T}} \succ 0$ *and* $c(x) = 0$ *at every site in a continuous spatial readout, the normalized field* $A(x)n(x)/\|A(x)n(x)\|$ *is homotopic to* n *by the explicit positive-definite path* $[(1-t)I + tA(x)]n$*. Thus* zero reference states *are needed to preserve* H *under this strong physical promise, even though* three *nondegenerate directional preparations are generically necessary to identify the full symmetric gain up to scale. The two targets—recovering a topological integer and reconstructing a detector—must not be conflated.*

a. Exact reproducible certificates and no-go controls. The pure-Python standard-library verifier `verifier/verify_direction_only_v18.py` computes all four ranks and minors in Eq. (AA4) using rational Gauss–Jordan elimination, reconstructs a nontrivial rational affine channel from six unscaled bearing representatives without supplying its coefficients to the inverse, and recovers five rational pure target states with no amplitude input. It also verifies the lower-count nullspaces, an axis-aligned rank-deficient six-reference design, the $\eta > 1$ two-to-one obstruction, and the $\eta = 1$ zero. For the six-reference identity channel a fixed 11×11 pivot block has the *exact* induced inverse norm $C = 41/4$. With synthetic bearing perturbation 10^{-5}, the checker verifies $e = 1/18750$, $Ce < 1$, and the parameter-error inequality (AA8). Its full-matrix rank and residual are reported; no unverified real-device noise assumptions enter this deterministic test. A separate end-to-end amplitude-free script `verifier/verify_v18_blind_field.py` reads the archived 1,040-vertex rational field without its expected Hopf charge, applies spatially varying nonsymmetric affine matrices and small biases, strips the positive scale separately from each of the six probe responses and each

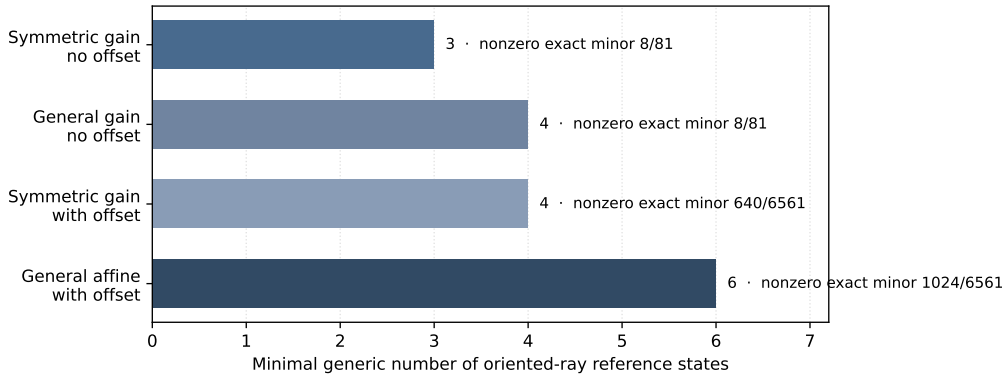


FIG. 35. Exactly certified minimal number of *direction-only* pure reference preparations for four nested affine readout families. Bars are minimal generic identifiability counts for the full model up to positive scale; they do not count target-state repetitions or prove optimal total shot resources.

target response, reconstructs an affine model from the six resulting rays, and checks that all 1,040 target vectors are recovered *exactly*. This extends the inverse-algebra test to a complete previously certified charge-four PL benchmark; the charge-four link computation itself remains a distinct inherited certificate rather than a new experiment. The same nominal 12×12 bearing matrix has squared Frobenius norm $196/9$ and a nonzero 11×11 minor $1024/6561$. Cauchy–Binet therefore supplies a (very conservative) rational lower bound $9/282475249$ on its smallest positive singular value. Neither this bound nor the exact ranks establishes a world-first invention: the results are a transparent finite-dimensional projective identifiability specialization within the existing theory of Bloch-affine channels [57, 58].

Appendix AB: AI-guided rational matrix construction with exact error and erasure certificates

Purpose and research context. Appendix G proves that six generically placed pure reference directions suffice for projective calibration of a pointwise affine Bloch response when detector amplitudes are discarded. Here we improve an explicit *finite-dimensional reconstruction matrix*, rather than asserting a new generic-identifiability theorem. The discovery protocol combines numerical candidate search with exact rational certificate checking, in the spirit of AlphaTensor, FunSearch and AlphaEvolve; those earlier AI systems have independent discoveries in arithmetic complexity and program search. OpenAI's matrix-multiplication family also provides formally specified asymptotic bounds, but these do not imply an improvement for our eleven-dimensional inverse matrix. The optimization objective, certificate and physical assumptions must be kept separate from historical

claims of mathematical priority [59–62].

For a known pure input $r_j \in \mathbb{S}^2$ and observed oriented output ray $d_j \parallel Ar_j + c$, fix the first nonzero component index $p(j)$ for the ideal reference ray, and define for each $k \neq p(j)$ the linear constraint

$$d_{j,p(j)}(Ar_j + c)_k - d_{j,k}(Ar_j + c)_{p(j)} = 0. \tag{AB1}$$

The twelve equations for six references are homogeneous in the twelve entries of (A, c). We fix the projective gauge $A_{33} = 1$ and use eleven unknown parameters in row-major order with column A_{33} omitted. With ideal data $A = I_3, c = 0, d_j = r_j$, denote the resulting 12×11 matrix by $B(R)$, and write $P_{(s)}(R)$ for the square submatrix obtained by removing equation $s \in \{1, \ldots, 12\}$. We use the unscaled component equations (AB1), without post-hoc row normalization; the objective is therefore a precisely defined *protocol-dependent* condition number.

Theorem 43 (An explicit rational six-reference improvement)**.** *Consider the six rational pure states*

$$\begin{aligned} r_1 &= (4/5,\ 0,\ -3/5),\\ r_2 &= (0,\ 4/5,\ -3/5),\\ r_3 &= (4/5,\ 0,\ 3/5),\\ r_4 &= (-2/3,\ -1/3,\ -2/3),\\ r_5 &= (-2/3,\ -2/3,\ 1/3),\\ r_6 &= (0,\ 4/5,\ 3/5). \end{aligned} \tag{AB2}$$

The rational matrix $B(R)$ has rank eleven. Every *one of its twelve eleven-equation submatrices is invertible. The best exact inverse-ℓ_∞ operator norm within this finite family is*

$$C_{\text{new}} = \min_{s=1}^{12} \|P_{(s)}(R)^{-1}\|_{\infty\to\infty} = \frac{127525}{31104}, \tag{AB3}$$

attained by omitting equation nine, whose determinant is

$$\det P_{(9)}(R) = -\frac{130459631616}{152587890625}. \tag{AB4}$$

For the six reference states used in the baseline v18 verifier, only six of the twelve leave-one-equation-out blocks are invertible, and their smallest inverse norm is $C_{\text{old}} = 41/4$. Hence the exact improvement for this fixed gauge, row normalization and submatrix-based solver is

$$\frac{C_{\text{old}}}{C_{\text{new}}} = \frac{318816}{127525} > \frac{249}{100}. \tag{AB5}$$

This is a constructive comparison, not a claim of global optimality over all choices of reference states, gauges, estimators or experimental noise models.

Proof. Every entry of (AB2) has squared coordinates summing to one. For the stated deterministic pivot convention, build the twelve rows by (AB1). Clearing denominators makes every matrix entry integral. For each of the twelve choices of omitted row, exact Gauss–Jordan inversion decides whether the determinant is zero. The provided independent verifiers check all cases, not just the selected minor. They find twelve nonzero determinants for the new construction and six for the old. In the selected new block they check the rational identities $PP^{-1} = P^{-1}P = I_{11}$ in all 121 entries, the determinant (AB4), and the eleven exact sums of absolute values of its inverse rows. Their maximum is precisely (AB3); comparing the remaining eleven blocks proves the finite minimum. All quantities are exact fractions and the scripts require no numerical optimizer. As a separate negative control, duplicating a reference makes every potential eleven-row block singular. □

Theorem 44 (An explicit rational perturbation certificate)**.** *Use the minimizing block in Theorem 43. Suppose each component of every measured (fixed-scale) bearing representative is perturbed by at most ϵ, while the ideal reference inputs remain exact. Assume the original pivot components retain their nonzero signs, and fix the same gauge $A_{33} = 1$. If*

$$0 \le \epsilon < \frac{5832}{127525}, \tag{AB6}$$

then the perturbed selected matrix $P + E$ is invertible. Writing z_0 for the eleven true identity-response coefficients and $\widehat{z}$ for the perturbed solution, one has

$$\|\widehat{z} - z_0\|_\infty \le \frac{C_{\mathrm{new}}(K + L)\epsilon}{1 - C_{\mathrm{new}}K\epsilon}, \qquad K = \frac{16}{3}, \quad L = \frac{2}{3}. \tag{AB7}$$

For the same conservative row perturbation method the old six-reference matrix has $K = 16/3, L = 1$ and requires $\epsilon < 3/164$. The invertibility threshold improves by the exact factor $318816/127525$; the full parameter-error bounds differ also through L.

Proof. Each selected row depends only on two components of one bearing. A componentwise perturbation of size ϵ changes the absolute row sum of the eleven-parameter matrix by at most $2(\|r_j\|_1 + 1)\epsilon$. By direct exact evaluation of the six vectors the maximum is $K\epsilon = 16\epsilon/3$. The gauge column A_{33} changes the right hand side by at most $\max_j |r_{j,3}|\epsilon = L\epsilon = 2\epsilon/3$. Thus $\|E\|_{\infty\to\infty} \le K\epsilon$ and $\|\Delta q\|_\infty \le L\epsilon$. The Neumann series provides $\|(P + E)^{-1}\| \le C_{\mathrm{new}}/(1 - C_{\mathrm{new}}K\epsilon)$ when (AB6) holds. Subtracting the two linear systems and using $\|z_0\|_\infty = 1$ proves (AB7). The old threshold follows from the same argument with its exact inverse norm. For genuine measured

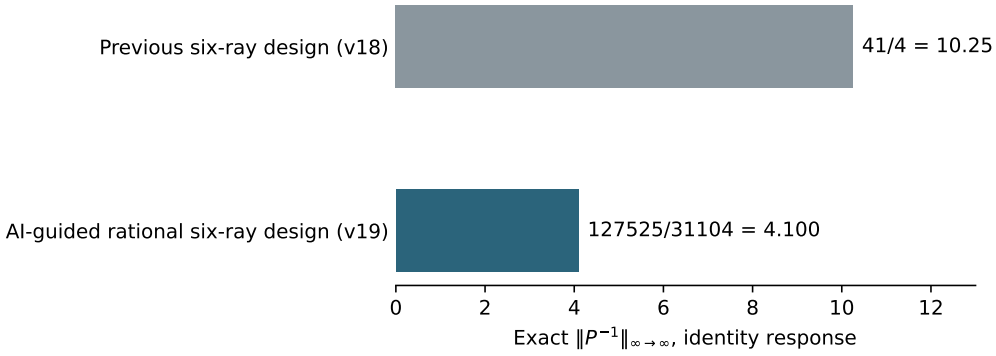


FIG. 36. Exactly certified inverse-ℓ_∞ operator norms for two specific six-ray affine calibration designs at the identity response. The new reference geometry reduces the worst-case linear inverse amplification for this fixed solver; the calculation does not establish global optimality, shot complexity or a physical detector benchmark.

unit bearings, the pre-normalization or direction-estimation procedure must independently supply a valid componentwise error envelope in this row representation; otherwise no probabilistic or experimental statement follows from this deterministic bound. □

a. Exact audit and limits. The standalone standard-library checker `verifier/verify_ai_matrix_design_v19.py` reconstructs the two matrices, tests all 24 candidate row deletions, validates 121 inverse entries for each nonsingular block, runs 36 rational direction-noise trials, and rejects a duplicated-reference configuration. A second script uses independent SymPy matrix operations to rederive both optimal norms and determinants. Discovery used a 22,000-candidate seeded numerical exploration of rational pure-state six-tuples; its best numerical suggestions were subsequently subjected to this independent exact certification. The search is not an exhaustive proof of best possible experimental design. Applications to a general unknown response $(A(x), c(x))$ require a newly checked local rank and inverse bound at *every* spatial vertex; the displayed constants are certified at the identity response only. A verified spatial regularity assumption remains necessary to transfer any PL Hopf integer to an unknown continuum field.

Appendix AC: Sharp all-equation inverse design for bounded directional errors

The six-reference direction-only calibration of Appendix G produces twelve scalar linear constraints on eleven gauge-fixed affine-response parameters. Appendix H selects one invertible eleven-row submatrix and reconstructs all unknowns from that same selection. The remaining scalar equa-

tion can instead be used *differently for each parameter*, without nonlinear optimization or numerical inverses. We identify the best possible linear left inverse for an adversarial componentwise-error model, including unequal observation budgets.

Let $B \in \mathbb{Q}^{(n+1)\times n}$ have rank n and let $w = (w_1, \dots, w_{n+1}) \in \mathbb{Q}^{n+1}_{>0}$. For an exact left inverse $L \in \mathbb{Q}^{n\times(n+1)}$ satisfying $LB = I_n$, define

$$\mathcal{C}_w(L) = \max_{1\le i\le n} \sum_{j=1}^{n+1} w_j |L_{ij}|. \tag{AC1}$$

Thus $\mathcal{C}_w(L)$ is exactly the worst-case amplification from independent interval uncertainties $|e_j| \le w_j\epsilon$ in the input vector to $\|Le\|_\infty/\epsilon$. For each row index s such that deleting row s of B yields an invertible $n \times n$ matrix $B_{\widehat{s}}$, let $L^{(s)}$ be its inverse embedded in $n+1$ columns by placing a zero in column s.

Theorem 45 (Finite exact minimax left inverse)**.** *For every full-column-rank $(n+1)\times n$ matrix B and every strictly positive vector w, the optimal value over* all real *linear left inverses is*

$$\boxed{\min_{LB=I_n} \mathcal{C}_w(L) = \max_{1\le i\le n} \min_{s:\det B_{\widehat{s}}\neq 0} \sum_{j=1}^{n+1} w_j |L^{(s)}_{ij}|.} \tag{AC2}$$

A minimizer can be assembled row by row from (possibly different) $L^{(s)}$. In particular, for rational data it is rational, and the optimal value is certified by at most $n+1$ finite exact matrix inversions. These claims do not assert that the selected reference states are globally optimal, or that the resulting estimator is minimax among nonlinear procedures.

Proof. The ith row ℓ of a left inverse must satisfy $B^{\mathsf{T}}\ell = e_i$. Since $\operatorname{rank} B = n$ and B has $n+1$ rows, $\ker(B^{\mathsf{T}})$ is one-dimensional. Fix a feasible $\ell^{(0)}$ and a nonzero vector h spanning that kernel. Every feasible row is $\ell(t) = \ell^{(0)} + th$, $t \in \mathbb{R}$. The function $\phi_i(t) = \sum_j w_j|\ell^{(0)}_j + th_j|$ is coercive, convex and piecewise affine, so one of its minima lies at a breakpoint $t = -\ell^{(0)}_s/h_s$ with $h_s \neq 0$. At that point $\ell_s = 0$. The cofactor expansion for a one-dimensional left nullspace gives $h_s \neq 0$ if and only if $B_{\widehat{s}}$ is nonsingular. Consequently the unique solution of $B^{\mathsf{T}}\ell = e_i$ with $\ell_s = 0$ is row i of $L^{(s)}$; all optimal row candidates are represented in the finite right-hand minimum of (AC2). The rows of L obey independent linear constraints, so minimizing their maximum weighted ℓ_1 cost means minimizing each row separately. Taking the maximum of these minima proves the identity. The argument includes any vanishing cofactors by excluding the corresponding singular submatrices. □

Proposition 11 (Exact improvement of two published synthetic designs)**.** *Fix the projective gauge $A_{33} = 1$, the bearing equations and pivot convention of Appendix H, and uniform weights $w_j = 1$.*

Let B_{18} be the rational six-probe matrix of Appendix G and B_{19} that of Appendix H. Exhaustive rational enumeration of the twelve deletions and the construction of Theorem 45 give

$$\min_s \|L_{18}^{(s)}\|_{\infty\to\infty} = \frac{41}{4}, \qquad \min_{LB_{18}=I} \|L\|_{\infty\to\infty} = 10, \tag{AC3}$$

$$\min_s \|L_{19}^{(s)}\|_{\infty\to\infty} = \frac{127525}{31104}, \qquad \min_{LB_{19}=I} \|L\|_{\infty\to\infty} = \frac{14563561}{3950208}. \tag{AC4}$$

In the second line, the exact gain relative to the old single-deletion estimator is

$$\frac{127525/31104}{14563561/3950208} = \frac{16195675}{14563561} > \frac{11}{10}, \tag{AC5}$$

while the joint improvement relative to the original six-probe estimator is $\frac{41/4}{14563561/3950208} = \frac{40489632}{14563561} > \frac{27}{10}$. *For the new probes, the zero-based indices of deleted equations for the eleven parameter rows are*

$$(6, 6, 0, 2, 0, 1, 4, 6, 6, 10, 6). \tag{AC6}$$

The maximal row cost occurs at parameter index four (zero-based). The baseline and improved left inverses are rational and satisfy $LB = I_{11}$ in all 121 entries. The optimum is exact for the fixed matrices and linear estimators with the stated componentwise norm.

Proof. Form the twelve rational rows from the six stated reference vectors, for each omitted row invert the remaining matrix by exact rational Gauss–Jordan elimination when nonsingular, and embed a zero in that column. The independent checker computes every one of the 11×12 row costs. It verifies the two exact values and selected row indices of (AC3)–(AC6), the complete $LB = I_{11}$ identity and nonnegative gaps from each candidate to the minimal cost in its row. Theorem 45 then upgrades this finite computation from a candidate comparison to a global minimum within the entire space of left inverses for each fixed B. □

Corollary 20 (Error-aware recovery without discarding equations)**.** *Suppose the true affine parameters z obey $Bz = q$ with known componentwise error bounds on observed $(\widehat{B}, \widehat{q})$: $|\widehat{q}_j - q_j| \le \epsilon f_j$ and $\sum_k |\widehat{B}_{jk} - B_{jk}|\,|z_k| \le \epsilon e_j$. If $\widehat{B}$ has full column rank, let $\widehat{L}$ be the weighted minimax left inverse from Theorem 45 for $w_j = f_j + e_j > 0$, and set $\widehat{z} = \widehat{L}\widehat{q}$. Then*

$$\|\widehat{z} - z\|_\infty \le \epsilon\, \mathcal{C}_w(\widehat{L}). \tag{AC7}$$

The full twelve observed rows may be inconsistent due to noise; no fictitious exact rank-eleven fit is imposed. The estimator and its deterministic certificate remain computable from rational observations. The errors and the structural affine-readout model must still be justified by independent calibration.

Proof. Since $\widehat{L}\widehat{B} = I_n$, $\widehat{z} - z = \widehat{L}[(\widehat{q} - q) - (\widehat{B} - B)z]$. Applying the weighted induced norm to the independently justified per-row error bounds proves (AC7). □

Reproducibility and limits. The minimal checker `verifier/verify_rowwise_minimax_v20.py` uses only exact rational arithmetic and verifies both matrices, all valid leave-one-out blocks, the new rowwise left inverses, unequal rational weights, and a deliberately singular input. It performs 36 fixed-design perturbation checks and six tests in which the left inverse is reconstructed from the perturbed observed matrix. The numerical constants concern synthetic calibration data, not a physical device or a newly proven open conjecture. The finite minimax identity follows from classical convex piecewise-linear optimization; the application here is an explicit precision improvement for the existing certification pipeline.

ACKNOWLEDGMENTS

Substantial assistance from OpenAI ChatGPT (GPT-6, October 2026) was used for mathematical exploration, drafting, code and figure generation, and literature searches. The human authors are responsible for verifying the proofs, claims of priority, data provenance and appropriate research attribution before submission. The work reports synthetic mathematical examples; no independently recorded laboratory data were analysed. Formal Lean verification and external peer review are not claimed.

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